STAT 38100
Measure-Theoretic Probability I

Winter 2026

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Measure Spaces 1

1.1 Introduction

Example 1.1.1 (Non-measurable event).

Consider a uniform probability measure (measure of 1) on $[0, 2\pi)$. For example,

$$\mathbb{P}\left[[0,1]\right] = \frac{1}{2\pi}, \quad \mathbb{P}\left[[0,1] \cup [2,3]\right] = \frac{1}{\pi}.$$

Define the function T by

$$T(x) = \begin{cases} x+1, & x+1 < 2\pi \\ x+1-2\pi, & x+1 \geq 2\pi \end{cases}.$$

Thinking of a circle, the operator moves the point by one unit. With this operator, we can define an equivalence relation: we will say $x \sim y$ iff $T(x) = T(y)$ or $T(y) = T(x)$.

The equivalence relation leads to a partition $(A_\lambda : \lambda)$; i.e.,

$$A_\lambda \cap A'_\lambda = \varnothing, \;\; \lambda \neq \lambda', \quad \cup_\lambda A_\lambda = [0, 2\pi)$$

such that $x, y \in A_\lambda \iff x \sim y$.

For every λ, select an element $x_\lambda \in A_\lambda$ and collect them in the *representative set* $X = \{x_\lambda : \lambda\}$.[a] Define the translation

$$X + i = \{x_\lambda + i : \lambda\}.$$

Note that $\{X + i \bmod 2\pi : i \in \mathbb{Z}\}$ is a partition of $[0, 2\pi)$.

We know that

$$1 = \mathbb{P}\left[[0,2\pi)\right] = \sum_{i\in\mathbb{Z}} \mathbb{P}\left[(X+i)\right]$$
$$= \sum_{i\in\mathbb{Z}} \mathbb{P}\left[X\right].$$

Then what is $\mathbb{P}[X]$? It is not well-defined; X is not a measurable event.

[a] Here we use the Axiom of Choice.

We need to restrict ourselves to measurable subsets to have our notion of probability be well-defined; hence measure theory is necessary.

Definition 1.1.2 (σ-algebra).

Let Ω denote the sample space and $2^\Omega := \{\text{subsets of } \Omega\}$.
Given Ω, $\mathscr{F} \subseteq 2^\Omega$ is a σ-field (σ-algebra) if

(i) $\varnothing \in \mathscr{F}$

(ii) If $A \in \mathscr{F}$, then $A^c \in \mathscr{F}$

(iii) If $A_n \in \mathscr{F}$, then $\cup_n A_n \in \mathscr{F}$ (countable union).

Trivial examples include 2^Ω and $\{\varnothing, \Omega\}$.

Definition 1.1.3 (Generated σ-field).

Given $\mathscr{E} \subseteq 2^\Omega$, define $\sigma(\mathscr{E})$ to be the smallest σ-field containing $\mathscr{E}$. In other words, for any σ-field $\mathscr{F} \supseteq \mathscr{E}$, then $\sigma(\mathscr{E}) \subseteq \mathscr{F}$.
Note that $\sigma(\mathscr{E})$ is well-defined, because we can put

$$\sigma(\mathscr{E}) = \bigcap \{\mathscr{F} \subseteq 2^\Omega : \mathscr{F} \text{ is a } \sigma\text{-field and } \mathscr{F} \supseteq \mathscr{E}\}.$$

We use the following lemma, whose proof is immediate.

Lemma 1.1.4.

If $\mathscr{F}_\lambda$ is a σ-field for each λ, then $\cap_\lambda \mathscr{F}_\lambda$ is a σ-field.

Definition 1.1.5 (Monotone sequences of sets and their limits).

We say that A_n is an increasing sequence ($A_n \uparrow$) if

$$A_1 \subseteq A_2 \subseteq \dots$$

and similarly for decreasing sequences of sets.
If $A_n \uparrow$, define $\lim_{n\to\infty} A_n := \cup_n A_n$, and if $A_n \downarrow$, define $\lim_{n\to\infty} A_n := \cap_n A_n$.

We can define the limit of an arbitrary, non-monotone sequence.

Definition 1.1.6 ($\liminf$ and $\limsup$ of sequences).

For arbitrary A_n, define

$$\liminf_{n\to\infty} A_n = \cup_k \cap_{n\geq k} A_n, \quad \limsup_{n\to\infty} A_n = \cap_k \cup_{n\geq k} A_n.$$

If $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$, then the limit exists and is the common value.

Remark 1.1.7. Thinking hard, these definitions make sense. Alternatively,

$$\begin{aligned} x \in \liminf_{n\to\infty} A_n &\iff \exists\, k \text{ s.t. } \forall\, n \geq k,\ x \in A_n \\ &\iff x \text{ belongs to all but finitely many } A_n \\ x \in \limsup_{n\to\infty} A_n &\iff \forall\, k, \exists\, n \geq k \text{ s.t. } x \in A_n \\ &\iff x \text{ belongs to infinitely many } A_n. \end{aligned}$$

Furthermore, $\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$.

Definition 1.1.8 (π-system, monotone class, λ-system, field).

We say that $\mathscr{P}$ is a π-system if it is closed under intersection; i.e., for any $A, B \in \mathscr{P}$, $A \cap B \in \mathscr{P}$.
We say that $\mathscr{M}$ is a monotone class if for any monotone sequence (A_n), $\lim_{n\to\infty} A_n \in \mathscr{M}$. Monotone classes are closed under the limit operation.
We say that $\mathscr{L}$ is a λ-system if

(i) $\Omega \in \mathscr{L}$,

(ii) If $A, B \in \mathscr{L}$ such that $A \supseteq B$, then $A \setminus B \in \mathscr{L}$, and

(iii) If (A_n) is increasing sequence in $\mathscr{L}$, then $\lim_{n\to\infty} A_n \in \mathscr{L}$. $\mathscr{L}$ is closed under countable unions, but only for increasing sequences.

$\mathscr{A}$ is a field (algebra) if

(i) $\varnothing \in \mathscr{A}$,

(ii) $\forall\, A \in \mathscr{A}$, $A^c \in \mathscr{A}$, and

(iii) $\forall\, A, B \in \mathscr{A}$, $A \cup B \in \mathscr{A}$.

The first and fourth definitions deal with finite operations; the second and third include limits.

This assumption is weaker than the one in the definition of a σ-field.

Theorem 1.1.9.

(Monotone class.) If $\mathscr{A}$ is a field, then $m(\mathscr{A}) = \sigma(\mathscr{A})$.

Proof of Theorem 1.1.9. The inclusion $m(\mathscr{A}) \subseteq \sigma(\mathscr{A})$ is straightforward. By definition, $\sigma(\mathscr{A})$ contains $\mathscr{A}$. Also, $\sigma(\mathscr{A})$ is a monotone class. Since $m(\mathscr{A})$ is the smallest monotone class containing $\mathscr{A}$, we get $m(\mathscr{A}) \subseteq \sigma(\mathscr{A})$.

$m(\mathscr{A})$ is defined to be the smallest monotone class containing $\mathscr{A}$.

For the reverse inclusion, it suffices to show that $m(\mathscr{A})$ is a σ-field containing $\mathscr{A}$. It contains $\mathscr{A}$ by definition.

(i) Since $\mathscr{A}$ is a field, $\varnothing \in \mathscr{A} \subseteq m(\mathscr{A})$.

(ii) (Closure under complements.) Define

$$M_0 := \{A \in m(\mathscr{A}) : A^c \in m(\mathscr{A})\}.$$

We claim M_0 is a monotone class containing $\mathscr{A}$.

(a) $\mathscr{A} \subseteq M_0$. If $A \in \mathscr{A}$, then $A^c \in \mathscr{A} \subseteq m(\mathscr{A})$, hence $A \in M_0$.

(b) M_0 is a monotone class. Suppose $A_n \uparrow A$ with each $A_n \in M_0$. Then $A_n^c \downarrow A^c$, and since each $A_n^c \in m(\mathscr{A})$ and $m(\mathscr{A})$ is a monotone class, we have $A^c \in m(\mathscr{A})$. Also $A \in m(\mathscr{A})$ because $A_n \uparrow A$. Hence $A \in M_0$. The argument for $A_n \downarrow A$ is identical (then $A_n^c \uparrow A^c$).

Thus M_0 is a monotone class containing $\mathscr{A}$, so by minimality $m(\mathscr{A}) \subseteq M_0$. In particular, $m(\mathscr{A})$ is closed under complements.

(iii) (Closure under countable unions.) First we show closure under finite unions.

Define

$$M_1 := \{A \in m(\mathscr{A}) : \forall\, B \in \mathscr{A},\ A \cup B \in m(\mathscr{A})\}.$$

Clearly $M_1 \subseteq m(\mathscr{A})$. We check that M_1 is a monotone class containing $\mathscr{A}$.

(a) $\mathscr{A} \subseteq M_1$. If $A \in \mathscr{A}$ and $B \in \mathscr{A}$, then $A \cup B \in \mathscr{A} \subseteq m(\mathscr{A})$ (since $\mathscr{A}$ is a field). Hence $A \in M_1$, so $\mathscr{A} \subseteq M_1$.

(b) M_1 is a monotone class. Suppose $A_n \uparrow A$ with $A_n \in M_1$ and fix a $B \in \mathscr{A}$. Then $A_n \cup B \uparrow A \cup B$. Since each $A_n \cup B \in m(\mathscr{A})$ and $m(\mathscr{A})$ is a monotone

class, $A \cup B \in m(\mathscr{A})$. B was arbitrary in $\mathscr{A}$, so $A \in M_1$. The decreasing case is the same.

Therefore M_1 is a monotone class containing $\mathscr{A}$, hence $m(\mathscr{A}) \subseteq M_1$. Equivalently,

$$\forall A \in m(\mathscr{A}),\ \forall B \in \mathscr{A}, \qquad A \cup B \in m(\mathscr{A}).$$

Now define

$$M_2 := \{A \in m(\mathscr{A}) : \forall B \in m(\mathscr{A}),\ A \cup B \in m(\mathscr{A})\}.$$

Again $M_2 \subseteq m(\mathscr{A})$. We show M_2 is a monotone class containing $\mathscr{A}$.

(a) $\mathscr{A} \subseteq M_2$. Fix $A \in \mathscr{A}$ and $B \in m(\mathscr{A})$. From the conclusion about M_1 (with roles swapped), $B \cup A \in m(\mathscr{A})$, hence $A \cup B \in m(\mathscr{A})$. Thus $A \in M_2$.

(b) M_2 is a monotone class. If $A_n \uparrow A$ with $A_n \in M_2$ and $B \in m(\mathscr{A})$, then $A_n \cup B \uparrow A \cup B$. Since each $A_n \cup B \in m(\mathscr{A})$ and $m(\mathscr{A})$ is a monotone class, $A \cup B \in m(\mathscr{A})$, so $A \in M_2$. The decreasing case is identical.

Thus M_2 is a monotone class containing $\mathscr{A}$, so $m(\mathscr{A}) \subseteq M_2$. In particular, $m(\mathscr{A})$ is closed under finite unions.

Finally, let $(A_n)_{n \geq 1} \subseteq m(\mathscr{A})$ and set $U_m := \bigcup_{n=1}^m A_n$. By finite-union closure, $U_m \in m(\mathscr{A})$ for each m, and $U_m \uparrow \bigcup_{n=1}^\infty A_n$. Since $m(\mathscr{A})$ is a monotone class, $\bigcup_{n=1}^\infty A_n \in m(\mathscr{A})$.

We have shown $m(\mathscr{A})$ is a σ-field containing $\mathscr{A}$, hence $\sigma(\mathscr{A}) \subseteq m(\mathscr{A})$. Combined with the first inclusion, $m(\mathscr{A}) = \sigma(\mathscr{A})$. □

Theorem 1.1.10.

(π-λ.) If $\mathscr{P}$ is a π-system, then

$$l(\mathscr{P}) = \sigma(\mathscr{P}).$$

This and the previous theorem give us two different ways to generate a σ-field. We can start with something more complicated (field) or less complicated (a π-system). In this course, the monotone class theorem will be more commonly used. The choice comes down to convenience.

1.2 σ-fields

Definition 1.2.1 (Topology).

Let Ω be a set. A *topology* on Ω is a collection $\mathcal{T} \subseteq 2^\Omega$ such that:

(i) $\varnothing \in \mathcal{T}$ and $\Omega \in \mathcal{T}$,

(ii) if $U, V \in \mathcal{T}$, then $U \cap V \in \mathcal{T}$,

(iii) if $\{U_i\}_{i \in I} \subseteq \mathcal{T}$, then $\bigcup_{i \in I} U_i \in \mathcal{T}$.

The sets in $\mathcal{T}$ are called *open sets*, and $(\Omega, \mathcal{T})$ is a *topological space*.

Definition 1.2.2 (Measurable space).

A *measurable space* is a pair $(\Omega, \mathscr{F})$ where Ω is a set and $\mathscr{F} \subseteq 2^\Omega$ is a σ-algebra on Ω, i.e.:

(i) $\Omega \in \mathscr{F}$,

(ii) if $A \in \mathscr{F}$ then $A^c \in \mathscr{F}$,

(iii) if $A_1, A_2, \ldots \in \mathscr{F}$ then $\bigcup_{n=1}^\infty A_n \in \mathscr{F}$.

A measure space is a measurable space equipped with a measure.

Definition 1.2.3 (Borel σ-algebra).

If $(\Omega, \mathcal{T})$ is a topological space, the *Borel σ-algebra* on Ω is

$$\mathscr{B}(\Omega) := \sigma(\mathcal{T}),$$

the smallest σ-algebra containing all open sets. In particular, for $\mathbb{R}$ with its usual topology,

$$\mathscr{B}_\mathbb{R} := \mathscr{B}(\mathbb{R}) = \sigma\big(\{U \subseteq \mathbb{R} : U \text{ is open}\}\big) = \sigma\big(\{[a, \infty) : a \in \mathbb{R}\}\big).$$

Definition 1.2.4 (Measure).

Suppose $\mathscr{E}$ is a set collection such that $\varnothing \in \mathscr{E} \subseteq 2^\Omega$. We say $\mu : \mathscr{E} \to [0, \infty]$ is a measure if:

(i) $\mu(\varnothing) = 0$

(ii) For disjoint $A_n \in \mathscr{E}$ such that $\cup_n A_n \in \mathscr{E}$,

$$\mu(\cup_n A_n) = \sum_n \mu(A_n).$$

In particular, if $\mu(\Omega) < \infty$, μ is called a finite measure. If $\mu(\Omega) = 1$, then it is a probability measure. If for any event $A \in \mathscr{E}$, there exists $A_n \in \mathscr{E}$ such that $\cup_n A_n \supseteq A$ and $\mu(A_n) < \infty$ for each n, we call μ is a σ-finite measure. Any set in a σ-finite measure is the countable union of sets each of finite measure.

Definition 1.2.5 (Probability space).

We call $(\Omega, \mathscr{F}, \mu)$ a probability space if $\mu(\Omega) = 1$.

Remark 1.2.6 (Properties of measure). Suppose μ is a measure on a σ-field $\mathscr{E}$.[1] Then

(i) (Monotonicity.) If $A \subseteq B$, then $\mu(A) \leq \mu(B)$.

(ii) (Continuity from below.) If (A_n) is an increasing sequence ($A_n \subseteq A_{n+1}$), then

$$\lim_{n\to\infty} \mu(A_n) = \mu\Big(\lim_{n\to\infty} A_n\Big) \qquad \text{where} \quad \lim_{n\to\infty} A_n := \bigcup_{n=1}^{\infty} A_n.$$

(iii) (Continuity from above.) If (A_n) is a decreasing sequence ($A_{n+1} \subseteq A_n$) and $\mu(A_1) < \infty$, then

$$\mu\Big(\lim_{n\to\infty} A_n\Big) = \lim_{n\to\infty} \mu(A_n) \qquad \text{where} \quad \lim_{n\to\infty} A_n := \bigcap_{n=1}^{\infty} A_n.$$

Proof of Theorem 1.1.10. (i) If $A \subseteq B$, then $B = A \amalg (B \setminus A)$, so

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

by nonnegativity.

(ii) Assume $A_n \uparrow$ and set $A := \bigcup_{n=1}^{\infty} A_n$. Define

$$D_1 := A_1, \qquad D_m := A_m \setminus A_{m-1} \quad (m \geq 2),$$

(with the convention $A_0 := \varnothing$). Then the D_m are pairwise disjoint and for each n,

$$A_n = \amalg_{m=1}^{n} D_m, \qquad \text{and} \qquad A = \amalg_{m=1}^{\infty} D_m.$$

By countable additivity,

$$\mu(A) = \sum_{m=1}^{\infty} \mu(D_m).$$

On the other hand, for each n,

$$\mu(A_n) = \mu(\amalg_{m=1}^{n} D_m) = \sum_{m=1}^{n} \mu(D_m),$$

so taking $n \to \infty$ gives

$$\lim_{n\to\infty} \mu(A_n) = \lim_{n\to\infty} \sum_{m=1}^{n} \mu(D_m) = \sum_{m=1}^{\infty} \mu(D_m) = \mu(A) = \mu\left(\bigcup_{n=1}^{\infty} A_n\right).$$

[1] Note now μ is not a premeasure on an arbitrary set collection; it is defined on a σ-field for these properties.

(iii) Assume $A_n \downarrow$ and $\mu(A_1) < \infty$. Consider $B_n := A_1 \setminus A_n$. Then (B_n) is increasing, so by (ii),

$$\mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \lim_{n\to\infty} \mu(B_n).$$

But

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A_1 \setminus A_n) = A_1 \cap \bigcup_{n=1}^{\infty} A_n^c = A_1 \cap \left(\bigcap_{n=1}^{\infty} A_n\right)^c = A_1 \setminus \bigcap_{n=1}^{\infty} A_n.$$

Also, since $A_n \subseteq A_1$ and $\mu(A_1) < \infty$,

$$\mu(B_n) = \mu(A_1 \setminus A_n) = \mu(A_1) - \mu(A_n).$$

Therefore,

$$\mu\left(A_1 \setminus \bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} (\mu(A_1) - \mu(A_n)) = \mu(A_1) - \lim_{n\to\infty} \mu(A_n).$$

Finally, since

$$\mu\left(A_1 \setminus \bigcap_{n=1}^{\infty} A_n\right) = \mu(A_1) - \mu\left(\bigcap_{n=1}^{\infty} A_n\right),$$

again using $\mu(A_1) < \infty$, we conclude

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n).$$

□

1.3 Constructing the Uniform Measure

Given a premeasure[2] on a small family of sets (a finite algebra/field), how can we get an extension to a useful σ-field?

We will first construct a measure μ on a field $\mathscr{A}$. Then we will extend μ from $\mathscr{A}$ to 2^Ω through outer measure as an approximation of μ. Finally, we will restrict the domain to $\mathscr{F}_\tau \subseteq 2^\Omega$ so that $\mathscr{F}_\tau$ is a σ-field and μ is a measure on that field.

Ultimately, we want to construct the uniform measure on $\Big([0,1], \mathscr{B}_{[0,1]}\Big)$.

[2] For example, a finitely/countably additive set function like one involving volumes of rectangles.

Example 1.3.1.

With $\Omega = [0, 1]$, put $\mathscr{A} = \{\text{finite unions of intervals}\}$. It is easy to construct a measure on $\mathscr{A}$ by considering lengths of disjoint intervals in its elements.

How do we extend the measure from $\mathscr{A}$ to $\sigma(\mathscr{A})$? If we are able to do so, it is unique? We first answer the second question.

Theorem 1.3.2.

Suppose $\mathscr{A}$ is a field and μ and ν are two finite measures on $\sigma(\mathscr{A})$ such that for every $A \in \mathscr{A}$, $\mu(A) = \nu(A)$. Then for every $B \in \sigma(\mathscr{A})$,

$$\mu(B) = \nu(B).$$

Proof of Theorem 1.3.2. Define

$$\mathscr{H} = \{A \in \sigma(\mathscr{A}) : \mu(A) = \nu(A)\}.$$

We know that $\mathscr{A} \subseteq \mathscr{H} \subseteq \sigma(\mathscr{A})$. Suppose we can show that $\mathscr{H}$ is a monotone class. Then it must contain the monotone class generated by $\mathscr{A}$, which is $\sigma(\mathscr{A})$ by the monotone class theorem. This would show equality. Let's prove it!

Suppose we have a monotone sequence A_n in $\mathscr{H}$. Then by definition of $\mathscr{H}$,

$$\begin{aligned} \mu(A_n) &= \nu(A_n) \\ \implies \lim_{n\to\infty} \mu(A_n) &= \lim_{n\to\infty} \nu(A_n) \\ \mu(\lim_{n\to\infty} A_n) &= \nu\left(\lim_{n\to\infty} A_n\right) \\ \implies \lim_{n\to\infty} A_n &\in \mathscr{H}, \end{aligned}$$

so $\mathscr{H}$ is a monotone class. $\square$

Note the existence of $\lim_{n\to\infty} \mu(A_n), \nu(A_n)$ is guaranteed in $[0, \infty]$ by the sequence being monotone.

Definition 1.3.3 (Outer measure).

We say that $\tau : 2^\Omega \to [0, \infty]$ is an outer measure if:

(i) $\tau(\varnothing) = 0$,

(ii) If $A \subseteq B$, then $\tau(A) \leq \tau(B)$, and

(iii) $\tau\left(\cup_n A_n\right) \leq \sum_n \tau(A_n)$.

The definition of outer measure differs from that of measure in three ways: (i) outer measure is defined on 2^Ω instead of a subset; (ii) outer measure must explicitly be monotone, while measure does not; (iii) outer measure is only required to be countably *sub*additive.

Theorem 1.3.4 (Construction of outer measure).

Suppose $\varnothing \in \mathscr{E} \subseteq 2^{\Omega}$ and $\mu : \mathscr{E} \to [0, \infty]$ is such that $\mu(\varnothing) = 0$. For every $A \in 2^{\Omega}$, define

$$\tau(A) = \inf\{\sum_n \mu(A_n) : A_n \in \mathscr{E},\ A \subseteq \cup_n A_n\}.$$

Then τ is an outer measure.

Proof of Theorem 1.3.4. (i) Since $m(\varnothing) = 0$ and $\varnothing \subseteq \varnothing \cup \varnothing \cup \ldots$,

$$\tau(\varnothing) \leq \sum_{n=1}^{\infty} m(\varnothing) = 0.$$

τ is an inf over nonnegative numbers, so $\tau(\varnothing) = 0$.

(ii) Any covering of B is also a covering of A.

(iii) For any $\epsilon > 0$ and A_n, there exists $\{B_{n,k}\}_k$ such that $B_{n,k} \in \mathscr{E}$, $\cup_k B_{n,k} \supseteq A_n$, and

$$\tau(A_n) + \frac{\epsilon}{2^n} \geq \sum_k \mu(B_{n,k}).$$

We know

$$\bigcup_n A_n \subseteq \bigcup_n \bigcup_k B_{n,k},$$

so

$$\tau\left(\bigcup_n A_n\right) \leq \sum_n \sum_k \mu(B_{n,k}) \leq \sum_n \tau(A_n) + \epsilon.$$

We can send $\epsilon \to 0$ to conclude.

□

Think of $\mathscr{E}$ as a field, even though the theorem does not require this. For instance, $\mathscr{E}$ might be all finite unions of rectangles.

If there does not exist a countable covering of A, convention is to put $\tau(A) = \infty$.

Now that we have an outer measure on 2^{Ω}, how do we restrict the domain to where we have countable additivity of disjoint sets?

Theorem 1.3.5 (Restriction to Carathéodory-measurable sets).

Let τ be an outer measure. Define

$$\mathscr{F}_\tau := \{A \subseteq \Omega : \forall\ B \subseteq \Omega, \tau(B) = \tau(B \cap A) + \tau(B \cap A^c)\}.$$

Then $(\Omega, \mathscr{F}_\tau, \tau)$ is a measure space.

Thinking of A as a knife that splits any set B into $A \cap B$ and $A^c \cap B$, we may think that A is a good knife if it divides all sets in Ω well.

Proof of Theorem 1.3.5. Step I ($\mathscr{F}_\tau$ is a σ-field). We see that $\varnothing \in \mathscr{F}_\tau$, as for any $B \supseteq \Omega$,

$$\tau(B) = \underbrace{\tau(B \cap \varnothing)}_{=\tau(\varnothing)=0} + \underbrace{\tau(B \cap \varnothing^c)}_{=\tau(B)}.$$

Furthermore, because if $\tau(B) = \tau(A \cap B) + \tau(A^c \cap B) = \tau(A^c \cap B) + \tau(A \cap B)$, $\mathscr{F}_\tau$ is closed under complementation.

We will now show that if $A_1, A_2 \in \mathscr{F}_\tau$, $A_1 \cap A_2 \in \mathscr{F}_\tau$. For any $B \in \Omega$,

There is no good reason for proving closure under intersection instead of union. The proof for showing closure under finite union is analogous, so it might be a good idea to do it as an exercise.

$$\begin{aligned}\tau(B) &= \tau(B \cap A_1) + \tau(B \cap A_1^c) \\ &= \tau(B \cap A_1 \cap A_2) + \underbrace{\tau(B \cap A_1 \cap A_2^c) + \tau(B \cap A_1^c \cap A_2) + \tau(B \cap A_1^c \cap A_2^c)}_{\text{goal: make into } B\cap(A_1\cap A_2)^c}.\end{aligned}$$

Showing the union of these three terms is equal to $B \cap (A_1 \cap A_2)^c$ is a bit difficult, so let's try decomposing the single set.

$$\begin{aligned}\tau\left(B \cap (A_1 \cap A_2)^c\right) &= \tau\left(B \cap (A_1^c \cup A_2^c)\right) \\ &= \underbrace{\tau\left(B \cap (A_2^c \cap A_1)\right)}_{\text{one of goal terms}} + \tau\left(B \cap (A_1^c)\right) \\ &= \tau\left(B \cap (A_2^c \cap A_1)\right) + \tau(B \cap A_1^c \cap A_2) + \tau(B \cap A_1^c \cap A_2^c),\end{aligned}$$

where we cut with A_1 for the second equality and by A_2 for the third.

Thus $\mathscr{F}_\tau$ is closed under finite intersection. Because we already showed it is closed under complementation, it is closed under finite union as well.

It just remains to show that $\mathscr{F}_\tau$ is closed under countable union. Suppose $A_n \in \mathscr{F}_\tau$. WLOG, we can assume that $\{A_n\}$ is pairwise disjoint. We know from $\mathscr{F}_\tau$ being closed under finite union that for any m, $\cup_{n=1}^m A_n \in \mathscr{F}_\tau$. For any $B \subseteq \Omega$,

$$\begin{aligned}\tau(B) &= \tau\left(B \cap \left(\bigcup_{n=1}^m A_n\right)\right) + \tau\left(B \cap \left(\bigcup_{n=1}^m A_n\right)^c\right) \\ &= \tau(B \cap A_1) + \tau\left(B \cap \left(\bigcup_{n=2}^m A_n\right)\right) + \tau\left(B \cap \left(\bigcup_{n=1}^m A_n\right)^c\right),\end{aligned}$$

since we can cut the first term by A_1 and the A_ns are disjoint. Iterate on this.

Then by monotonicity of outer measure,

$$\begin{aligned}
\tau(B) &= \sum_{n=1}^{m} \tau(B \cap A_n) + \tau\left(B \cap \left(\bigcup_{n=1}^{m} A_n\right)^c\right) \\
&\geq \sum_{n=1}^{m} \tau(B \cap A_n) + \tau\left(B \cap \underbrace{\left(\bigcup_{n=1}^{\infty} A_n\right)^c}_{\text{now smaller}}\right) \\
&\geq \sum_{n=1}^{\infty} \tau(B \cap A_n) + \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)^c\right) \\
&\geq \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)\right) + \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)^c\right),
\end{aligned}$$

where the last inequality follows from countable subadditivity of τ. However, also by countable (in this case, finite) subadditivity,

$$\tau(B) \leq \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)^c\right) + \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)^c\right),$$

so $\cup_{n=1}^{\infty} A_n$ cuts arbitrary $B \subseteq \Omega$ well, so $\cup_{n=1}^{\infty} A_n \in \mathscr{F}_\tau$ and $\mathscr{F}_\tau$ is closed under countable union.

Step II (τ is a measure on $\mathscr{F}_\tau$). We check $\tau(\varnothing) = 0$ because it is an outer measure. The only other condition for τ being a measure is countable additivity on disjoint sets.

Again by τ as an outer measure, for disjoint $\{A_n\}$,

$$\tau\left(\cup_n A_n\right) \leq \sum_n \tau(A_n).$$

For the other direction, take $B = \cup_n A_n$, so

$$\begin{aligned}
\tau(B) = \tau(\cup_n A_n) &\leq \sum_{n=1}^{\infty} \tau(A_n) \\
&= \sum_{n=1}^{\infty} \tau(A_n \cap B) + \tau\left(A_n \cap B^c\right) \\
&\leq \sum_{n=1}^{\infty} \tau(A_n) + \tau(0) \\
&= \sum_{n=1}^{\infty} \tau(A_n).
\end{aligned}$$

$\mathscr{F}_\tau$ being a σ-field on Ω and τ being a valid measure on $\mathscr{F}_\tau$ satisfies the definition of $(\Omega, \mathscr{F}_\tau, \tau)$ being a measure space, so we are done. □

Theorem 1.3.6 (Unique extension of measure).

Suppose μ is a finite measure on a field $\mathscr{A}$. Then there exists a unique extension of μ from $\mathscr{A}$ to $\sigma(\mathscr{A})$.

Proof of Theorem 1.3.6. For every $A \subseteq \Omega$, define

$$\tau(A) = \inf\{\sum_n \mu(B_n) : B_n \in \mathscr{A},\ \cup_n B_n \supseteq A\}.$$

By our earlier theorem, τ is an outer measure.

Defining $\mathscr{F}_\tau$ according to the Caratheodory criterion, we know $(\Omega, \mathscr{F}_\tau, \tau)$ is a measure space.

Since we already showed that a measure on a field is unique on a σ-algebra generated by that field, we just need to show that τ is actually an extension of μ to $\sigma(\mathscr{A})$. This entails showing that for any $A \in \mathscr{A}$, $\mu(A) = \tau(A)$ and that $\mathscr{F}_\tau \supseteq \sigma(\mathscr{A})$.

This is why we need μ to be a measure on a *field*.

Fix $A \in \mathscr{A}$. Since $A \subseteq A \cup \varnothing \cup \varnothing \cup \ldots$,

$$\tau(A) \leq \mu(A) + \varnothing + \varnothing + \cdots = \mu(A).$$

For the other direction of the inequality, note that for any covering B_n of A,

$$\begin{aligned} \mu(A) &= \mu\left(\cup_n (A \cap B_n)\right) \\ &\leq \sum_n \mu(A \cap B_n) \\ &\leq \sum_n \mu(B_n), \end{aligned}$$

where the second inequality follows from monotonicity of μ. Taking an inf yields

$$\mu(A) \leq \tau(A).$$

Since $\mathscr{F}_\tau$ is a σ-field, in order to show that $\mathscr{F}_\tau \supseteq \sigma(\mathscr{A})$, we just need to prove that $A \subseteq \mathscr{F}_\tau$. Hence, we want to show for every $A \in \mathscr{A}$ and every $B \subseteq \Omega$, $\tau(B) = \tau(B \cap A) + \tau(B \cap A^c)$. The $\leq$ direction is given by the subadditivity of τ with $(B \cap A) \cup (B \cap A^c)$

Fix $\epsilon > 0$. There exists $B_n \in \mathscr{A}$ such that $\cup_n B_n \supseteq B$ and

$$\tau(B) + \epsilon \geq \sum_n \mu(B_n).$$

Then

The last two inequalities use only outer-measure properties. At this point we do not yet know, for example, that $B \cap A \in \mathscr{F}_\tau$.

$$
\begin{aligned}
\tau(B) + \epsilon &\geq \sum_n \mu(B_n) \\
&= \sum_n \left[\mu(B_n \cap A) + \mu(B_n \cap A^c)\right] \text{ by additivity of measure} \\
&= \sum_n \left[\tau(B_n \cap A) + \tau(B_n \cap A^c)\right] \text{ since } B_n, A, A^c \in \mathscr{A}, \text{ where } \mu, \tau \text{ agree} \\
&\geq \tau\left((\cup_n B_n) \cap A\right) + \tau\left((\cup_n B_n) \cap A^c\right) \text{ by subadditivity of outer measure} \\
&\geq \tau\left(B \cap A\right) + \tau(B \cap A^c) \text{ by monotonicity of outer measure.}
\end{aligned}
$$

Sending $\epsilon \to 0$,

$$\tau(B) \geq \tau(B \cap A) + \tau\left(B \cap A^c\right),$$

so we have shown

$$\tau(B) = \tau(B \cap A) + \tau(B \cap A^c)$$

for arbitrary $A \in \mathscr{A}$ and $B \subseteq \Omega$, so $\mathscr{A} \subseteq \mathscr{F}_\tau$, and hence $\sigma(\mathscr{A}) \subseteq \mathscr{F}_\tau$. $\square$

Now we finally consider $\left([0,1], \mathscr{B}_{[0,1]}\right)$ and $\mathscr{A}$ as the set of finite unions of intervals and μ the length of an interval. By the theorem, we can extend μ to be a measure on $\mathscr{F} \supseteq \sigma(\mathscr{A}) = \mathscr{B}_{[0,1]}$.

Definition 1.3.7 (Complete measure space).

A measure space $(\Omega, \mathscr{F}, \mu)$ is complete if

$$A \in \mathscr{F} \text{ and } \mu(A) = 0 \implies \forall\, B \subseteq A, B \in \mathscr{F}, .$$

i.e. if the σ-field contains all subsets of measure zero sets.

Theorem 1.3.8.

Given an outer measure τ, the measure space $(\Omega, \mathscr{F}_\tau, \tau)$ constructed from the Caratheodory criterion is a complete measure space.

Proof of Theorem 1.3.8. We already showed that $(\Omega, \mathscr{F}_\tau, \tau)$ is a measure space. Suppose $\tau(A) = 0$. For any $B \in \Omega$,

$$\tau(B) \leq \tau(B \cap A) + \tau(B \cap A^c)$$

is true by subadditivity.

By monotonicity of outer measure,

$$\tau(B \cap A^c) \leq \tau(B).$$

Also, $\tau(B \cap A) \leq \tau(A) = 0$, so

$$\tau(B) \geq \tau(B \cap A) + \tau(B \cap A^c).$$

Thus $\tau(B) = \tau(B \cap A) + \tau(B \cap A^c)$, for every $B \in \Omega$, so $A \in \mathscr{F}_\tau$. $\square$

We receive the fact that $\left|\mathscr{B}_{[0,1]}\right| = |[0,1]|$, which allows us to claim that $\left([0,1], \mathscr{B}_{[0,1]}, \mu\right)$, where μ is the uniform measure, is not a complete measure space.[3]

Definition 1.3.9 (Cantor set).

Define $C_0 = [0,1]$, $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, and recursively $C_n = \frac{C_{n-1}}{3} \cup \left(\frac{2}{3} + \frac{C_{n-1}}{3}\right)$. Then define the Cantor set $C = \cap_{n=0}^{\infty} C_n$.

Immediately, $C \in \mathscr{B}_{[0,1]}$, and $\mu(C) = \lim_{n \to \infty} \left(\frac{2}{3}\right)^n = 0$. However, $|C| = |[0,1]|$ because C bijects with binary expansions of numbers in $[0,1]$. Thus

$$\left|2^C\right| = \left|2^{[0,1]}\right| > |[0,1]| = \left|\mathscr{B}_{[0,1]}\right|.$$

The inequality is because for any set S, $\left|2^S\right| > |S|$.

Because this inequality is strict, there exists $A \in 2^C$ such that $A \notin \mathscr{B}_{[0,1]}$.

What is the connection between the result $\mathscr{F}_\tau$ of Caratheodory extension and $\mathscr{B}_{[0,1]}$?

Theorem 1.3.10 (Completion).

Let $(\Omega, \mathscr{F}, \mu)$ be a measure space. Define

$$\tilde{\mathscr{F}} = \{A \cup N : A \in \mathscr{F} \text{ and } \exists\ B \in \mathscr{F} \text{ s.t. } \mu(B) = 0 \text{ and } B \supseteq N\}.$$

$\mathscr{F}$ is a σ-field. For any $A \cup N \in \tilde{\mathscr{F}}$, defining

$$\tilde{\mu}(A \cup N) = \mu(A),$$

$\left(\Omega, \tilde{\mathscr{F}}, \tilde{\mu}\right)$ is a complete and well-defined measure space. Moreover, for any $A \in \mathscr{F}$, $\tilde{\mu}(A) = \mu(A)$; the complete measure space is an extension of the original measure space.
We call $\tilde{\mathscr{F}}$ the *completion* of $\mathscr{F}$.

$\tilde{\mathscr{F}}$ just adds all subsets of sets in $\mathscr{F}$ with measure 0.

Proof of Theorem 1.3.10. We need to check that $\tilde{\mathscr{F}}$ is a σ-field.

(i) Since $\varnothing = \varnothing \cup \varnothing$, where $\varnothing \in \mathscr{F}$ and $\varnothing \subseteq \varnothing$, $\varnothing \in \tilde{\mathscr{F}}$

$\tilde{\mathscr{F}}$ is the smallest completion of $\mathscr{F}$.

[3]This implies that the measure space we constructed via Caratheodory is strictly larger than this Borel measure space.

(ii) Suppose $A \in \mathscr{F}$ and $N \subseteq B \in \mathscr{F}$ such that $m(B) = 0$.[4] We need to show $(A \cup N)^c \in \tilde{F}$. WE want to write $(A \cup N)^c$ as the union of something in $\mathscr{F}$ and a subset of a set of measure 0.

We can write[5]

$$(A \cup N)^c = (A \cup B)^c \cup (B \setminus (A \cup N)).$$

Since $A, B \in \mathscr{F} \implies (A \cup B)^c \in \mathscr{F}$ and $B \setminus (A \cup N) \subseteq B$ with $\mu(B) = 0$, we are done.

(iii) Suppose $A_n \in \mathscr{F}$, $N_n \subseteq B_n \in \mathscr{F}$, and $\mu(B_n) = 0$. We want to show $\cup_n (A_n \cup N_n) \in \tilde{\mathscr{F}}$.

Writing

$$\cup_n (A_n \cup N_n) = (\cup_n A_n) \cup (\cup_n N_n),$$

since $\cup_n N_n \subseteq \cup_n B_n$ and $\mu(\cup_n B_n) = 0$, we are done.

We want to show $\tilde{\mu}$ is well-defined. Suppose $A_1, A_2 \in \mathscr{F}$, $N_1 \subseteq B_1 \in \mathscr{F}$, $N_2 \subseteq B_2 \in \mathscr{F}$, $\mu(B_1) = \mu(B_2) = 0$, and $A_1 \cup N_1 = A_2 \cup N_2$. Can we show $\tilde{\mu}(A_1 \cup N_1) = \tilde{\mu}(A_2 \cup N_2)$?

By definition,

$$\begin{aligned}\tilde{\mu}(A_1 \cup B_1) &= \mu(A_1) \\ &= \mu(A_1 \cup B_1) \text{ because we can union with measure zero set} \\ &\geq \mu(A_2) = \tilde{\mu}(A_2 \cup N_2).\end{aligned}$$

Adding in B_1 comes from: (i) monotonicity giving $\mu(A_1) \leq \mu(A_1 \cup B_1)$ and (ii) subadditivity giving the other direction.

By symmetry, the other direction is true, so $\tilde{\mu}$ is well-defined.

Now we show $\tilde{\mu}$ is a measure on $\tilde{\mathscr{F}}$.

(i) $\tilde{\mu}(\varnothing) = \tilde{\mu}(\varnothing \cup \varnothing) = \mu(\varnothing) = 0$

(ii) Suppose $A_n \in \mathscr{F}$, $N_n \subseteq B_n \in \mathscr{F}$, $\mu(B_n) = 0$, and $A_n \cup N_n$ are pairwise disjoint.

Then

$$\begin{aligned}\tilde{\mu}(\cup_n (A_n \cup N_n)) &= \tilde{\mu}((\cup_n A_n) \cup (\cup_n N_n)) \\ &= \mu(\cup_n A_n) \\ &= \sum_n \mu(A_n) \text{ by disjointness} \\ &= \sum_n \tilde{\mu}(A_n \cup N_n),\end{aligned}$$

so $\tilde{\mu}$ is a measure.

Since $\tilde{\mu}(A) = \tilde{\mu}(A \cup \varnothing) = \mu(A)$, $\tilde{\mu}$ is an extension of μ. □

[4] With this sentence, we are supposing $A \cup N \in \tilde{\mathscr{F}}$.

[5] draw

Theorem 1.3.11.

Suppose μ is a finite measure on a field $\mathscr{A}$ of subsets of Ω and that τ is the outer measure constructed by μ. With $\mathscr{F}_\tau$ as the set of τ-measurable subsets of Ω according to the Caratheodory criterion, $(\Omega, \mathscr{F}_\tau, \tau)$ is the completion of $(\Omega, \sigma(\mathscr{A}), \tau)$.

We require the following lemma.

One example is taking $\mathscr{A}$ as all finite unions of intervals in the unit interval and $\mathscr{F}_\tau$ as the *completion* of the Borel σ-field.

Lemma 1.3.12.

Under the same setting (finite measure μ with induced outer measure τ), for any $A \in \mathscr{F}_\tau$, there exists $B \in \sigma(\mathscr{A})$ such that $B \supseteq A$ and $\tau(B) = \tau(A)$. Elements in $\mathscr{F}_\tau$ can be approximated above by elements in $\sigma(\mathscr{A})$.

Proof of Lemma 1.3.12. By definition of outer measure induced by a premeasure,

$$\tau(A) = \inf\{\sum_n \mu(B_n) : B_n \in \mathscr{A}, \cup_n B_n \subseteq A\}.$$

For every $m \in \mathbb{N}$, there exists a sequence $\{B_{m,n}\}_n \subseteq \mathscr{A}$ such that $\cup_n B_{m,n} \supseteq A$ and

$$\tau(A) + \frac{1}{n} \geq \sum_n \mu(B_{m,n}).$$

Write

$$B = \cap_m \cup_n B_{m,n} \supseteq A.$$

Then $B \in \sigma(\mathscr{A})$ because each $B_{m,n} \in \mathscr{A}$. We see

$$\begin{aligned}\tau(A) + \frac{1}{m} &\geq \sum_n \mu(B_{m,n}) \\ &= \sum_n \tau(B_{m,n}) \\ &\geq \tau\left(\cup_n B_{m,n}\right) \\ &\geq \tau(B).\end{aligned}$$

Sending $m \to \infty$, $\tau(A) \geq \tau(B)$. Since $A \subseteq B$, $\tau(A) \leq \tau(B)$. □

Proof. (Of the theorem.) Write $\mathscr{F} = \sigma(\mathscr{A})$. We want to show that $\tilde{\mathscr{F}} = \mathscr{F}_\tau$, where

$$\tilde{\mathscr{F}} = \{A \cup N : A \in \mathscr{F}, \exists\, B \in \mathscr{F} \text{ s.t. } B \supseteq N, \mu(B) = 0\}.$$

We know $\tilde{\mathscr{F}} \subseteq \mathscr{F}_\tau$ since $\tilde{\mathscr{F}}$ is the smallest completion of $\mathscr{F}$ and $\mathscr{F}_\tau$ is a completion of $\mathscr{F}$.

For $\mathscr{F}_\tau \subseteq \tilde{\mathscr{F}}$, fix $A \in \mathscr{F}_\tau$. We want to write A as the union of something in $\mathscr{F}$ and a subset of a measure zero set.

By the lemma, there exists $B \in \mathscr{F}$ such that $B \supseteq A$ and $\tau(B) = \tau(A)$. By finiteness of τ, this implies $\tau(B \setminus A) = 0$.

Apply the lemma again to $B \setminus A$: there exists $C \in \mathscr{F}$ such that $C \subseteq B \setminus A$ and $\tau(C) = \tau(B \setminus A) = 0$.

Draw, then write

$$A = (A \cap C^c) \cup (A \cap C)\,.$$

We don't know that $A \cap C^c \in \mathscr{F}$, because we just know $A \in \mathscr{F}_\tau$. However, we can write

$$A = (B \cap C^c) \cup (A \cap C).$$

$B, C \in \mathscr{F}$, so $B \cap C^c \in \mathscr{F}$. Furthermore, $A \cap C \subseteq C$, where $\tau(C) = 0$, so we are done. $\square$

Measurable Functions 2

2.1 Preliminaries

Definition 2.1.1 (Preimage).

Let $f : X \to Y$ be a function, and let $\mathscr{G}$ be a collection of subsets of Y (e.g., a σ-algebra on Y). For any $B \in \mathscr{G}$, define the preimage of B under f by

$$f^{-1}(B) := \{x \in X : f(x) \in B\}.$$

Definition 2.1.2 (Measurable function).

We say $f : (X, \mathscr{F}) \to (Y, \mathscr{G})$ is measurable if $f^{-1}\mathscr{G} \subseteq \mathscr{F}$; i.e. the preimage of any measurable set in the second measurable space is measurable. Alternatively, for any $B \in \mathscr{G}$, $f^{-1}B \in \mathscr{F}$.

Remark 2.1.3 (Rules of applying preimage). Some of these are shown on homework. We have that

f maps (sample space, σ-field) to (sample space, σ-field).

- $f^{-1}\varnothing = \varnothing, \ f^{-1}Y = X$
- $f^{-1}(B^c) = (f^{-1}B)^c$
- $A \subseteq B \implies f^{-1}A \subseteq f^{-1}B$
- $f^{-1}(\cup_\lambda A_\lambda) = \cup_\lambda f^{-1}A_\lambda$
- $f^{-1}(\cap_\lambda A_\lambda) = \cap_\lambda f^{-1}A_\lambda$

Definition 2.1.4 (Random variable).

$X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_\mathbb{R})$ is a random variable.

2.2 Properties of Measurable Functions

Lemma 2.2.1.

Suppose $f : (X, \mathscr{F}) \to (Y, \sigma(\mathscr{E}))$. f is measurable if $f^{-1}\mathscr{E} \subseteq \mathscr{F}$.

Proof of Lemma 2.2.1. Suppose $f^{-1}\mathscr{E} \subseteq \mathscr{F}$. We know that because $\mathscr{F}$ is a σ-field, $\sigma\left(f^{-1}\mathscr{E}\right) \subseteq \mathscr{F}$. We want to show that $f^{-1}\sigma(\mathscr{E}) \subseteq \mathscr{F}$. So, it suffices to show $\sigma\left(f^{-1}\mathscr{E}\right) = f^{-1}\sigma(\mathscr{E})$.

Since $\mathscr{E} \subseteq \sigma(\mathscr{E})$, it can be shown that $f^{-1}\sigma(\mathscr{E})$ is a σ-field containing $f^{-1}\mathscr{E}$. Since $\sigma\left(f^{-1}\mathscr{E}\right)$ is the smallest such σ-field, we have that

$$\sigma\left(f^{-1}\mathscr{E}\right) \subseteq f^{-1}\sigma\left(\mathscr{E}\right).$$

For the other direction, write $\mathscr{H} = \{B \subseteq Y : f^{-1}B \in \sigma\left(f^{-1}\mathscr{E}\right)\}$. Since $f^{-1}\mathscr{E} \subseteq \sigma(f^{-1}\mathscr{E})$, we have $\mathscr{E} \subseteq \mathscr{H}$. We can check that $\mathscr{H}$ is actually a σ-field, so $\sigma(\mathscr{E}) \subseteq \mathscr{H}$ and thus

$$f^{-1}\sigma(\mathscr{E}) \subseteq \sigma\left(f^{-1}\mathscr{E}\right).$$

□

Since $\mathscr{B}_{\mathbb{R}} = \sigma\left(\{(a, \infty) : a \in \mathbb{R}\}\right)$, a random variable $X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ is measurable if $\{X > a\}$ is measurable for all $a \in \mathbb{R}$ (equivalently, if $\{X \geq a\}$ is measurable for all $a \in \mathbb{R}$).

Proposition 2.2.2.

Applying a continuous operator to a measurable function produces a measurable function. For instance, for f, g measurable, $f + g$ is measurable. $\limsup$, $\liminf$ and such preserve measurability.

Proof of Proposition 2.2.2. We see that $\{f+g > a\}$ is equivalent to the statement that there exists some t such that $\{f > t\} \cap \{g > a - t\}$. We can take t to be rational such that $t \in (a - g, f)$, so

$$\{f + g > a\} = \bigcup_{t \in \mathbb{Q}} \left[\{f > t\} \cap \{g > a - t\}\right] \in \mathscr{F}.$$

Similarly,

$$\{\sup_k f_k \leq a\} = \bigcap_{k=1}^{\infty} \{f_k \leq a\} \in \mathscr{F}.$$

□

Lemma 2.2.3.

Suppose $f : (X, \mathscr{F}) \to (Y, \mathscr{G})$ and $g : (Y, \mathscr{G}) \to (Z, \mathscr{H})$ are measurable. Then $g \circ f$ is measurable.

Proof of Lemma 2.2.3. Take two preimages. □

2.3 Integration

Definition 2.3.1 (Indicator function).

Define the indicator function on A as

$$\mathbb{1}_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}.$$

Because

$$\{\mathbb{1}_A \leq a\} = \begin{cases} \Omega, & a \geq 1 \\ A^c, & 0 \leq a < 1 \\ \varnothing, & a < 0 \end{cases},$$

the indicator function $\mathbb{1}_A$ is measurable iff A is measurable.

Definition 2.3.2 (Simple function).

Suppose $\{A_1, \ldots, A_n\}$ is a finite partition of Ω. A simple function f is a linear combination of indicator functions on sets of the partition:

$$f = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i}, \quad a_i \in \mathbb{R}.$$

From simple functions, we can approximate measurable nonnegative functions, which we can use to get a general measurable function f as the difference of two nonnegative measurable functions: $f = f^+ - f^-$, where

$$\forall\, x \in \mathbb{R}, x^+ = \max(x, 0), \quad x^- = (-x)^+, \quad |x| = x^+ + x^-.$$

Lemma 2.3.3 (Approximation of nonnegative measurable functions).

For any nonnegative measurable function f, there exists a sequence (f_n) of nonnegative simple functions such that $f_n \uparrow f$ pointwise.
If f is bounded, then this convergence is uniform, i.e.

$$\sup_{\omega \in \Omega} |f_n(\omega) - f(\omega)| \to 0.$$

Proof of Lemma 2.3.3. Define

$$f_n = \sum_{k=0}^{n2^n - 1} \frac{k}{2^n} \mathbb{1}_{\frac{k}{2^n} \leq f < \frac{k+1}{2^n}} + n \mathbb{1}_{f \geq n}.$$

f_n divides the section where $f \leq n$ into $n2^n - 1$ intervals. Equivalently, we can write

$\wedge$ acts as a min operator.

$$f_n = \max\{\frac{k}{2^n} : \frac{k}{2^n} \leq f, k = 0, 1, 2, \ldots, n2^n - 1\} \wedge n.$$

We see that $f_n \leq f_{n+1}$ because $\frac{2k}{2^{n+1}}, \frac{2k+1}{2^{n+1}} \geq \frac{k}{2^n}$ and $n + 1 \geq n$.

If $f(\omega) \leq n$, then $|f_n(\omega) - f(\omega)| < \frac{1}{2^n}$, so we have pointwise convergence. If f is bounded, then eventually, we get a uniform bound on the approximation error. □

Remark 2.3.4 (Lebesgue integral). Suppose $f : (\Omega, \mathscr{F}, \mu) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ and that everything in these definitions is measurable when required. We will follow a similar process for constructing the Lebesgue integral.

(i) (Indicator.) Define $\int \mathbb{1}_A d\mu = \mu(A)$.

(ii) (Nonnegative simple function.) For $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$, where $a_i \geq 0$ and A_i are measurable, define

$$\int f d\mu = \sum_{i=1}^n a_i \mu(A_i).$$

(iii) (Nonnegative function.) For $f \geq 0$, define

$$\int f d\mu = \sup\{\int g d\mu : g \leq f, g \text{ is a nonnegative simple function}\}.$$

(iv) (General function.) Define

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu$$

when at least one of the integrals is finite.

Is the integral of a simple function well-defined? Since simple functions have multiple representations (consider $f = a$ on $A_1 \cup A_2$), we want to make sure $\int f d\mu = \sum a_i \mu(A_i)$ is well-defined. We can write

$$f = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i} = \sum_{j=1}^{n} b_j \mathbb{1}_{f=b_j}, \text{where } b_i \neq b_j,\ m \leq n,$$

which is a unique representation of the simple function.

For each i, j, either $A_i \subseteq \{f = b_j\}$ or $A_i \cap \{f = b_j\} = \varnothing$. Using the additivity of measure, write

$$\begin{aligned}
\int f d\mu &= \sum_{i=1}^{n} a_i \mu(A_i) \\
&= \sum_{i=1}^{n} a_j \sum_{j=1}^{m} \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^{m} \sum_{i=1}^{n} a_i \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^{m} \sum_{i:\ A_i \subseteq \{f=b_j\}} a_i \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^{m} \sum_{i: A_i \subseteq \{f=b_j\}} b_j \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^{m} \sum_{i=1}^{n} b_j \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^{m} b_j \mu\left(f = b_j\right).
\end{aligned}$$

Now that we know the integral of a simple function is well-defined, let's get basic properties.

Lemma 2.3.5.

Suppose $(f_n), f, g$ are nonnegative simple functions. Then

(i) $\int f d\mu \geq 0$

(ii) $\int (af)\, d\mu = a \int f d\mu$, for $a \geq 0$

(iii) $\int (f + g) d\mu = \int f d\mu + \int g d\mu$

(iv) $f \leq g \implies \int f d\mu \leq \int g d\mu$

(v) If $(f_n) \uparrow$ and $\lim_{n\to\infty} f_n \geq g$, then

$$\lim_{n\to\infty} \int f_n d\mu \geq \int g d\mu.$$

Proof of Lemma 2.3.5. (iii) For $f = \sum a_i \mathbb{1}_{A_i}$ and $g = \sum b_j \mathbb{1}_{B_j}$, we can write

$$f + g = \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + b_j) \mathbb{1}_{A_i \cap B_j},$$

so

$$\begin{aligned}
\int (f+g)\, d\mu &= \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + b_j) \mu\left(A_i \cap B_j\right) \\
&= \sum_{i=1}^{n} a_i \sum_{j=1}^{m} \mu\left(A_i \cap B_j\right) + \sum_{j=1}^{m} b_j \sum_{i=1}^{n} \mu\left(A_i \cap B_j\right) \\
&= \sum_{i=1}^{n} a_i \mu\left(A_i\right) + \sum_{j=1}^{m} b_j \mu(B_j) \\
&= \int f d\mu + \int g d\mu.
\end{aligned}$$

Note that even if the limit of (f_n) takes on a value of ∞, we are never actually evaluating the integral of the limit, so there is no cause for concern.

(iv) Write

$$\int g d\mu = \int \left((g - f) + f\right) d\mu,$$

where $g - f$ is a nonnegative simple function because $g \geq f$. We can then use property (iii) and property (i).

(v) Introduce for each $\alpha \in (0, 1)$ the set

$$A_n(\alpha) = \{f_n \geq \alpha g\}.$$

Then $A_n(\alpha) \uparrow \Omega$.

Write $g = \sum_{j=1}^{m} b_j \mathbb{1}_{B_j}$. Then by property (iv), we get the first and second

If $\alpha = 1$, some example like $f_n = 1 - \frac{1}{n} \uparrow 1 = g$, shows that $A_n(\alpha) \uparrow \Omega$ would not hold.

inequalities in

$$\begin{aligned}
\int f_n d\mu &\geq \int f_n \mathbb{1}_{A_n(\alpha)} d\mu \\
&\geq \int (\alpha g \mathbb{1}_{A_n(\alpha)}) d\mu \\
&= \alpha \int g \mathbb{1}_{A_n(\alpha)} d\mu \\
&= \sum_{j=1}^{m} b_j \mathbb{1}_{B_j \cap A_n(\alpha)} d\mu \\
&= \sum_{j=1}^{m} b_j \mu \left(B_j \cap A_n(\alpha)\right) \\
\implies \lim_{n\to\infty} \int f_n d\mu &\geq \lim_{n\to\infty} \alpha \sum_{j=1}^{m} b_j \mu \left(B_j \cap A_n(\alpha)\right) \\
&= \alpha \sum_{j=1}^{m} b_j \lim_{n\to\infty} \mu \left(B_j \cap A_n(\alpha)\right) \\
&= \alpha \sum_{j=1}^{m} b_j \mu \left(B_j\right) = \alpha \int g d\mu,
\end{aligned}$$

where the second-to-last equality follows from continuity of measure from below. Sending $\alpha \to 1^-$, we are done. The introduction of α allows us to reduce the switching of a limit and integral to the switching of a limit and a finite sum.

□

Is the integral of a nonnegative function well-defined? Recall that for nonnegative f, we define

$$\int f d\mu = \sup\{\int g d\mu : g \leq f, g \text{ nonnegative simple}\}.$$

If f itself is nonnegative and simple, does this definition coincide with our previous one?

Writing $f = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i}$ for a partition $\{A_i\}$, define

$$T(f) = \sum_{i=1}^{n} a_i \mu(A_i), \quad S(f) = \sup\{T(g) : g \leq f, g \text{ nonnegative simple }\}.$$

Is $T(f) = S(f)$?

For any $g \leq f$ where g is nonnegative and simple, we see $T(g) \leq T(f)$ by monotonicity of the integral of a nonnegative simple function. Taking a sup on both sides over all nonnegative simple minorants g of f, we get $S(f) \leq T(f)$. Furthermore, since $f \leq f$, $S(f) \geq T(f)$ and the integral of a nonnegative function extends the integral of nonnegative simple functions.

Lemma 2.3.6.

Assume f, g are nonnegative measurable functions. Suppose we have a sequence (f_n) of nonnegative simple functions that increases to f.

(i) Then
$$\int f d\mu = \lim_{n\to\infty} \int f_n d\mu$$

(ii) We can write
$$\int f d\mu = \lim_{n\to\infty} \sum_{k=1}^{n2^n-1} \frac{k}{2^n}\mu\left(\frac{k}{2^n} \le f < \frac{k+1}{2^n}\right) + n\mu\left(f \ge n\right).$$

(iii) For all $a \ge 0$,
$$\int (af) d\mu = a \int f d\mu.$$

(iv)
$$\int (f+g) d\mu = \int f d\mu + \int g d\mu.$$

(v) The integral of nonnegative functions is monotone:
$$f \le g \implies \int f d\mu \le \int g d\mu.$$

Proof of Lemma 2.3.6. (i) By definition of the integral of f (not monotonicity),

$$\int f d\mu \ge \int f_n d\mu.$$

This shows linearity of the integral of nonnegative functions, which is not immediate from the definition.

Taking a limit on both sides, we get

$$\int f d\mu \ge \lim_{n\to\infty} \int f_n d\mu.$$

By property (v) of the integral of nonnegative simple functions, for $g \le f$ where g is nonnegative and simple, we have

$$\lim_{n\to\infty} \int f_n d\mu \ge \int g d\mu.$$

Taking a sup over all nonnegative simple minorants g of f, we show

$$\lim_{n\to\infty} \int f_n d\mu \ge \int f d\mu.$$

(ii) Trivial.

(iii) There exists a sequence of simple $f_n \uparrow f$, so $af_n \uparrow af$. Then by repeatedly applying property (ii) of nonnegative simple functions,

$$\begin{aligned}\int (af) d\mu &= \lim_{n\to\infty} \int (af_n)\, d\mu \\ &= a \lim_{n\to\infty} \int f_n d\mu \\ &= a \int f d\mu.\end{aligned}$$

(iv) There exist simple $f_n \uparrow f$ and $g_n \uparrow g$, so $f_n + g_n \uparrow f + g$. Then by property (iii) of nonnegative simple functions,

$$\begin{aligned}\int (f+g) d\mu &= \lim_{n\to\infty} \int f_n + g_n d\mu \\ &= \lim_{n\to\infty} \int f_n d\mu + \lim_{n\to\infty} \int g_n d\mu \\ &= \int f d\mu + \int g d\mu.\end{aligned}$$

(v) Trivial, since any simple function below f is also below g.

□

How do we extend the definition to an integral of a general measurable function? Decompose $f = f^+ - f^-$. If $\min\{\int f^+ d\mu, \int f^- d\mu\} < \infty$, then we say the integral of f exists and define $\int f d\mu = \int f^+ d\mu - \int f^- d\mu$ Both integrals are finite iff $\int |f|\, d\mu < \infty$, since $|f| = f^+ + f^-$. In this case (the integral of f is finite), we call f integrable.

From now on, we only consider f that are integrable.

The integral of x^3 does not exist and x^3 is not integrable on the real line.

Definition 2.3.7 (Definite integral).

For any measurable A, define

$$\int_A f d\mu = \int (f \mathbb{1}_A)\, d\mu.$$

Definition 2.3.8 (Expectation of a random variable).

For a random variable $X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, we define the expectation of X as

$$\mathbb{E}[X] := \int X d\mathbb{P}.$$

We say the expectation exists if $\mathbb{E}[|X|] < \infty$ (X is integrable).[a]

[a] Note that the expectation exists if X is integrable. The integral of X existing is not

sufficient; it must be finite as well.

Lemma 2.3.9.

Suppose f, g are integrable.

(i) If $A \in \mathscr{F}$ such that $\mu(A) = 0$,

$$\int_A f d\mu = 0.$$

(ii) If $f \leq g$ almost everywhere ($\mu(f > g) = 0$), then

$$\int f d\mu \leq \int g d\mu.$$

(iii) If $f = g$ almost everywhere, then

$$\int f d\mu = \int g d\mu.$$

Proof of Lemma 2.3.9. (i) First, consider $f = \sum_i a_i \mathbb{1}_{A_i}$ to be a nonnegative simple function. Then

$$\int_A f d\mu = \int \left(\sum_i a_i \mathbb{1}_{A_i \cap A} \right) d\mu = \sum_i a_i \mu\left(A_i \cap A\right) = 0,$$

since $\mu(A_i \cap A) \leq \mu(A) = 0$.

Now consider f nonnegative. Then there exists nonnegative simple function sequence $f_n \uparrow f$, so $f_n \mathbb{1}_A \uparrow f \mathbb{1}_A$. Then

$$\int_A f d\mu = \lim_{n \to \infty} \int_A f_n d\mu = 0.$$

For a general f, put $f = f^+ - f^-$, so

$$f \mathbb{1}_A = f^+ \mathbb{1}_A - f^- \mathbb{1}_A.$$

Taking an integral on both sides with linearity of the integral on nonnegative functions, we see the right hand side is 0.

(ii) Suppose $f \leq g$ almost everywhere. Let

$$A = \{x : f \leq g\}, \quad A^c = \{x : f > g\}.$$

Then $\mu\left(A^c\right) = 0$. Suppose for now that $f, g \geq 0$.

We can write

$$\begin{aligned}\int f d\mu &= \int_A f d\mu + \int_{A^c} f d\mu \\ &= \int f \mathbb{1}_A d\mu + \int f \mathbb{1}_{A^c} d\mu \\ &\le \int_A g d\mu + \int_{A^c} g d\mu \\ &= \int g d\mu.\end{aligned}$$

where the inequality follows from monotonicity of integrals of nonnegative functions.

For general f, g, write $f = f^+ - f^-$ and $g = g^+ - g^-$. If $f \le g$, then $f^+ \le g^+$ and $-f \ge -g$, so

$$(-f)^+ \ge (-g)^+ .$$

then $f^- \ge g^-$.

Using monotonicity for the integral of nonnegative functions,

$$\int f^+ d\mu \le \int g^+ d\mu, \quad \int f^- d\mu \ge \int g^- d\mu.$$

we conclude by subtracting the second inequality from the first.

(iii) Follows from symmetry with (ii).

□

Lemma 2.3.10 (Linearity of integral).

Suppose f, g are integrable. Then

(i) $\int (af) d\mu = a \int f d\mu$, for all $a \in \mathbb{R}$.

(ii) $\int (f+g) d\mu = \int f d\mu + \int g d\mu$

Proof of Lemma 2.3.10. (i) If $a \ge 0$, we see that

$$(af)^+ = \max\{0, af\} = a \max\{0, f\} = af^+.$$

Similarly, $(af)^- = af^-$. This gives us the second equality in the following:

$$\begin{aligned}\int (af)d\mu &= \int (af)^+ d\mu - \int (af)^- d\mu \\ &= a\left(\int f^+ d\mu - \int f^- d\mu\right) \\ &= a\int f d\mu.\end{aligned}$$

If $a < 0$, then $(af)^+ = ((-a)(-f))^+ = -a(-f)^+ = -af^-$. Similarly, $(af)^- = -af^+$. We show linearity in the same way.

(ii) Note the three decompositions we have:

- $f = f^+ - f^-$
- $g = g^+ - g^-$
- $(f+g) = (f+g)^+ - (f+g)^-$.

We also know that

$$\begin{aligned} (f+g)^+ - (f+g)^- &= f^+ - f^- + g^+ - g^- \\ \text{Rearrange: } (f+g)^+ + f^- + g^- &= (f+g)^- + f^+ + g^+. \end{aligned}$$

We have a nonnegative function on either side; integrate, apply linearity of integral of nonnegative functions, and rearrange again to get

$$\int (f+g)^+ d\mu - \int (f+g)^- d\mu = \int f^+ d\mu - \int f^- d\mu + \int g^+ d\mu - \int g^- d\mu.$$

Use linearity of integrals of nonnegative functions again to conclude!

$\square$

2.4 Convergence Theorems

Theorem 2.4.1 (Levi's monotone convergence).

Suppose $(f_n), f \geq 0$ and $f_n \uparrow f$ almost everywhere. Then

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

Proof of Theorem 2.4.1. We have shown the theorem but with f_n as simple functions. We want to approximate each f_n by a sequence of nonnegative simple functions.

For the proof, consider $f_n \uparrow f$ everywhere. At the end, we can add in the measure zero set.

For each f_n, there exist nonnegative simple functions $(f_{n,k})_k$ such that $f_{n,k} \uparrow f_n$ as $k \to \infty$.

Define $g_k(x) := \max_{1\leq n\leq k} f_{n,k}(x)$. We see that each g_k is a nonnegative simple function. Furthermore, because $f_{n,k} \leq f_{n,k+1}$, (g_k) is increasing.

$f_{n,k+1}$ is not considered in g_k, so the addition of $f_{n+1,k+1}$ poses no problem.

We see that $g_k \leq f_k$, so $\lim\limits_{k\to\infty} g_k \leq \lim\limits_{k\to\infty} f_k = f$.

Fix $i \in \mathbb{N}$. For every $k \geq i$, since

$$g_k(x) := \max_{1\leq n\leq k} f_{n,k}(x),$$

we have $g_k(x) \geq f_{i,k}(x)$ for all x. Taking $k \to \infty$ and using $f_{i,k} \uparrow f_i$ gives

$$\lim_{k\to\infty} g_k(x) \geq \lim_{k\to\infty} f_{i,k}(x) = f_i(x), \quad \forall x.$$

Hence $\lim_{k\to\infty} g_k \geq f_i$ pointwise for every i. Therefore, pointwise,

$$\lim_{k\to\infty} g_k \geq \sup_{i\in\mathbb{N}} f_i = f,$$

and we have equality in $\lim\limits_{k\to\infty} g_k = f$.

Since (g_k) is increasing, by the monotone convergence theorem for simple functions in Lemma (2.3.6),

$$\lim_{k\to\infty} \int g_k d\mu = \int f d\mu.$$

By monotonicity of the integral of nonnegative functions, $\int f_n d\mu \leq \int f d\mu$. We can take a limit to see $\lim_{n\to\infty} \int f_n d\mu \leq \int f d\mu$. Because $g_n \leq f_n$, we see

$$\lim_{n\to\infty} \int g_n d\mu \leq \lim_{n\to\infty} \int f_n d\mu \leq \int f d\mu.$$

The outer terms are equal, so

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

□

Theorem 2.4.2 (Fatou's lemma).

We can take lim inf outside integrals of nonnegative functions, up to an inequality:

$$\int \left(\liminf_{n\to\infty} f_n\right) d\mu \leq \liminf_{n\to\infty} \int f_n d\mu.$$

Proof of Theorem 2.4.2. Write

$$\begin{aligned}\liminf_{n\to\infty} f_n &= \lim_{k\to\infty}\left(\inf_{n\geq k} f_n\right)\\ &= \lim_{k\to\infty} g_k.\end{aligned}$$

Note that g_k is a monotone increasing sequence of nonnegative functions, so Levi's monotone convergence theorem applies and we get

$$\lim_{k\to\infty}\int \inf_{n\geq k} f_n d\mu = \lim_{k\to\infty}\int g_k d\mu = \int \lim_{k\to\infty} g_k d\mu = \int \liminf_{n\to\infty} f_n d\mu.$$

Since $\inf_{j\geq k} f_j \leq f_n$ for all $n \geq k$, we get

$$\int \inf_{n\geq k} f_n d\mu \leq \inf_{n\geq k}\int f_n d\mu, \forall\ n \geq k.$$

Taking a limit,

$$\lim_{k\to\infty}\int \inf_{n\geq k} f_n d\mu \leq \liminf_{n\to\infty}\int f_n d\mu.$$

Making the substitution with the result from Levi's MCT, we have

$$\int \liminf_{n\to\infty} f_n d\mu \leq \liminf_{n\to\infty}\int f_n d\mu.$$

□

The following theorem is the most general of the convergence theorems.

Theorem 2.4.3 (Lebesgue's Dominated Convergence Theorem).

Suppose $f_n \to f$ and there exists an integrable function g such that $|f_n| \leq g$ for all $n \in \mathbb{N}$. Then

$$\lim_{n\to\infty}\int f_n d\mu = \int f d\mu.$$

Finding the *control function* g is often possible, but difficult.

Proof of Theorem 2.4.3. By hypothesis, $-g \leq f_n \leq g$ for every $n \in \mathbb{N}$.

Since $f_n + g \geq 0$, by Fatou's lemma,

$$\begin{aligned}\int \liminf_{n\to\infty} f_n + g d\mu &\leq \liminf_{n\to\infty}\int (f_n + g) d\mu\\ \int \liminf_{n\to\infty} f_n d\mu + \int g d\mu &\leq \liminf_{n\to\infty}\int f_n d\mu + \int g d\mu \text{ by linearity}\\ \int \liminf_{n\to\infty} f_n d\mu &\leq \liminf_{n\to\infty}\int f_n d\mu \text{ since } \int g d\mu < \infty.\end{aligned}$$

We symmetrically derive the other desired inequality, using the fact that

$g - f_n \geq 0$:

$$\int \liminf_{n\to\infty} g - f_n d\mu \leq \liminf_{n\to\infty} \int g - f_n d\mu$$
$$\int g d\mu + \int \liminf_{n\to\infty} -f_n \leq \int g d\mu + \liminf_{n\to\infty} \int -f_n d\mu$$
$$\int \limsup_{n\to\infty} f_n \geq \int \limsup_{n\to\infty} \int f_n d\mu.$$

Chaining together these two inequalities,

$$\limsup_{n\to\infty} \int f_n d\mu \leq \int \limsup_{n\to\infty} f_n d\mu = \int f d\mu = \int \liminf_{n\to\infty} f_n d\mu \leq \liminf_{n\to\infty} \int f_n d\mu.$$

Since $\liminf_{n\to\infty} \int f_n d\mu \leq \limsup_{n\to\infty} \int f_n d\mu$, we see that equality holds and thus the limit exists:

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

□

Corollary 2.4.4 (Bounded Convergence Theorem).

Suppose $f_n \to f$, $|f_n| \leq M < \infty$ almost everywhere, and $\mu(\Omega) < \infty$. Then

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

Proof of Corollary 2.4.4. Take $g = M$, which is integrable by finiteness of $\mu(\Omega)$.

□

Independence 3

3.1 Preliminaries

Consider a measure space $(\Omega, \mathscr{F}, \mathbb{P})$.

Notation 3.1.1.

For a random variable $X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, when we write $\{X \in B\}$, we are really referring to the set as

$$\{X \in B\} = \{\omega \in \Omega : X(\omega) \in B\}.$$

Similarly,

$$\{X \leq a\} = \{\omega \in \Omega : X(\omega) \leq a\},$$

$$\{X \in A, Y \in B\} = \{\omega \in \Omega : X(\omega) \in A, Y(\omega) \in B\},$$

and so on.
Perhaps most importantly,

$$\mathbb{P}\left[X \in B\right] = \mathbb{P}\left[\omega \in \Omega : X(\omega) \in B\right].$$

Definition 3.1.2 (Independent events).

We say that the events $A_1, \ldots, A_n \in \mathscr{F}$ are independent if for every $\Lambda \subseteq \{1, \ldots, n\}$,

$$\mathbb{P}\left[\bigcap_{i \in \Lambda} A_i\right] = \prod_{i \in \Lambda} \mathbb{P}\left[A_i\right].$$

We say $\mathscr{E}_1, \ldots, \mathscr{E}_n \subseteq \mathscr{F}$ are independent if for any $A_1 \in \mathscr{E}_1, \ldots, A_n \in \mathscr{E}_n$, $(A_1, \ldots, A_n)$ are independent.

Definition 3.1.3 (Generated σ-field (of a random variable)).

Let $X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ be a random variable. We define $\sigma(X)$ to be the smallest σ-field such that

$$X : (\Omega, \sigma(X), \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$$

is measurable.
Since $X^{-1}\mathscr{B}_{\mathbb{R}}$ is a σ-field containing X (preimage of a σ-field is also a σ-field),

$$\sigma(X) \subseteq X^{-1}\mathscr{B}_{\mathbb{R}} = X^{-1}\sigma\left(\{\{x > a\} : a \in \mathbb{Q}\}\right).$$

Conversely, suppose

$$X : (\Omega, \sigma\left(X\right), \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$$

is measurable. Then $X^{-1}(B) \in \sigma\left(X\right)$ for every $B \in \mathscr{B}_{\mathbb{R}}$. Hence,

$$X^{-1}\left(\mathscr{B}_{\mathbb{R}}\right) \subseteq \sigma\left(X\right).$$

Therefore,

$$\sigma(X) = X^{-1}\mathscr{B}_{\mathbb{R}}.$$

Example 3.1.4.

Suppose $A \in \mathscr{F}$ and $X = \mathbb{1}_A$. Then

$$\sigma(X) = \{\varnothing, \Omega, A, A^c\}.$$

Definition 3.1.5 (Independent random variables).

We say that $X_1, \ldots, X_n : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ are independent if $\sigma\left(X_1\right), \ldots, \sigma(X_n)$ are independent.

Why does this definition make sense? For $X \perp\!\!\!\perp Y$ and for any $A, B \in \mathscr{B}_{\mathbb{R}}$, we want

$$\mathbb{P}\left[X \in A, Y \in B\right] = \mathbb{P}\left[X \in A\right]\mathbb{P}\left[Y \in B\right].$$

We see that

$$\begin{aligned} \mathbb{P}\left[X \in A\right] &= \mathbb{P}\left[X^{-1}A\right], \\ \mathbb{P}\left[Y \in B\right] &= \mathbb{P}\left[Y^{-1}B\right], \\ \mathbb{P}\left[X \in A, Y \in B\right] &= \mathbb{P}\left[X^{-1}A \cap Y^{-1}B\right] \end{aligned} \qquad .$$

We need $\mathbb{P}$ to be defined on and satisfy the independence condition on sets of this last type.

Lemma 3.1.6.

If $X \perp\!\!\!\perp Y$ and $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y].$$

Proof of Lemma 3.1.6. (i) If $X = \mathbb{1}_A, Y = \mathbb{1}_B$, then $XY = \mathbb{1}_{A\cap B}$. Hence

$$\begin{aligned}\mathbb{E}[XY] &= \mathbb{E}[\mathbb{1}_{A\cap B}] \\ &= \mathbb{P}[A\cap B] \\ &= \mathbb{P}[A]\,\mathbb{P}[B] \text{ by independence} \\ &= \mathbb{E}[X]\,\mathbb{E}[Y].\end{aligned}$$

(ii) For $\{A_i\}, \{B_j\}$ partitions of Ω, if $X = \sum_i a_i \mathbb{1}_{A_i}$ and $Y = \sum_j b_j \mathbb{1}_{B_j}$, then we are tempted to write

$$\begin{aligned}XY &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{1}_{A_i\cap B_j} \\ \mathbb{E}[XY] &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{P}[A_i\cap B_j] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{P}[A_i]\,\mathbb{P}[B_j]\end{aligned}$$

and continue, but we don't know A_i and B_j are independent, since this requires $A_i \in \sigma(X)$ and $B_j \in \sigma(Y)$. This is not guaranteed if the a_i, b_j are not unique. (For example, if $X(\omega) = 1$ for $\omega \in A_1$ and $\omega \in A_2$, then $A_1 \notin \sigma(X)$, since A_1 being in $\sigma(X)$ is not necessary for X to be measurable — only $A_1 \cup A_2 \in \sigma(X)$ is required.

Instead, we should decompose as

$$X = \sum_{i=1}^{n} a_i \mathbb{1}_{X=a_i}, \quad Y = \sum_{j=1}^{m} b_j \mathbb{1}_{Y=b_j}.$$

Then we can actually continue to get

$$\begin{aligned}\mathbb{E}[XY] &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{P}[X = a_i, Y = b_j] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{P}[X = a_i]\,\mathbb{P}[Y = b_j] \\ &= \left(\sum_{i=1}^{n} a_i \mathbb{P}[X = a_i]\right)\left(\sum_{j=1}^{m} b_j \mathbb{P}[Y = b_j]\right) \\ &= \mathbb{E}[X]\,\mathbb{E}[Y].\end{aligned}$$

(iii) Now consider $X, Y \geq 0$. Write

$$\begin{aligned}X_n &= \sum_{k=0}^{n2^n-1} \frac{k}{2^n}\mathbb{1}_{\frac{k}{2^n} \leq X < \frac{k+1}{2^n}} + n\mathbb{1}_{X \geq n} \\ Y_n &= \sum_{k=0}^{n2^n-1} \frac{k}{2^n}\mathbb{1}_{\frac{k}{2^n} \leq Y < \frac{k+1}{2^n}} + n\mathbb{1}_{Y \geq n}.\end{aligned}$$

Then $X_n \uparrow X$, $Y_n \uparrow Y$. Since $\sigma(X_n) \subseteq \sigma(X)$ and $\sigma(Y_n) \subseteq \sigma(Y)$ imply that $X_n \perp\!\!\!\perp Y_n$,

$$\mathbb{E}[X_n Y_n] = \mathbb{E}[X_n]\,\mathbb{E}[Y_n].$$

Taking $n \to \infty$ concludes.

> Consider $X = 3\mathbb{1}_{A_1} + 4\mathbb{1}_{A_2} - \mathbb{1}_{A_3}$ for disjoint A_1, A_2, A_3, and find $\sigma(X)$. This provides intuition for what $\sigma(X)$ generally is.

(iv) For general X, Y, put $X = X^+ - X^-$ and $Y = Y^+ - Y^-$, so

$$XY = X^+Y^+ + X^-Y^- - X^+Y^- - X^-Y^+.$$

Since $\sigma(X^+) \subseteq \sigma(X)$ and so on, we see that each term in the sum satisfies independence; we can take the expectation termwise. Using linearity of the integral to recombine X^+ with X^- and similarly for Y, we get independence of X and Y.

□

Definition 3.1.7 (Identically distributed random variables).

We say that $X_1, \ldots, X_n$ are identically distributed if

$$\mathbb{P}[X_j \in A] = \mathbb{P}[X_i \in A]$$

for all i, j and $A \in \mathscr{F}$.

Law of Large Numbers 4

4.1 Preliminaries

Lemma 4.1.1 (Borel-Cantelli).

If $\sum_n \mathbb{P}[A_n] < \infty$, then

$$\mathbb{P}[A_n \text{ infinitely often}] = \mathbb{P}\left[\bigcap_{m=1}^{\infty}\bigcup_{n=m}^{\infty} A_n\right] = \mathbb{P}\left[\limsup_{n\to\infty} A_n\right] = 0.$$

Proof of Lemma 4.1.1. By finiteness of the entire sum, for any $\epsilon > 0$, there exists $m \in \mathbb{N}$ such that

$$\sum_{n=m}^{\infty} \mathbb{P}[A_n] < \epsilon.$$

Write

$$\begin{aligned}\mathbb{P}[A_n \text{ i.o.}] &= \mathbb{P}\left[\bigcap_{m=1}^{\infty}\bigcup_{n=m}^{\infty} A_n\right]\\ &\le \mathbb{P}\left[\bigcup_{n=m}^{\infty} A_n\right]\\ &\le \sum_{n=m}^{\infty} \mathbb{P}[A_n] < \epsilon.\end{aligned}$$

Sending $\epsilon \to 0$ yields the result. □

We need the following lemma to bound the second moments in one formulation of the LLN.

Lemma 4.1.2 (Existence of lower moments).

Suppose $\mathbb{E}[|X|^r] < \infty$. Then $\mathbb{E}[|X|^s] < \infty$ for all $s < r$.

Proof of Lemma 4.1.2. For all $t \geq 0$,

$$t^s \leq \max\{t^r, 1\} \leq t^r + 1.$$

So

$$\mathbb{E}\left[|X|^s\right] \leq \mathbb{E}\left[|X|^r\right] + 1 < \infty.$$

$\square$

We also need the following lemma.

Lemma 4.1.3 (Maximal inequality).

Suppose $Z_1, \ldots, Z_N$ are independent. Then

$$\mathbb{P}\left[\max_{i \leq N} |Z_1 + \cdots + Z_i| > \epsilon_1 + \epsilon_2\right] \leq \beta \mathbb{P}\left[|Z_1 + \cdots + Z_N| > \epsilon_1\right],$$

where $\beta = \frac{1}{\min_{i \leq N} \mathbb{P}[|Z_{i+1} + \cdots + Z_N| \leq \epsilon_2]}$

Proof of Lemma 4.1.3. Define $S_i = Z_1 + \cdots + Z_i$ and $T_i = Z_{i+1} + \cdots + Z_N$. We see that

$$\mathbb{P}\left[\max_{i \leq N} |S_i| > t\right] \leq \sum_{i \leq N} \mathbb{P}\left[|S_i| > t\right].$$

This is not helpful, because applying union bound is best when the terms are independent — if they are highly dependent, then union bound throws out a lot of information (think of $A_1 = A_2 = A_3 = \cdots$.)

Define the stopping time

$$\tau = \begin{cases} \text{first time } |S_i| > \epsilon_1 + \epsilon_2 \\ N \text{ if } \max_{i \leq N} |S_i| \leq \epsilon_1 + \epsilon_2 \end{cases}.$$

Then we can write

$$\begin{aligned}
\mathbb{P}\left[\max_{i \leq N} |S_i| > \epsilon_1 + \epsilon_2\right] &= \mathbb{P}\left[|S_\tau| > \epsilon_1 + \epsilon_2\right] \\
&= \sum_{i=1}^{N} \mathbb{P}\left[|S_\tau| > \epsilon_1 + \epsilon_2, \tau = i\right] \\
&= \sum_{i=1}^{N} \mathbb{P}\left[|S_i| > \epsilon_1 + \epsilon_2, \tau = i\right] \\
&= \sum_{i=1}^{N} \frac{\mathbb{P}\left[|S_i| > \epsilon_1 + \epsilon_2, \tau = i, |T_i| \leq \epsilon_2\right]}{\mathbb{P}\left[|T_i| \leq \epsilon_2\right]}.
\end{aligned}$$

Note that τ is random and S_i is random, so S_τ is very random! That is why we decouple in the second equality.

This last equality is allowed because $\{|S_i| > \epsilon_1 + \epsilon_2, \tau = i\}$ is independent from $|T_i| \le \epsilon_2$.[1]

Continuing,

$$\begin{aligned}
\mathbb{P}\left[\max_{i\le N} |S_i| > \epsilon_1 + \epsilon_2\right] &\le \frac{1}{\min_i \mathbb{P}\left[|T_i| \le \epsilon_2\right]} \sum_{i=1}^{N} \mathbb{P}\left[|S_i| > \epsilon_1 + \epsilon_2, \tau = i, |T_i| \le \epsilon_2\right] \\
&\le \frac{1}{\min_i \mathbb{P}\left[|T_i| \le \epsilon_2\right]} \sum_{i=1}^{N} \mathbb{P}\left[|S_N| > \epsilon_1, \tau = i\right] \\
&= \frac{1}{\min_i \mathbb{P}\left[|T_i| \le \epsilon_2\right]} \mathbb{P}\left[|S_N| > \epsilon_1\right].
\end{aligned}$$

□

4.2 Strong and Stronger Laws

Theorem 4.2.1 (Strong law with uniformly bounded second moments).

Suppose $\{X_i\}$ is a sequence of independent random variables with $\mathbb{E}\left[X_i\right] = 0$ for each i and

$$\sup_i \mathbb{E}\left[X_i^2\right] < \infty.$$

Then

$$\frac{X_1 + \cdots + X_n}{n} \to 0$$

almost surely.

Proof of Theorem 4.2.1. First, consider the stronger condition

$$\sup_i \mathbb{E}\left[X_i^4\right] < \infty.$$

Write $S_n = \sum_{i=1}^{n} X_i$. It suffices to show that

$$\sum_n \mathbb{P}\left[\frac{|S_n|}{n} > \epsilon\right] < \infty$$

since by Borel-Cantelli we would have

$$\mathbb{P}\left[\frac{|S_n|}{n} > \epsilon \text{ i.o.}\right] = 0,$$

[1] Note the nuance with the independence of $\tau = i$ and $|T_i| \le \epsilon_2$. Also, we know that these events are independent because they are measurable functions of independent random variables (see the Appendix).

which implies

$$\mathbb{P}\left[\limsup_{n\to\infty}\frac{|S_n|}{n}\leq\epsilon\right]=1,$$

from which we can send ϵ to 0.

By the previous lemma, $\sup_i \mathbb{E}\left[X_i^2\right]<\infty$ as well. We have by Chebyshev's inequality that

$$\mathbb{P}\left[\frac{|S_n|}{n}>\epsilon\right]\leq\frac{\mathbb{E}\left[S_n^2\right]}{n^2\epsilon^2}.$$

This part of the proof provides a reason for why the easiest case is with a uniform bound on the fourth moments, rather than on the second moments.

Writing

$$\begin{aligned}\mathbb{E}\left[S_n^2\right]&=\mathbb{E}\left[(X_1+\ldots+X_n)(X_1+\ldots+X_n)\right]\\&=\mathbb{E}\left[X_1^2+X_2^2+\ldots+X_n^2+\text{ terms in }X_iX_j\right]\\&=\sum_{i=1}^n\mathbb{E}\left[X_i^2\right],\end{aligned}$$

where the last equality follows from independence of X_i and X_j for $i\neq j$. Thus,

$$\mathbb{P}\left[\frac{|S_n|}{n}>\epsilon\right]\leq\frac{\sum_{i=1}^n\mathbb{E}\left[X_i^2\right]}{n^2\epsilon^2}\leq\frac{\sup_i\mathbb{E}\left[X_i^2\right]}{\epsilon^2}\cdot\frac{1}{n}.$$

Summing across n, we do *not* get convergence. If we try to use the third power in Chebyshev's inequality, we cannot decompose $\mathbb{E}\left[|X_i|^3\right]$.

We compute

$$\begin{aligned}S_n^4=&X_1^4+X_2^4+\ldots+X_n^4\\&+\text{terms in }X_i^3X_j,\ i\neq j\\&+\underbrace{\text{terms of the form }X_i^2X_j^2}_{\binom{n}{2}\binom{4}{2}\text{ terms}}.\end{aligned}$$

By assumption, there exists $M>0$ such that $\mathbb{E}\left[X_i^4\right]<M$ for every i. We see by Cauchy–Schwarz that

$$\mathbb{E}\left[X_1^2X_2^2\right]\leq\sqrt{\mathbb{E}\left[X_1^4\right]}\sqrt{\mathbb{E}\left[X_2^4\right]}\leq M.$$

Here, the inner product is defined

$$\langle X_i,X_j\rangle=\mathbb{E}\left[X_iX_j\right].$$

Cauchy–Schwarz gives

$$\mathbb{E}\left[X_iX_j\right]\leq\sqrt{\mathbb{E}\left[X_i^2\right]}\sqrt{\mathbb{E}\left[X_j^2\right]}.$$

Note that the "odd" terms vanish from independence giving $\mathbb{E}\left[X_i^3X_j\right]=\mathbb{E}\left[X_i^3\right]\cdot 0=0$. Computing $\binom{n}{2}=\frac{n!}{2!(n-2)!}=\frac{n(n-1)}{2}$, we have the bound

$$\begin{aligned}\mathbb{E}\left[S_n^4\right]&\leq nM+\binom{4}{2}\binom{n}{2}M\\&\leq 100000n^2M.\end{aligned}$$

By Chebyshev's inequality,

$$\mathbb{P}\left[|S_n| > n\epsilon\right] \leq \frac{\mathbb{E}\left[S_n^4\right]}{n^4\epsilon^4} \leq \frac{100000M}{\epsilon^4 n^2}.$$

Summing across n,

$$\sum_{n=1}^{\infty} \mathbb{P}\left[|S_n| > n\epsilon\right] \leq \frac{100000M}{\epsilon^4} \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

We have our result to which we can apply Borel-Cantelli.

Now suppose the weaker condition $\sup_i \mathbb{E}\left[X_i^2\right] = \sigma^2 < \infty$. Define blocks $B_1, B_2, \ldots$ such that

$$B_k = \{i \in \mathbb{N} : n_{k-1} < i \leq n_k\}$$

for $n_k = 2^k$ (this particular choice of n_k is not important to the proof).

We want to show that

$$\max_{n \in B_k} \frac{|S_n|}{n} \xrightarrow{k \to \infty} 0$$

almost surely. We can get this via Borel-Cantelli by showing for any $\epsilon > 0$,

$$\sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n \in B_k} \frac{|S_n|}{n} > \epsilon\right] < \infty.$$

We know thanks to our foresight in Lemma (4.1.3) that

$$\begin{aligned}
\sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n \in B_k} \frac{|S_n|}{n} > \epsilon\right] &\leq \sum_{k=1}^{\infty} \mathbb{P}\left[\frac{1}{n_{k-1}} \max_{n \in B_k} |S_n| > \epsilon\right] \\
&\leq \sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n \leq n_k} |S_n| > n_{k-1}\epsilon\right] \\
&\leq \sum_{k=1}^{\infty} \beta_k \mathbb{P}\left[|S_{n_k}| > \frac{n_{k-1}\epsilon}{2}\right], \\
\beta_k &= \frac{1}{\min_{n \leq n_k} \mathbb{P}\left[|S_{n_k} - S_n| \leq \frac{n_{k-1}\epsilon}{2}\right]}.
\end{aligned}$$

Analyzing each term of the product $\beta_k \mathbb{P}\left[|S_{n_k}| > \frac{n_{k-1}\epsilon}{2}\right]$ separately,

$$\begin{aligned}
\frac{1}{\beta_k} &= \min_{n \le n_k} \mathbb{P}\left[|S_{n_k} - S_n| \le \frac{n_{k-1}\epsilon}{2}\right] \\
&= \min_{n \le k} \left(1 - \mathbb{P}\left[|S_{n_k} - S_n| > \frac{n_{k-1}\epsilon}{2}\right]\right) \\
&= 1 - \max_{n \le n_k} \mathbb{P}\left[|S_{n_k} - S_n| > \frac{n_{k-1}\epsilon}{2}\right] \\
&\ge 1 - \frac{4\mathbb{E}\left[|S_{n_k} - S_n|^2\right]}{n_{k-1}^2 \epsilon^2} \\
&\ge 1 - \max_{n \le n_k} \frac{4(n_k - n)\sigma^2}{n_{k-1}^2 \epsilon^2} \\
&\ge 1 - \frac{4 n_k \sigma^2}{n_{k-1}^2 \epsilon^2} \\
&\ge \frac{1}{2} \text{ for } k \ge K
\end{aligned}$$

and

$$\mathbb{P}\left[|S_{n_k}| > \frac{n_{k-1}\epsilon}{2}\right] \le \frac{4\mathbb{E}\left[S_{n_k}^2\right]}{n_{k-1}^2 \epsilon^2} \le \frac{4 n_k \sigma^2}{n_{k-1}^2 \epsilon^2}.$$

Hence,

$$\mathbb{P}\left[\max_{n \le n_k} |S_n| > n_{k-1}\epsilon\right] \le 2 \frac{4 n_k \sigma^2}{n_{k-1}^2 \epsilon^2} = \frac{8 n_k \sigma^2}{n_{k-1}^2 \epsilon^2}$$

for $k \ge K$.

Since the probabilities for $k < K$ are uniformly bounded by 1,

$$\begin{aligned}
\sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n \le n_k} |S_n| > n_{k-1}\epsilon\right] &\le K - 1 + \sum_{k \ge K} \frac{8 n_k \sigma^2}{n_{k-1}^2 \epsilon^2} \\
&= K - 1 + O\left(\sum_k \frac{1}{2^k}\right) < \infty.
\end{aligned}$$

□

Theorem 4.2.2 (Kolmogorov's Law of Large Numbers).

Suppose $\{X_i\}$ is a sequence of independent random variables, $\mathbb{E}[X_i] = 0$ for each X_i, and $\sum_i \frac{\mathbb{E}[X_i^2]}{i^2} < \infty$. Then $\frac{X_1 + \cdots + X_n}{n} \to 0$ almost surely.

Proof of Theorem 4.2.2. Left to homework. □

Theorem 4.2.3 (Strong Law of Large Numbers).

Suppose $\{X_i\}$ is a sequence of independent and identically distributed random variables with $\mathbb{E}[|X_1|] < \infty$. Then

$$\frac{X_1 + \cdots + X_n}{n} \to \mathbb{E}[X_1]$$

almost surely.

Proof of Theorem 4.2.3. By replacing X_i with $X_i - \mathbb{E}[X_1]$, it suffices to prove the centered case, so assume $\mathbb{E}[X_i] = 0$.

We proceed by truncation, splitting the sum $\frac{1}{n}\sum_{i=1}^{n} X_n$ into the difference of bad and good behavior, the difference of good behavior and its expectation, and the expectation of good behavior. Define

$$Y_i = X_i \mathbb{1}_{|X_i| \leq i}.$$

Then

$$\begin{aligned} \mu_i := \mathbb{E}[Y_i] &= \mathbb{E}\left[X_i \mathbb{1}_{|X_i| \leq i}\right] \\ &= \mathbb{E}\left[X \mathbb{1}_{|X| \leq i}\right] \text{ by identical distribution} \\ &= -\mathbb{E}\left[X \mathbb{1}_{|X| > i}\right] \text{ by } \mathbb{E}[X_i] = 0. \end{aligned}$$

Decompose

$$\frac{1}{n}\sum_{i=1}^{n} X_i = \underbrace{\frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i)}_{(I)} + \underbrace{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mu_i)}_{(II)} + \underbrace{\frac{1}{n}\sum_{i=1}^{n}\mu_i}_{(III)}.$$

Starting with the last term,

$$\begin{aligned} |(III)| = \left|\frac{1}{n}\sum_{i=1}^{n}\mu_i\right| &= \left|\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[X \mathbb{1}_{|X|>i}\right]\right| \\ &\leq \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X| \mathbb{1}_{|X|>i}\right] \\ &= \mathbb{E}\left[|X| \frac{1}{n}\sum_{i=1}^{n}\mathbb{1}_{|X|>i}\right]. \end{aligned}$$

The indicator sum is a counting function:

$$\begin{aligned}
\sum_{i=1}^{n} \mathbb{1}_{|X|>i} &\leq \sum_{i=1}^{\infty} \mathbb{1}_{|X|>i} \\
&= \sum_{i=1}^{\infty} \sum_{j=i}^{\infty} \mathbb{1}_{j<|X|\leq j+1} \\
&= \sum_{j=1}^{\infty} \sum_{i=1}^{j} \mathbb{1}_{j<|X|\leq j+1} \\
&= \sum_{j=1}^{\infty} j \mathbb{1}_{j<|X|\leq j+1} \\
&\leq |X| \sum_{j=1}^{\infty} \mathbb{1}_{j<|X|\leq j+1} = |X|,
\end{aligned}$$

Naively, we may try to plug this bound on the count to the bound on $|(III)|$ and get

$$|(III)| \leq \mathbb{E}\left[|X| \frac{1}{n} |X|\right],$$

which may be infinite because we don't have control over the second moment. However, noting that $\frac{1}{n} \sum\limits_{j=1}^{n} \mathbb{1}_{|X|>i} \leq 1$, we get

$$|(III)| \leq \mathbb{E}\left[|X| \min\{\frac{1}{n} |X|, 1\}\right].$$

Because $|X|$ dominates $|X| \min\{\frac{1}{n} |X|, 1\}$ and $\mathbb{E}[|X|] < \infty$, we apply the DCT to see

$$\begin{aligned}
\lim_{n\to\infty} |(III)| &\leq \lim_{n\to\infty} \mathbb{E}\left[|X| \min\left\{\frac{1}{n} |X|, 1\right\}\right] \\
&= \mathbb{E}\left[\lim_{n\to\infty} |X| \min\left\{\frac{1}{n} |X|, 1\right\}\right]. \\
&\xrightarrow{n\to\infty} 0
\end{aligned}$$

For (I), consider with Borel-Cantelli in mind that

$$\begin{aligned}
\sum_{i=1}^{\infty} \mathbb{P}[X_i \neq Y_i] &\leq \sum_{i=1}^{\infty} \mathbb{P}[|X| > i] \\
&= \mathbb{E}\left[\sum_{i=1}^{\infty} \mathbb{1}_{|X|>i}\right] \\
&\leq \mathbb{E}[|X|] < \infty,
\end{aligned}$$

so

$$\mathbb{P}[X_i \neq Y_i \text{ i.o.}] = 0.$$

Then there exists $\Omega_0 \subseteq \Omega$ such that $\mathbb{P}[\Omega_0] = 1$ and for every $\omega \in \Omega_0$, there exists

$i(\omega)$ such that

$$X_i(\omega) = Y_i(\omega)$$

for all $i > i(\omega)$. For each $\omega \in \Omega_0$,

$$(I) = \frac{1}{n}\sum_{i=1}^{n}(X_i(\omega) - Y_i(\omega)) = \frac{1}{n}\sum_{i=1}^{i(\omega)}(X_i(\omega) - Y_i(\omega)) \to 0,$$

where the convergence follows from the sum being fixed.

Since $\mathbb{P}[\Omega_0] = 1$,

$$(I) = \frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i) \to 0$$

almost surely.

Examining (II), we want to use Kolmogorov's Law of Large Numbers. Note that $Y_i^2 = (Y_i - \mu_i + \mu_i)^2 = (Y_i - \mu_i)^2 + \mu_i^2 + 2(Y_i - \mu_i)(\mu_i)$. The third term vanishes in expectation. This gives us the Kolmogorov condition:

The second inequality holds since $\sum_{i=j}^{\infty} \frac{1}{i^2} \leq C\frac{1}{j}$ for some absolute constant C.

$$\begin{aligned}
\sum_{i=1}^{\infty}\frac{\mathbb{E}\left[(Y_i - \mu_i)^2\right]}{i^2} &= \sum_{k=1}^{\infty}\frac{\mathbb{E}\left[Y_i^2\right]}{i^2} - \frac{\mathbb{E}\left[\mu_i^2\right]}{i^2} \\
&\leq \sum_{i=1}^{\infty}\frac{\mathbb{E}\left[Y_i^2\right]}{i^2} \\
&= \sum_{i=1}^{\infty}\frac{1}{i^2}\mathbb{E}\left[X^2 \mathbb{1}_{|X|\leq i}\right] \\
&= \sum_{i=1}^{\infty}\frac{1}{i^2}\mathbb{E}\left[X^2 \sum_{j=1}^{i}\mathbb{1}_{j-1<|X|\leq j}\right] \\
&= \sum_{j=1}^{\infty}\sum_{i=j}^{\infty}\frac{1}{i^2}\mathbb{E}\left[X^2 \mathbb{1}_{j-1<|X|\leq j}\right] \\
&\leq C\sum_{j=1}^{\infty}\frac{1}{j}\mathbb{E}\left[X^2 \mathbb{1}_{j-1<|X|\leq j}\right] \\
&\leq C\sum_{j=1}^{\infty}\mathbb{E}\left[|X|\, \mathbb{1}_{j-1<|X|\leq j}\right] \\
&\leq C\mathbb{E}\left[|X|\right] < \infty,
\end{aligned}$$

so we can apply Kolmogorov's LLN to get

$$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mu_i) \to 0$$

almost surely.

All three terms converge to 0 almost surely, so their sum does as well. □

4.3 Proof Techniques

When proving almost sure convergence for averages of partial sums, it helps to decide early which obstruction is preventing a direct Borel–Cantelli argument.

This section was not done in lecture; I wrote it after to organize the different techinques we learned, since they were all unfamiliar.

1. If
$$\sum_{n=1}^{\infty} \mathbb{P}\left[|S_n| > \epsilon a_n\right] < \infty$$
for every $\epsilon > 0$, then Borel–Cantelli already gives $\frac{S_n}{a_n} \to 0$ almost surely.

2. If the direct sum diverges but a subsequence could be manageable, choose block endpoints n_k and try to prove
$$\sum_{k=1}^{\infty} \mathbb{P}\left[|S_{n_k}| > \epsilon a_{n_k}\right] < \infty.$$
Geometric choices such as $n_k = 2^k$ work well, as we can then block maxima with the maximal inequality. For
$$B_k = \{n_{k-1} < n \le n_k\},$$
one tries to bound
$$\mathbb{P}\left[\max_{n \in B_k} \frac{|S_n|}{a_n} > \epsilon\right].$$
When $a_n = n$, blocking is especially good because
$$\max_{n \in B_k} \frac{|S_n|}{n} \le \frac{1}{n_{k-1}} \max_{n \in B_k} |S_n|.$$

3. If the original variables do not have enough finite moments, truncate:
$$Y_i = X_i \mathbb{1}_{|X_i| \le b_i}.$$
Then split the average into three parts:
$$\frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i) + \frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathbb{E}[Y_i]) + \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[Y_i].$$
The first term handles rare large jumps, the second is the centered "good" part, and the third is the bias introduced by truncation.

4. After truncation, set
$$Z_i = Y_i - \mathbb{E}[Y_i].$$
If
$$\sum_{i=1}^{\infty} \frac{\mathbb{E}\left[Z_i^2\right]}{i^2} < \infty,$$

then Kolmogorov's LLN applies to $\frac{1}{n}\sum_{i=1}^{n} Z_i$. The truncation threshold b_i can depend on i.

Central Limit Theorem 5

5.1 Weak Convergence

We will eventually prove the following.

Theorem 5.1.1 (Central Limit Theorem).

Let $\{X_i\}$ be iid with $\mathbb{E}[X_i] = 0$ and $\operatorname{Var}(X_i) = 1$. Then

$$\sqrt{n}\frac{X_1 + \cdots + X_n}{n} \xrightarrow{d} N(0,1).$$

But what does $\xrightarrow{d}$ mean? It means $\rightsquigarrow$.

Consider a metric space $(\mathscr{X}, d)$ and a random variable

$$X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathscr{X}, \mathscr{B}(\mathscr{X})).$$

For every $B \in \mathscr{B}(\mathscr{X})$, define

$$P(B) = \mathbb{P}\left[X^{-1}(B)\right] = \mathbb{P}[X \in B].$$

We showed on the midterm that P is a probability measure on $(\mathscr{X}, \mathscr{B}(\mathscr{X}))$. For $f : (\mathscr{X}, \mathscr{B}(\mathscr{X})) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, we equivalently write all of the following

$$Pf = \int f dP = \int_{\mathscr{X}} f(x) dP(x) = \int_{\Omega} f(X(\omega)) d\mathbb{P}(\omega) = \int f(X) d\mathbb{P} = \mathbb{P}f = \mathbb{E}[f(X)].$$

We say $X \sim P$, where P is the distribution of X.

Definition 5.1.2 (Bounded Lipschitz function).

Define the class of bounded Lipschitz functions on $\mathscr{X}$ as

$$\mathrm{BL}(\mathscr{X}) = \left\{ f : \mathscr{X} \to \mathbb{R} : \sup_x |f(x)| < \infty, \sup_{x \neq y} \frac{|f(x) - f(y)|}{d(x,y)} < \infty \right\}.$$

Definition 5.1.3 (Weak convergence).

Suppose P_n, P are probability measures on $(\mathscr{X}, \mathscr{B}(\mathscr{X}))$. We say $P_n \to P$ weakly if $P_n l \to Pl$ for all $l \in \mathrm{BL}(\mathscr{X})$.

Equivalently, $\mathbb{E}[l(X_n)] \to \mathbb{E}[l(X)]$ for all $X_n \sim P_n$ and $X \sim P$.

Note that $X_1, X_2, \ldots$ can all live on different probability spaces, but they all must map to the same measure space. Each X_n then induces its own probability measure on the common measure space. Essentially,

$$\begin{gathered} X_1 : (\Omega_1, \mathscr{F}_1, \mathbb{P}_1) \xrightarrow{X_1} (\mathscr{X}, \mathscr{B}(\mathscr{X})), \quad X_1 \sim P_1 = \mathbb{P} \circ X_1^{-1} \\ X_2 : (\Omega_2, \mathscr{F}_2, \mathbb{P}_2) \xrightarrow{X_2} (\mathscr{X}, \mathscr{B}(\mathscr{X})), \quad X_2 \sim P_2 = \mathbb{P} \circ X_2^{-1} \\ \vdots \end{gathered}$$

Definition 5.1.4 (Lower semi-continuity, upper semi-continuity).

We say that $g : \mathscr{X} \to \mathbb{R}$ is lower semi-continuous if

$$\{x \in \mathscr{X} : g(x) > t\} = \{g > t\}$$

is open in the topology of $\mathscr{X}$ for every $t \in \mathbb{R}$.
$f : \mathscr{X} \to \mathbb{R}$ is upper semi-continuous if $-f$ is lower semi-continuous (equivalently, $\{f < t\}$ is open for every $t \in \mathbb{R}$).

Example 5.1.5.

$g = \mathbb{1}_{(1,2)}$ is lower semi-continuous, but $h = \mathbb{1}_{[1,2]}$ is not. All continuous g are lower semi-continuous. Any indicator function of an open set is LSC.

Lemma 5.1.6.

Suppose g is lower semi-continuous and $g \geq -M$. Then there exist $l_n \in \mathrm{BL}(\mathscr{X})$ such that $l_n \uparrow g$.

Proof of Lemma 5.1.6. WLOG, assume $g \geq 0$. Define $h_t = t\mathbb{1}_{g>t}$. Then for each

$x \in \mathscr{X}$, $\sup_t h_t(x) = g(x)$, as

$$\sup_t h_t(x) = \sup_t t\mathbb{1}_{g(x)>t} = g(x).$$

These step functions are not Lipschitz.

Define $l_{k,t}(x) = t \wedge (k \cdot d(x, \{g \le t\}))$ with the metric between a point x and set B defined as

$$d(x, B) = \inf_{y \in B} d(x, y)$$

using the metric d on $\mathscr{X}$. We see that $l_{k,t} \in \mathrm{BL}(\mathscr{X})$ and

$$\begin{aligned}\sup_k l_{k,t}(x) &= t\mathbb{1}_{d(x,\{g\le t\})>0} \\ &= t\mathbb{1}_{x\notin\{g\le t\}} \\ &= t\mathbb{1}_{g(x)>t} = h_t(x).\end{aligned}$$

Combining our two results,

$$\sup_{k,t} l_{k,t} = g.$$

We have approximated g with the supremum of bounded Lipschitz functions, but not rigorously.

Because $\mathbb{Q}_+$ is countable and dense in $[0, \infty)$, the countable family $\{l_{k,t} : k \in \mathbb{N},\ t \in \mathbb{Q}_+\}$ still has supremum g. Enumerate it as $f_1, f_2, \ldots$, and replace f_n by $\max_{1\le i\le n} f_i$; finite maxima preserve the bounded Lipschitz property, so we obtain $l_n \uparrow g$. $\square$

Lemma 5.1.7.

Suppose $P_n \to P$. Then

(i) For any lower semi-continuous g that is bounded below,

$$\liminf_{n\to\infty} P_n g \ge Pg.$$

(ii) For any upper semi-continuous f that is bounded above,

$$\limsup_{n\to\infty} P_n f \le Pf.$$

Proof of Lemma 5.1.7. Note that it suffices to prove (i), since (ii) follows from the definition of upper semi-continuity. We know by our approximation lemma that there exists $\{l_i\} \subseteq \mathrm{BL}(\mathscr{X})$ such that $l_i \uparrow g$. For every $i \in \mathbb{N}$,

$$\begin{aligned}\liminf_{n\to\infty} P_n g &\ge \liminf_{n\to\infty} P_n l_i \\ &= Pl_i\end{aligned}$$

by definition of weak convergence of $P_n \to P$. Sending $i \to \infty$, we get by the monotone convergence theorem that $Pl_i \to Pg$. $\square$

Example 5.1.8.

While the directions of the inequalities seem counterintuitive when we look at where the inf and sup are, the direction is determined by the LSC/USC property of f and g.
We can see a simple example of the inequalities holding strictly. Let $X_n : \frac{1}{n}$ and $X := 0$ almost surely, and let P_n, P denote their distributions. Then $P_n \to P$ weakly since for every bounded continuous l,

$$\mathbb{E}\left[l(X_n)\right] = l\left(\frac{1}{n}\right) \to l(0) = \mathbb{E}\left[l(X)\right].$$

Now take

$$g = \mathbb{1}_{(0,\infty)}, \qquad f = \mathbb{1}_{\{0\}}.$$

Then g is lower semi-continuous and bounded below, while f is upper semi-continuous and bounded above. We see

$$P_n g = \mathbb{P}\left(X_n > 0\right) = 1, \qquad Pg = \mathbb{P}\left(X > 0\right) = 0,$$

so $\limsup_{n\to\infty} P_n g \leq Pg$ is false; and

$$P_n f = \mathbb{P}\left(X_n = 0\right) = 0, \qquad Pf = \mathbb{P}\left(X = 0\right) = 1,$$

so $\liminf_{n\to\infty} P_n f \geq Pf$ is false.

Corollary 5.1.9.

If $P_n \to P$, then

(i) For any open B, $\liminf\limits_{n\to\infty} P_n(B) \geq P\left(B\right)$

(ii) For any closed G, $\limsup\limits_{n\to\infty} P_n(G) \leq P(G)$.

Proof of Corollary 5.1.9. For (i), consider $f := \mathbb{1}_B$, which is LSC from B being open. Then

$$\liminf_{n\to\infty} P_n(B) = \liminf_{n\to\infty} P_n f \geq Pf = P(B).$$

For (ii), we can consider the lemma with $g := \mathbb{1}_G$, which is USC from G being closed. $\square$

Notation 5.1.10 (Lower semi-continuous minorant).

We write
$$\mathring{f} := \sup\{g \le f : g \text{ is LSC}\} = \sup_{g \in \mathscr{G}} g$$
and
$$\overline{f} = \inf\{g \ge f : g \text{ is USC}\}.$$

Note that $\mathring{f}$ itself is LSC since
$$\left\{\mathring{f} > t\right\} = \left\{\sup_{g\in\mathscr{G}} g > t\right\} = \bigcup_{g\in\mathscr{G}} \{g > t\}$$
is open. Hence $\mathring{f}$ is LSC.

Similarly, $\overline{f}$ is USC. On the homework, we show that $\mathring{\mathbb{1}}_B = \mathbb{1}_{\mathring{B}}$ and also have $\overline{\mathbb{1}}_B = \mathbb{1}_{\overline{B}}$. This justifies the notation.

Since the interior of a set is the largest open subset, $\mathring{f}$ is the function analogue: the largest lower semi-continuous function not exceeding f.

Theorem 5.1.11.

Suppose $|f| \le M$. Then
$$\mathring{f}(x) = \overline{f}(x) \iff f \text{ is continuous at } x.$$

Proof of Theorem 5.1.11. Recall f is continuous at x if for every $\epsilon > 0$, there exists open $U \ni x$ such that for every $y \in U$, $|f(x) - f(y)| < \epsilon$.

($\implies$) From $\mathring{f}(x) = f(x)$, $U = \left\{y : \mathring{f}(y) > f(x) - \epsilon\right\}$ is open. Then $\mathring{f}(x) = f(x) > f(x) - \epsilon$, so $x \in U$. For every $y \in U$, $f(y) \ge \mathring{f}(y) > f(x) - \epsilon$.

From $\overline{f}(x) = f(x)$, put
$$V = \left\{y : \overline{f}(y) < f(x) + \epsilon\right\}.$$

This set is open and contains x. For every $y \in V$, $f(y) \le \overline{f}(y) < f(x) + \epsilon$.

Define $W = U \cap V$, which is the intersection of open sets that contain x, so $W \ni x$ is open. For every $y \in W$, $|f(x) - f(y)| < \epsilon$, so f is continuous at x.

($\impliedby$) We want to find for any $\epsilon \in (0,1)$ a LSC $g \le f$ and USC $h \ge f$ such that $h(x) - g(x) \le 2\epsilon$, since then we can let $\epsilon \downarrow 0$ and conclude. We can expand the inequality given to us by continuity to see
$$f(x) - \epsilon \le f(y) \le f(x) + \epsilon$$
for all $y \in U$, where $U \ni x$ is open. We can upgrade this inequality to all of $\mathscr{X}$ from U by fixing some U then seeing
$$g := (f(x) - \epsilon)\,\mathbb{1}_U - (M+1)\mathbb{1}_{U^c} \le f(y) \le (f(x) + \epsilon)\,\mathbb{1}_U + (M+1)\,\mathbb{1}_{U^c} =: h.$$

Since $M \geq |f|$ and $1 \geq \epsilon$, the upper bound is upper semi-continuous and the lower bound is lower semi-continuous. Also,

$$g(x) = f(x) - \epsilon, \qquad h(x) = f(x) + \epsilon.$$

Hence

$$f(x) - \epsilon = g(x) \leq \mathring{f}(x) \leq \overline{f}(x) \leq h(x) = f(x) + \epsilon.$$

Sending $\epsilon \to 0$ gives

$$\mathring{f}(x) = \overline{f}(x).$$

□

Note that the $\implies$ direction does not require boundedness of f.

Remark 5.1.12. From this, we can write the points of $\mathscr{X}$ where f is continuous as

$$\begin{aligned} C_f &= \left\{x : \mathring{f}(x) = \overline{f}(x)\right\} \\ &= \left\{\mathring{f} = \overline{f}\right\}. \end{aligned}$$

Because $\mathring{f}$ and $\overline{f}$ are measurable (their definitions characterize the preimage of open sets), $\mathring{f} - \overline{f}$ is measurable. Hence, the preimage of $\{0\}$ under $\mathring{f} - \overline{f}$ is measurable; i.e., $C_f \in \mathscr{B}(\mathscr{X})$.

We also see that

$$P(C_f) = 1 \iff P\overline{f} = P\mathring{f} \iff P\left(\mathring{f} - \overline{f}\right) = 0.$$

Theorem 5.1.13.

Suppose $P_n \to P$ weakly. Then

(i) $P_n f \to Pf$ for every f such that $|f| \leq M$ and $P(C_f) = 1$

(ii) In particular, applying (i) to indicator functions yields that $P_n B \to PB$ for every $B \in \mathscr{B}(\mathscr{X})$ such that $P(\partial B) = 0$.

Proof of Theorem 5.1.13. Recall from Lemma (5.1.7) that

$$\begin{aligned} P\overline{f} &\geq \limsup_{n\to\infty} P_n f \\ &\geq \liminf_{n\to\infty} P_n f \\ &\geq P\mathring{f}. \end{aligned}$$

Now applying Theorem (5.1.11) yields $P\overline{f} = P\mathring{f}$, so all of the inequalities hold with equality. In particular,

$$\limsup_{n\to\infty} P_n f = \liminf_{n\to\infty} P_n f = \lim_{n\to\infty} P_n f = P\overline{f} = P\mathring{f} = Pf.$$

With $B \in \mathscr{B}(\mathscr{X})$ and $f := \mathbb{1}_B$, we see that f is discontinuous at ∂B. If $P(\partial B) = 0$, then $P(C_f) = 1$, so

$$P_n(B) = P_n \mathbb{1}_B \to P\mathbb{1}_B = P(B).$$

□

Theorem 5.1.14 (Portmanteau).

The following are equivalent:

(i) $P_n l \to Pl$ for every $l \in \mathrm{BL}(\mathscr{X})$

(ii) $\liminf_{n\to\infty} P_n g \geq Pg$ for every lower semi-continuous $g \geq -M$

(iii) $\limsup_{n\to\infty} P_n f \leq Pf$ for every upper semi-continuous $f \leq M$

(iv) $\liminf_{n\to\infty} P_n B \geq PB$ for every open B

(v) $\limsup_{n\to\infty} P_n G \leq PG$ for every closed G

(vi) $P_n f \to Pf$ for every $|f| \leq M$ such that $P(C_f) = 1$

(vii) $P_n B \to PB$ for every $B \in \mathscr{B}(\mathscr{X})$ such that $P(\partial B) = 0$.

Proof of Theorem 5.1.14. The implications proved above already showed that (i) implies (ii) and (iii), (iv) and (v), and (vi) and (vii), so we only need to show (vii) $\implies$ (i).

Proving the desired result for every bounded continuous f is more than enough, since Lipschitz continuous functions are a subclass of bounded continuous functions.

By splitting into positive and negative parts, we may assume $0 \leq f \leq 1$. Note that

$$f(x) = \int_0^1 \mathbb{1}_{f(x)>t}\, dt$$

$$\begin{aligned} Pf = \int_{\mathscr{X}} f(x)\, dP(x) &= \int_{\mathscr{X}} \int_0^1 \mathbb{1}_{f(x)>t} dt dP(x) \\ &= \int_0^1 \int_{\mathscr{X}} \mathbb{1}_{f(x)>t} dP(x) dt \\ &= \int_0^1 P(f > t) dt \end{aligned}$$

by Fubini.

Similarly,

$$P_n f = \int_0^1 P_n(f > t) dt.$$

We can relate functions to sets by noting

$$\begin{aligned}
\partial \{f > t\} &= \overline{\{f > t\}} \setminus \{f \overset{\circ}{>} t\} \\
&= \overline{\{f > t\}} \setminus \{f > t\} \\
&\subseteq \{f \geq t\} \setminus \{f > t\} \\
&= \{f = t\}.
\end{aligned}$$

Hence, if $P(f = t) = 0$, then $P(\partial \{f > t\}) = 0$, so by (vii), we have

$$P_n(f > t) \to P(f > t).$$

Now define $T = \{t : P(f = t) > 0\}$. We claim that T is countable. To see why, write

$$\begin{aligned}
T &= \{t : P(f = t) > 0\} \\
&= \bigcup_n \left\{ t : P(f = t) \geq \frac{1}{n} \right\}.
\end{aligned}$$

Since the cardinality of $\left\{ t : P(f = t) \geq \frac{1}{n} \right\}$ is at most n, T is countable. Thus, the integrals of $P(f > t)$ and $P_n(f > t)$ over T are zero.

This lets us write

$$\begin{aligned}
Pf &= \int_0^1 P(f > t) dt = \int_{T^c} P(f > t) dt \\
P_n f &= \int_0^1 P_n(f > t) dt = \int_{T^c} P_n(f > t) dt.
\end{aligned}$$

For every $t \in T^c$, we have convergence of the integrands, and they are all bounded by 1. By the DCT,

$$P_n f \to Pf.$$

$\square$

Definition 5.1.15 (Products).

For $(\Omega_1, \mathscr{F}_1, \mu_1)$ and $(\Omega_2, \mathscr{F}_2, \mu_2)$ where μ_1, μ_2 are finite measures, define

$$\Omega_1 \times \Omega_2 = \{(\omega_1, \omega_2) : \omega_1 \in \Omega_1, \omega_2 \in \Omega_2\}$$
$$\mathscr{F}_1 \times \mathscr{F}_2 = \sigma\left(\{\underbrace{A_1 \times A_2}_{\text{rectangle}}\} : A_1 \in \mathscr{F}_1, A_2 \in \mathscr{F}_2\right)$$

.

Note that the product measure is first defined on the field of rectangles, then extended to the generated σ-field.

Theorem 5.1.16 (Fubini).

Suppose μ_1, μ_2 are finite.

(i) There exists a unique μ on $(\Omega_1 \times \Omega_2, \mathscr{F}_1 \times \mathscr{F}_2)$ such that for every $A_1 \in \mathscr{F}_1$ and $A_2 \in \mathscr{F}_2$,

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2).$$

(ii) Suppose $f : (\Omega_1 \times \Omega_2, \mathscr{F}_1 \times \mathscr{F}_2) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ is measurable and $\int |f|\, d\mu < \infty$. Then

$$\begin{aligned} \int_{\Omega_1 \times \Omega_2} f d\mu &= \int_{\Omega_2} \int_{\Omega_1} f(\omega_1, \omega_2) d\mu_1(\omega_1) d\mu_2(\omega_2) \\ &= \int_{\Omega_2} \int_{\Omega_1} f(\omega_1, \omega_2) d\mu_2(\omega_2) d\mu_1(\omega_1). \end{aligned}$$

The Portmanteau theorem (5.1.14) above gave us several equivalent characteriztions of weak convergence of measures defined on an arbitrary measurable space $(\mathscr{X}, \mathscr{B}_{\mathscr{X}})$. We have the following equivalent characterizations when the measurable space is $(\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, the second of which is the usual notion of convergence in distribution.

Theorem 5.1.17 (Weak Convergence on $\mathbb{R}$).

The following are equivalent:

(i) $P_n l \to Pl$ for every $l \in \mathrm{BL}(\mathbb{R})$

(ii) $P_n(-\infty, t] \to P(-\infty, t]$ for every t such that $P(\{t\}) = 0$

(iii) $P_n f \to Pf$ for every $f \in C^\infty(\mathbb{R})$, where

$$C^\infty(\mathbb{R}) = \left\{ f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty < \infty, \left\| f^{(k)} \right\|_\infty < \infty, \ \forall\, k \right\}.$$

Proof of Theorem 5.1.17. Homework is to show equivalence of (i) and (ii). (i) $\implies$ (iii) because $C^\infty(\mathbb{R}) \subseteq \mathrm{BL}(\mathbb{R})$, while (iii) $\implies$ (i) requires more work.

Fix $l \in \mathrm{BL}(\mathbb{R})$. We know l is continuous, but do not know it is infinitely differentiable, nor do we have control over the derivatives, if they exist. Define

$$l_\sigma(x) = \mathbb{E}\left[l(x + \sigma Z)\right], \quad Z \sim N(0,1).$$

We can check that $l_\sigma \in C^\infty(\mathbb{R})$ with induction.

l_σ is the Gaussian smoothing of l, i.e. convolution with the $N(0, \sigma^2)$ density.

We have that

$$\begin{aligned} |l_\sigma(x) - l(x)| &= \left|\mathbb{E}\left[l(x + \sigma Z)\right] - l(x)\right| \\ &\le \mathbb{E}\left[|l(x + \sigma Z) - l(x)|\right] \\ &\le K\mathbb{E}\left[\sigma\,|Z|\right] \\ &= \sigma K \mathbb{E}\left[|Z|\right]. \end{aligned}$$

For every $\epsilon > 0$, there exists $\sigma > 0$ such that

$$\sup_x |l_\sigma(x) - l(x)| \le \epsilon,$$

which yields the uniform bounds

$$P_n\left(l - l_\sigma\right) \le \epsilon, \quad P\left(l - l_\sigma\right) \le \epsilon.$$

By the triangle inequality,

$$|P_n l - Pl| \le |P_n l_\sigma - P l_\sigma| + 2\epsilon \to 0 + 2\epsilon,$$

so

$$\limsup_{n \to \infty} |P_n l - Pl| \le 2\epsilon.$$

We send $\epsilon \to 0$ to conclude. ☐

5.2 Proving the Central Limit Theorem

We will use Lindeberg swapping, then see a secondary proof using Stein's method.

Because linear combinations of Gaussian random variables are Gaussian, we know that if $Z_i \sim N(0,1)$,

$$\frac{Z_1 + \cdots + Z_n}{\sqrt{n}} \sim N(0,1),$$

as the first and second moments match. Lindeberg swapping will try to get us from $\frac{X_1+\cdots+X_n}{\sqrt{n}}$ to $\frac{Z_1+\cdots+Z_n}{\sqrt{n}}$.

First, we provide intuition for the specific statement of the CLT.

Aside 5.2.1. Consider $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. Then the CLT tells us

$$\frac{\sum_{i=1}^{n}(X_i - p)}{\sqrt{np\,(1-p)}} \rightsquigarrow N(0,1).$$

But what if p depends on n, for example $p_n \to 0$? Then the previous iid statement no longer applies verbatim. If instead $np_n \to \lambda$, then $\sum_{i=1}^{n} X_i \rightsquigarrow \text{Poisson}(\lambda)$. What really happens is that we have

$$\begin{aligned}
n = 1 : X_{11} &\sim \text{Bernoulli}(p_1)\\
n = 2 : X_{21}, X_{22} &\overset{\text{iid}}{\sim} \text{Bernoulli}(p_2)\\
&\vdots\\
n \ : X_{n_1}, \ldots, X_{nn} &\overset{\text{iid}}{\sim} \text{Bernoulli}(p_n).
\end{aligned}$$

We see that we need to consider this triangular array in our statement of the CLT. We want to say

$$\frac{\sum_{i=1}^{n}(X_{ni} - p_n)}{\sqrt{np_n(1-p_n)}} \sim P_n, \quad P_n \rightsquigarrow N(0,1).$$

Definition 5.2.2 (Triangular array).

An array is triangular if it is of the form

$$\begin{aligned}
&\xi_{1,1}, \xi_{1,2}, \ldots, \xi_{1,k(1)}\\
&\xi_{2,1}, \xi_{2,2}, \ldots, \xi_{2,k(2)}\\
&\vdots\\
&\xi_{n,1}, \xi_{n,2}, \ldots, \xi_{n,k(n)}
\end{aligned}$$

with $k(n) \to \infty$ as $n \to \infty$. We have independence within each row and do not care about dependence across rows.

Theorem 5.2.3 (Central Limit).

Suppose we have a triangular array of random variables $\xi_{n,i}$ such that

(i) $\sum_{i=1}^{k(n)} \mathbb{E}\left[\xi_{n,i}\right] \to \mu$,

(ii) $\sum_{i=1}^{k(n)} \operatorname{Var}\left(\xi_{n,i}\right) \to \sigma^2$, and

(iii) $\sum_{i=1}^{k(n)} \mathbb{E}\left[\left|\xi_{n,i}\right|^3\right] \to 0$.

Then

$$\sum_{i=1}^{k(n)} \xi_{n,i} \rightsquigarrow N\left(\mu, \sigma^2\right).$$

Note that the summations are defined on each row, so the limit takes us *down* the array.

Lemma 5.2.4.

Let $\xi_1, \ldots, \xi_k$ and $\eta_1, \ldots, \eta_k$ be independent random variables such that each η_i is Gaussian and, for every i,

$$\mathbb{E}\left[\xi_i\right] = \mathbb{E}\left[\eta_i\right], \qquad \mathbb{E}\left[\xi_i^2\right] = \mathbb{E}\left[\eta_i^2\right], \qquad \mathbb{E}\left[\left|\xi_i\right|^3\right] < \infty.$$

Then for every $f \in C^3(\mathbb{R})$ with $\left\|f'''\right\|_\infty < \infty$,

$$\left|\mathbb{E}\left[f\left(\sum_{i=1}^k \xi_i\right)\right] - \mathbb{E}\left[f\left(\sum_{i=1}^k \eta_i\right)\right]\right| \le C \left\|f'''\right\|_\infty \sum_{i=1}^k \mathbb{E}\left[\left|\xi_i\right|^3\right],$$

where $C > 0$ is a universal constant.

Proof of Lemma 5.2.4. Defining

$$\begin{aligned}
X_i &= \eta_1 + \eta_2 + \cdots + \eta_{i-1} + \xi_{i+1} + \cdots + \xi_k, \\
Y_i &= \xi_i, \\
Z_i &= \eta_i,
\end{aligned}$$

the ith swap replaces $X_i + Y_i$ with $X_i + Z_i$. By the triangle inequality, it suffices to control one swap and then sum over i.

Let $M = \frac{1}{6}\left\|f'''\right\|_\infty$. Taylor's theorem gives

$$\begin{aligned}
f(X_i + Y_i) &= f(X_i) + f'(X_i)Y_i + \frac{1}{2} f''(X_i) Y_i^2 + R(X_i, Y_i), \\
R(X_i, Y_i) &= \frac{1}{6} f'''(x^*) Y_i^3, \quad x^* \text{ between } X_i \text{ and } X_i + Y_i,
\end{aligned}$$

so $|R(X_i, Y_i)| \le M\,|Y_i|^3$. The same bound holds with Z_i in place of Y_i.

By independence of X_i from both Y_i and Z_i,

$$\begin{aligned}
\mathbb{E}\left[f(X_i+Y_i)\right] &= \mathbb{E}\left[f(X_i)\right] + \mathbb{E}\left[f'(X_i)\right]\mathbb{E}\left[Y_i\right] \\
&\quad + \frac{1}{2}\mathbb{E}\left[f''(X_i)\right]\mathbb{E}\left[Y_i^2\right] + \mathbb{E}\left[R(X_i, Y_i)\right], \\
\mathbb{E}\left[f(X_i+Z_i)\right] &= \mathbb{E}\left[f(X_i)\right] + \mathbb{E}\left[f'(X_i)\right]\mathbb{E}\left[Z_i\right] \\
&\quad + \frac{1}{2}\mathbb{E}\left[f''(X_i)\right]\mathbb{E}\left[Z_i^2\right] + \mathbb{E}\left[R(X_i, Z_i)\right].
\end{aligned}$$

Since $\mathbb{E}\left[Y_i\right] = \mathbb{E}\left[Z_i\right]$ and $\mathbb{E}\left[Y_i^2\right] = \mathbb{E}\left[Z_i^2\right]$, the first three Taylor terms cancel, and therefore

$$\begin{aligned}
\left|\mathbb{E}\left[f(X_i+Y_i)\right] - \mathbb{E}\left[f(X_i+Z_i)\right]\right| &\le \left|\mathbb{E}\left[R(X_i, Y_i)\right]\right| + \left|\mathbb{E}\left[R(X_i, Z_i)\right]\right| \\
&\le M\mathbb{E}\left[|Y_i|^3\right] + M\mathbb{E}\left[|Z_i|^3\right].
\end{aligned}$$

Write $Z_i \sim N(\mu, \sigma^2) = \mu + \sigma X$ for $X \sim N(0,1)$. We have the bound

$$\begin{aligned}
\mathbb{E}\left[|Z_i|^3\right] &= \mathbb{E}\left[|\mu + \sigma X|^3\right] \\
&\le 4\left(|\mu|^3 + \sigma^3\,|X|^3\right) \\
&\le C_1\left(|\mu|^3 + \sigma^3\right) \\
&\le C_2\left(\mathbb{E}\left[|Y_i|\right]^3 + \mathbb{E}\left[Y_i^2\right]^{\frac{3}{2}}\right) \\
&\le C_3\mathbb{E}\left[|Y_i|^3\right],
\end{aligned}$$

where the second to last step moves the absolute value inside the expectation and the last step follows from Jensen's inequality with convex $f(x) = x^{\frac{3}{2}}$. Hence

$$\left|\mathbb{E}\left[f(X_i+Y_i)\right] - \mathbb{E}\left[f(X_i+Z_i)\right]\right| \le C\left\|f'''\right\|_\infty \mathbb{E}\left[|\xi_i|^3\right].$$

Summing over $i = 1, \ldots, k$ yields

$$\left|\mathbb{E}\left[f\left(\sum_{i=1}^k \xi_i\right)\right] - \mathbb{E}\left[f\left(\sum_{i=1}^k \eta_i\right)\right]\right| \le C\left\|f'''\right\|_\infty \sum_{i=1}^k \mathbb{E}\left[|\xi_i|^3\right].$$

$\square$

Proof of Theorem (5.2.3). Define

$$\mu_n = \sum_{i=1}^{k(n)} \mathbb{E}\left[\xi_{n,i}\right]$$
$$\sigma_n^2 = \sum_{i=1}^{k(n)} \operatorname{Var}\left(\xi_{n,i}\right).$$

For every $f \in C^\infty(\mathbb{R})$ with $\|f'''\|_\infty < \infty$,

$$\left| \mathbb{E}\left[f\left(\sum_{i=1}^{k(n)} \xi_{n,i} \right) \right] - \mathbb{E}\left[f\left(N(\mu, \sigma^2) \right) \right] \right| \leq \text{(I)} + \text{(II)},$$

where

$$\text{(I)} = \left| \mathbb{E}\left[f\left(\sum_{i=1}^{k(n)} \xi_{n,i} \right) \right] - \mathbb{E}\left[f\left(N\left(\mu_n, \sigma_n^2\right) \right) \right] \right|,$$
$$\text{(II)} = \left| \mathbb{E}\left[f\left(N\left(\mu_n, \sigma_n^2\right) \right) \right] - \mathbb{E}\left[f\left(N\left(\mu, \sigma^2\right) \right) \right] \right|.$$

By moment-matching, there exist independent $y_{n,1}, \ldots, y_{n,k(n)}$ that are Gaussian and match the first two moments of $\xi_{n,1}, \ldots, \xi_{n,k(n)}$, so[1]

$$\sum_{i=1}^{k(n)} y_{n,i} \sim N\left(\mu_n, \sigma_n^2\right).$$

By assumption (iii) and Lemma (5.2.4), we have

$$\text{(I)} \leq C \left\|f'''\right\|_\infty \sum_{i=1}^{k(n)} \mathbb{E}\left[\left|\xi_{n,i}\right|^3 \right] \to 0.$$

We can write

$$\begin{aligned}
\text{(II)} &= \left| \mathbb{E}\left[f\left(\mu_n + \sigma_n Z\right) \right] - \mathbb{E}\left[f\left(\mu + \sigma Z\right) \right] \right|, \quad Z \sim N(0,1), \\
&\leq L \mathbb{E}\left[\left|\mu_n - \mu\right| + \left|\sigma_n - \sigma\right| \left|Z\right| \right] \\
&= L \left|\mu_n - \mu\right| + C_1 \mathbb{E}\left[|Z| \right] \left|\sigma_n - \sigma\right| \\
&\to 0,
\end{aligned}$$

where the convergence follows from assumptions (i) and (ii). □

[1] Because the sum of normal random variables has mean equal to the sum of the means and variance equal to the sum of the variances. We also use that the sum of independent normal random variables is again normal.

Example 5.2.5.

Returning to our example that motivated the triangular matrix consideration, suppose

$$X_{n,1}, \ldots, X_{n,n} \overset{\text{iid}}{\sim} \text{Bernoulli}(p_n).$$

We want to show that

$$\frac{\sum_{i=1}^{n} (X_{n,i} - p_n)}{\sqrt{np_n(1-p_n)}} \rightsquigarrow N(0,1).$$

Define

$$\xi_{n,i} = \frac{X_{n,i} - p_n}{\sqrt{np_n(1-p_n)}}.$$

We see that the first two conditions are satisfied and the third when $np_n(1-p_n) \to \infty$:

(i) $\sum_{i=1}^{n} \mathbb{E}[\xi_{n,i}] = n \cdot 0 = 0$

(ii) $\sum_{i=1}^{n} \mathrm{Var}(\xi_{n,i}) = 1$

(iii) We see

$$\begin{aligned}
\sum_{i=1}^{n} \mathbb{E}\left[\left|\frac{|X_{n,i} - p_n|}{\sqrt{np_n(1-p_n)}}\right|^3\right] &= n\frac{p_n(1-p_n)^3 + (1-p_n)p_n^3}{(np_n(1-p_n))^{1.5}} \\
&= n\frac{p_n(1-p_n)}{n^{1.5}}\frac{(1-p_n)^2 + p_n^2}{(p_n(1-p_n))^{1.5}}, \\
&\asymp \frac{1}{\sqrt{np_n(1-p_n)}}
\end{aligned}$$

which goes to 0 if $np_n(1-p_n) \to \infty$

Corollary 5.2.6.

Suppose $X_i \overset{\text{iid}}{\sim} \mathbb{P}$ where $\mathbb{P}$ does not depend on n, $\mathbb{E}[X_i] = 0$, and $\mathrm{Var}(X_i) = 1$. Then

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \rightsquigarrow N(0,1).$$

In particular, we do not require a finite third moment.

Proof of Corollary 5.2.6. We proceed by truncation. Put $\xi_{n,i} = \frac{X_i}{\sqrt{n}}\mathbb{1}_{|X_i| \leq \sqrt{n}}$.

Then

$$\begin{aligned}\left|\sum_{i=1}^{n} \mathbb{E}\left[\xi_{n,i}\right]\right| &= \left|\sqrt{n}\mathbb{E}\left[X\mathbb{1}_{|X|\le\sqrt{n}}\right]\right| \\ &= \left|\sqrt{n}\mathbb{E}\left[X\mathbb{1}_{|X|>\sqrt{n}}\right]\right| \\ &\le \mathbb{E}\left[X^2\mathbb{1}_{|X|>\sqrt{n}}\right] \\ &\to 0,\end{aligned}$$

where in the convergence we use a dominating function of X^2.

To check the second condition, consider

$$\begin{aligned}\sum_{i=1}^{n} \operatorname{Var}\left(\xi_{n,i}\right) &= n\operatorname{Var}\left(\frac{X}{\sqrt{n}}\mathbb{1}_{|X|\le\sqrt{n}}\right) \\ &= \operatorname{Var}\left(X\mathbb{1}_{|X|\le\sqrt{n}}\right) \\ &= \mathbb{E}\left[X^2\mathbb{1}_{|X|\le\sqrt{n}}\right] - \left(\mathbb{E}\left[X\mathbb{1}_{|X|\le\sqrt{n}}\right]\right)^2.\end{aligned}$$

We analyze these terms separately. We see from our previous work that

$$\begin{aligned}\mathbb{E}\left[X^2\mathbb{1}_{|X|\le\sqrt{n}}\right] &= \mathbb{E}\left[X^2\right] - \mathbb{E}\left[X^2\mathbb{1}_{|X|>\sqrt{n}}\right] \\ &\to 1 + 0 = 1,\end{aligned}$$

while

$$\begin{aligned}\left|\mathbb{E}\left[X\mathbb{1}_{|X|\le\sqrt{n}}\right]\right| &= \mathbb{E}\left[X\mathbb{1}_{|X|>\sqrt{n}}\right] \\ &\le \mathbb{E}\left[|X|\,\mathbb{1}_{|X|>\sqrt{n}}\right] \to 0\end{aligned}$$

by the DCT with a dominating function of $|X|$. Hence, the second condition is satisfied.

For the third condition, we examine

$$\begin{aligned}\sum_{i=1}^{n} \mathbb{E}\left[\left|\xi_{n,i}\right|^3\right] &= n\mathbb{E}\left[\left(\frac{|X|}{\sqrt{n}}\right)^3\mathbb{1}_{|X|\le\sqrt{n}}\right] \\ &= \mathbb{E}\left[\frac{|X|^3}{\sqrt{n}}\mathbb{1}_{|X|\le\sqrt{n}}\right] \\ &= \mathbb{E}\left[X^2\frac{|X|}{\sqrt{n}}\mathbb{1}_{|X|\le\sqrt{n}}\right] \\ &\le \mathbb{E}\left[X^2\min\left\{\frac{|X|}{\sqrt{n}},1\right\}\right] \to 0\end{aligned}$$

by X^2 being a dominating function.

Therefore, we can invoke the CLT to get $\sum_{i=1}^{n} \xi_{n,i} \rightsquigarrow N(0,1)$.

For any $f \in C^\infty(\mathbb{R})$, we want to show that

$$\left|\mathbb{E}\left[f\left(\sum_{i=1}^n \xi_{n,i}\right)\right] - \mathbb{E}\left[f\left(\sum_{i=1}^n \frac{X_i}{\sqrt{n}}\right)\right]\right| \to 0,$$

as we can use the triangle inequality with our CLT result.

Put

$$S_n = \sum_{i=1}^n \xi_{n,i}, \qquad T_n = \sum_{i=1}^n \frac{X_i}{\sqrt{n}}.$$

Then

$$\begin{aligned}
\left|\mathbb{E}\left[f\left(S_n\right)\right] - \mathbb{E}\left[f\left(T_n\right)\right]\right| &= \left|\mathbb{E}\left[\left(f\left(S_n\right) - f\left(T_n\right)\right)\mathbb{1}_{S_n \neq T_n}\right]\right| \\
&\le 2\left\|f\right\|_\infty \mathbb{P}\left[S_n \neq T_n\right] \\
&\le 2\left\|f\right\|_\infty \sum_{i=1}^n \mathbb{P}\left[\xi_{n,i} \neq \frac{X_i}{\sqrt{n}}\right] \\
&\le 2\left\|f\right\|_\infty \sum_{i=1}^n \mathbb{P}\left[\left|X_i\right| > \sqrt{n}\right] \\
&= 2\left\|f\right\|_\infty \mathbb{E}\left[n\mathbb{1}_{|X|>\sqrt{n}}\right] \\
&\le 2\left\|f\right\|_\infty \mathbb{E}\left[X^2\mathbb{1}_{|X|>\sqrt{n}}\right] \to 0.
\end{aligned}$$

where the convergence follows from X^2 being a dominating function and the indicator giving us pointwise convergence to 0. Consequently,

$$\begin{aligned}
\left|\mathbb{E}\left[f\left(T_n\right)\right] - \mathbb{E}\left[f\left(N(0,1)\right)\right]\right| \le{}& \left|\mathbb{E}\left[f\left(T_n\right)\right] - \mathbb{E}\left[f\left(S_n\right)\right]\right| \\
&+ \left|\mathbb{E}\left[f\left(S_n\right)\right] - \mathbb{E}\left[f\left(N(0,1)\right)\right]\right| \to 0^{\cdot}
\end{aligned}$$

□

Theorem 5.2.7 (Lindeberg Central Limit Theorem).

Suppose we have a triangular array $\{X_{n,i}\}$ such that

(i) $\mathbb{E}\left[X_{n,i}\right] = 0$

(ii) $\sum\limits_{i=1}^{k(n)} \operatorname{Var}\left(X_{n,i}\right) = 1$

(iii) (Lindeberg condition) For every $\epsilon > 0$,

$$L_n(\epsilon) := \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon}\right] \xrightarrow{n\to\infty} 0.$$

Then

$$\sum_{i=1}^{k(n)} X_{n,i} \rightsquigarrow N(0,1).$$

We require a lemma.

Note that the first two conditions are exact equalities, while in the general CLT, we only require convergence. This is what we give up for the weakening of the third condition.

Lemma 5.2.8.

Suppose for every $\epsilon > 0$,

$$H_n(\epsilon) = \frac{L_n(\epsilon)}{\epsilon^2} \to 0.$$

Then there exists a sequence $(\epsilon_n) \to 0$ such that $H_n(\epsilon_n) \to 0$.

Proof of Lemma 5.2.8. For any $k \in \mathbb{N}$, there exists n_k such that if $n \geq n_k$,

$$H_n(\frac{1}{k}) \leq \frac{1}{k}.$$

From this, we can pick an increasing subsequence (n_k).

Putting $\epsilon_n = \frac{1}{k}$ for $n_k \leq n < n_{k+1}$, we get that

$$H_n(\epsilon_n) \leq \frac{1}{k} \to 0.$$

□

We cannot simply put $\epsilon_n = \frac{1}{n}$: the hypothesis says $H_n(\epsilon) \to 0$ for each *fixed* $\epsilon > 0$, so we cannot substitute a sequence that changes with n directly.

Proof of Theorem (5.2.7). Define $H_n(\epsilon) = \frac{L_n(\epsilon)}{\epsilon^2}$. For any fixed $\epsilon > 0$, we see $H_n(\epsilon) \to 0$. By Lemma (5.2.8), there exists a sequence $\epsilon_n \to 0$ such that $H_n(\epsilon_n) \to 0$.

Define

$$\xi_{n,i} = X_{n,i}\mathbb{1}_{|X_{n,i}| \leq \epsilon_n}.$$

Our process is similar to the proof of the previous CLT corollary.

We check the first condition:

$$\begin{aligned}\left|\sum_{i=1}^{k(n)} \mathbb{E}\left[\xi_{n,i}\right]\right| &= \left|\sum_i \mathbb{E}\left[X_{n,i}\mathbb{1}_{|X_{n,i}|>\epsilon_n}\right]\right| \\ &\leq \frac{1}{\epsilon_n}\left|\sum_i \mathbb{E}\left[X_{n,i}^2\mathbb{1}_{|X_{n,i}|>\epsilon_n}\right]\right| \\ &= \epsilon_n H_n(\epsilon_n) \to 0,\end{aligned}$$

since both ϵ_n and $H_n \to 0$.

For the variance condition, note

$$\begin{aligned}
\sum_{i=1}^{k(n)} \operatorname{Var}(\xi_{n,i}) &= \sum_i \mathbb{E}\left[\xi_{n,i}^2\right] - \sum_i \left(\mathbb{E}[\xi_{n,i}]\right)^2 \\
\sum_i \mathbb{E}\left[\xi_{n,i}^2\right] &= \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|\le \epsilon_n}\right] \\
&= \sum_i \mathbb{E}\left[X_{n,i}^2\right] - \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|> \epsilon_n}\right] \\
&= 1 - L_n(\epsilon_n) = 1 - \epsilon_n^2 H_n(\epsilon_n) \to 1 - 0 = 1 \\
\sum_i \mathbb{E}[\xi_{n,i}]^2 &= \sum_i \left(\mathbb{E}\left[X_{n,i} \mathbb{1}_{|X_{n,i}|\le \epsilon_n}\right]\right)^2 \\
&= \sum_i \left(\mathbb{E}\left[X_{n,i} \mathbb{1}_{|X_{n,i}|> \epsilon_n}\right]\right)^2 \text{ by } \mathbb{E}[X_{n,i}] = 0,\ a^2 = (-a)^2 \\
&\le \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|> \epsilon_n}\right] = L_n(\epsilon_n) \to 0.
\end{aligned}$$

Hence the sum of the variances goes to 1.

For the third moment condition, we see

$$\begin{aligned}
\sum_i \mathbb{E}\left[|\xi_{n,i}|^3\right] &= \sum_i \mathbb{E}\left[|X_{n,i}|^3 \mathbb{1}_{|X_{n,i}|\le \epsilon_n}\right] \\
&\le \epsilon_n \sum_i \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|\le \epsilon_n}\right] \\
&= \epsilon_n \left(1 - L_n(\epsilon_n)\right) \le \epsilon_n \to 0,
\end{aligned}$$

so we apply the CLT to get $\sum_i \xi_{n,i} \rightsquigarrow N(0,1)$.

As in the previous proof, we write

$$\begin{aligned}
\left|\mathbb{E}\left[f\left(\sum_i \xi_{n,i}\right)\right] - \mathbb{E}\left[f\left(\sum_i X_{n,i}\right)\right]\right| &\le 2\|f\|_\infty \mathbb{P}\left[\sum_i \xi_{n,i} \ne \sum_i X_{n,i}\right] \\
&\le 2\|f\|_\infty \sum_i \mathbb{P}\left[|X_{n,i}| > \epsilon_n\right] \\
&= 2\|f\|_\infty \mathbb{E}\left[\sum_i \mathbb{1}_{|X_{n,i}|>\epsilon_n}\right] \\
&\le 2\|f\|_\infty \mathbb{E}\left[\sum_i \frac{X_{n,i}^2}{\epsilon_n^2} \mathbb{1}_{|X_{n,i}|>\epsilon_n}\right] \\
&= 2\|f\|_\infty H_n(\epsilon_n) \to 0.
\end{aligned}$$

By the triangle inequality, we see that

$$\left|\mathbb{E}\left[f\left(\sum_i X_{n,i}\right)\right] - \mathbb{E}\left[f\left(N(0,1)\right)\right]\right| \to 0,$$

so

$$\sum_{i=1}^{k(n)} X_{n,i} \rightsquigarrow N(0,1).$$

□

Theorem 5.2.9 (Lindeberg CLT (arbitrary variance)).

Suppose for each n the random variables $X_{n,1}, \ldots, X_{n,k(n)}$ are independent and satisfy $\mathbb{E}\left[X_{n,i}\right] = 0$. Let

$$S_n^2 := \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2\right],$$

and assume $S_n > 0$ for all sufficiently large n. If for every $\epsilon > 0$,

$$\frac{1}{S_n^2} \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}| > \epsilon S_n}\right] \to 0,$$

then

$$\frac{\sum_{i=1}^{k(n)} X_{n,i}}{S_n} \rightsquigarrow N(0,1).$$

Proof of Theorem 5.2.9. We standardize the array by setting

$$Y_{n,i} = \frac{X_{n,i}}{S_n}.$$

Then $\mathbb{E}\left[Y_{n,i}\right] = 0$ and $\sum_i \operatorname{Var}(Y_{n,i}) = 1$. The Lindeberg condition for $\{Y_{n,i}\}$ follows directly from the hypothesis, with cutoff ϵS_n. The conclusion follows from Theorem (5.2.7).

Note that there is no problem with (i) losing independence within the array's rows by dividing by S_n nor (ii) having cutoff ϵS_n instead of ϵ, because the term S_n is a deterministic value not dependent on realized values of $X_{n,i}$. □

Theorem 5.2.10 (Lyapunov CLT).

Suppose $X_{n,1}, \ldots, X_{n,k(n)}$ are independent such that $\mathbb{E}\left[X_{n,i}\right] = 0$ and for

$S_n = \sqrt{\sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2\right]}$, there exists $\delta > 0$ such that

$$\frac{1}{S_n^{2+\delta}} \sum_{i=1}^{k(n)} \mathbb{E}\left[|X_{n,i}|^{2+\delta}\right] \to 0,$$

then

$$\frac{X_{n,1} + \cdots + X_{n,k(n)}}{S_n} \rightsquigarrow N(0,1).$$

This is a rescaled version of Theorem (5.2.9).

Proof of Theorem 5.2.10. For any $\epsilon > 0$, we see that

$$\begin{aligned} X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon S_n} &= X_{n,i}^2 \mathbb{1}_{|X_{n,i}|^\delta > \epsilon^\delta S_n^\delta} \\ &\leq \frac{|X_{n,i}|^{2+\delta}}{\epsilon^\delta S_n^\delta} \\ \frac{1}{S_n^2} \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon S_n}\right] &\leq \frac{1}{\epsilon^\delta} \frac{1}{S_n^{2+\delta}} \sum_{i=1}^{k(n)} \mathbb{E}\left[|X_{n,i}|^{2+\delta}\right] \to 0. \end{aligned}$$

The previous theorem's condition is satisfied. □

Remark 5.2.11. The Lindeberg condition is *sufficient* for the CLT conclusion, but not in general *necessary*: a triangular array can satisfy the CLT without satisfying the Lindeberg condition. However, if the individual variances are uniformly negligible relative to the total variance, the two are equivalent.

Example 5.2.12 (Lindeberg's condition is not necessary).

Consider the triangular array with $k(n) = 1$ and $X_{n,1} \sim N(0,1)$ for all n. Since $X_{n,1}$ is already standard normal,

$$\sum_{i=1}^{k(n)} X_{n,i} = X_{n,1} \sim N(0,1) \rightsquigarrow N(0,1),$$

so the CLT conclusion holds trivially.
However, the Lindeberg condition fails. Since $\mathbb{E}[X_{n,1}] = 0$ and $\operatorname{Var}(X_{n,1}) = 1$, conditions (i) and (ii) of Theorem (5.2.7) are satisfied. But for any fixed $\epsilon > 0$,

$$L_n(\epsilon) = \mathbb{E}\left[X_{n,1}^2 \mathbb{1}_{|X_{n,1}|>\epsilon}\right] = \mathbb{E}\left[Z^2 \mathbb{1}_{|Z|>\epsilon}\right] > 0$$

is a positive constant independent of n, so $L_n(\epsilon) \not\to 0$.
Note also that $\mathbb{E}\left[|X_{n,1}|^3\right] = \mathbb{E}\left[|Z|^3\right]$ is a positive constant, so condition (iii) of the general triangular CLT (Theorem 5.2.3) fails as well; the CLT holds here because the distributions are already Gaussian, not because any of our sufficient conditions are met.
The key reason Lindeberg fails is that the uniform infinitesimality condition $\max_{k=1,\dots,k(n)} \frac{\sigma_k^2}{s_n^2} \to 0$ is violated: with $k(n) = 1$ the only term contributes all of $s_n^2 = 1$ to itself, so $\frac{\sigma_1^2}{s_n^2} = 1 \not\to 0$.

Theorem (5.2.13) was not done in class, nor was Example (5.2.12). After the preceeding remark was made in class, I found answers to the resulting question.

Theorem 5.2.13 (Necessity of Lindeberg's condition under uniform infinitesimality).

Suppose for each n the random variables $X_{n,1}, \dots, X_{n,k(n)}$ are independent with $\mathbb{E}[X_{n,i}] = 0$, and let $\sigma_{n,i}^2 = \mathrm{Var}(X_{n,i})$ and $s_n^2 = \sum_{i=1}^{k(n)} \sigma_{n,i}^2 > 0$. If

$$\max_{i=1,\dots,k(n)} \frac{\sigma_{n,i}^2}{s_n^2} \to 0 \quad \text{as } n \to \infty,$$

then the Lindeberg condition

$$\frac{1}{s_n^2} \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}| > \epsilon s_n}\right] \to 0 \quad \text{for every } \epsilon > 0$$

holds if and only if

$$\frac{\sum_{i=1}^{k(n)} X_{n,i}}{s_n} \rightsquigarrow N(0,1).$$

Proof of Theorem 5.2.13. The sufficiency direction is Theorem (5.2.9). We prove necessity.

By normalizing, set $\tilde{X}_{n,i} = X_{n,i}/s_n$, so $\mathbb{E}\left[\tilde{X}_{n,i}\right] = 0$, $\sum_i \tilde{\sigma}_{n,i}^2 = 1$ (where $\tilde{\sigma}_{n,i}^2 = \sigma_{n,i}^2/s_n^2$), $\max_i \tilde{\sigma}_{n,i}^2 \to 0$, and $Y_n = \sum_i \tilde{X}_{n,i} \rightsquigarrow N(0,1)$.

Denote $\phi_{n,i}(t) = \mathbb{E}\left[e^{it\tilde{X}_{n,i}}\right]$. By the continuity theorem, $\prod_i \phi_{n,i}(t) \to e^{-t^2/2}$ for every t.

Step 1. *Under uniform infinitesimality,* $\sum_i \mathbb{E}\left[1 - \cos(t\tilde{X}_{n,i})\right] \to \frac{t^2}{2}$.

Using $|e^{ix} - 1 - ix| \le x^2/2$, we get $|\phi_{n,i}(t) - 1| \le \frac{t^2}{2}\tilde{\sigma}_{n,i}^2$, so $|\phi_{n,i}(t) - 1| \to 0$ uniformly in i. For large n, each $|\phi_{n,i}(t) - 1| \le \frac{1}{2}$, and using $|\log(1+z) - z| \le |z|^2$

for $|z| \le \frac{1}{2}$:

$$\begin{aligned}\left|\sum_i \log \phi_{n,i}(t) - \sum_i (\phi_{n,i}(t) - 1)\right| &\le \sum_i |\phi_{n,i}(t) - 1|^2 \\ &\le \max_i |\phi_{n,i}(t) - 1| \cdot \sum_i |\phi_{n,i}(t) - 1| \\ &\le \frac{t^2 \max_i \tilde{\sigma}_{n,i}^2}{2} \cdot \frac{t^2}{2} \sum_i \tilde{\sigma}_{n,i}^2 \\ &= \frac{t^4 \max_i \tilde{\sigma}_{n,i}^2}{4} \to 0.\end{aligned}$$

Hence $\sum_i (\phi_{n,i}(t) - 1) \to -t^2/2$.

Since $\mathbb{E}\left[\tilde{X}_{n,i}\right] = 0$, we write $\phi_{n,i}(t) - 1 = \mathbb{E}\left[e^{it\tilde{X}_{n,i}} - 1 - it\tilde{X}_{n,i}\right]$. Then

$$\sum_i (\phi_{n,i}(t) - 1) = \sum_i \mathbb{E}\left[e^{it\tilde{X}_{n,i}} - 1 - it\tilde{X}_{n,i} + \frac{t^2 \tilde{X}_{n,i}^2}{2}\right] - \frac{t^2}{2}.$$

Since $\sum_i (\phi_{n,i}(t) - 1) \to -t^2/2$, taking real parts and using $\mathrm{Re}(e^{ix} - 1 - ix + x^2/2) = \cos(x) - 1 + x^2/2$ yields

$$\sum_i \mathbb{E}\left[1 - \cos(t\tilde{X}_{n,i})\right] = \frac{t^2}{2} + o(1).$$

Step 2. *Conclude $L_n(\epsilon) \to 0$.*

Fix $\epsilon > 0$. Split the expectation at ϵ:

$$\frac{t^2}{2} + o(1) = \sum_i \mathbb{E}\left[(1 - \cos(t\tilde{X}_{n,i}))\mathbb{1}_{|\tilde{X}_{n,i}| \le \epsilon}\right] + \sum_i \mathbb{E}\left[(1 - \cos(t\tilde{X}_{n,i}))\mathbb{1}_{|\tilde{X}_{n,i}| > \epsilon}\right].$$

For the first sum, we use $1 - \cos(u) \le u^2/2$:

$$\sum_i \mathbb{E}\left[(1 - \cos(t\tilde{X}_{n,i}))\mathbb{1}_{|\tilde{X}_{n,i}| \le \epsilon}\right] \le \frac{t^2}{2} \sum_i \mathbb{E}\left[\tilde{X}_{n,i}^2 \mathbb{1}_{|\tilde{X}_{n,i}| \le \epsilon}\right] = \frac{t^2}{2}(1 - L_n(\epsilon)).$$

For the second sum, we use $1 - \cos(u) \le 2$ and Markov's inequality:

$$\sum_i \mathbb{E}\left[(1 - \cos(t\tilde{X}_{n,i}))\mathbb{1}_{|\tilde{X}_{n,i}| > \epsilon}\right] \le 2 \sum_i \mathbb{P}\left(|\tilde{X}_{n,i}| > \epsilon\right) \le \frac{2}{\epsilon^2} \sum_i \mathbb{E}\left[\tilde{X}_{n,i}^2 \mathbb{1}_{|\tilde{X}_{n,i}| > \epsilon}\right] = \frac{2L_n(\epsilon)}{\epsilon^2}.$$

Combining,

$$\frac{t^2}{2} + o(1) \le \frac{t^2}{2}(1 - L_n(\epsilon)) + \frac{2L_n(\epsilon)}{\epsilon^2} = \frac{t^2}{2} - L_n(\epsilon)\left(\frac{t^2}{2} - \frac{2}{\epsilon^2}\right).$$

Rearranging,

$$L_n(\epsilon)\left(\frac{t^2}{2}-\frac{2}{\epsilon^2}\right)\le o(1).$$

Choosing $t > 2/\epsilon$ makes the prefactor $\frac{t^2}{2}-\frac{2}{\epsilon^2}>0$, so $L_n(\epsilon)\to 0$. □

5.3 Stein's Method

This proof technique provides an alternative to characteristic functions and Lindeberg swapping.

Lemma 5.3.1 (Stein's identity).

Suppose $Z \sim N(0,1)$. If f is absolutely continuous such that

$$\mathbb{E}\left[f(Z)\right]<\infty, \quad \mathbb{E}\left[Zf(Z)\right]<\infty,$$

we have

$$\mathbb{E}\left[f'(Z)\right]=\mathbb{E}\left[Zf(Z)\right].$$

This identity is the defining identity of $N(0,1)$.

Proof of Lemma 5.3.1. Let $\phi(t)=\frac{1}{\sqrt{2\pi}}e^{-t^2/2}$ denote the standard normal density, so $\mathbb{E}[g(Z)]=\int_{-\infty}^{\infty} g(t)\phi(t)\,dt$ for any measurable g. We begin with

$$\mathbb{E}\left[f'(Z)\right]=\int_{-\infty}^{\infty} f'(t)\phi(t)\,dt$$

Since $\phi(-\infty)=0$, the FTC gives $\phi(t)=\int_{-\infty}^{t}\phi'(x)\,dx$. Substituting:

$$=\int_{-\infty}^{\infty} f'(t)\int_{-\infty}^{t}\phi'(x)\,dx\,dt$$

Apply Fubini on the region $\{(x,t): x<t\}$:

$$=\int_{-\infty}^{\infty}\phi'(x)\int_{x}^{\infty} f'(t)\,dt\,dx.$$

We would like to apply the FTC to the inner integral: $\int_x^\infty f'(t)\,dt = f(\infty)-f(x)$. However, $f(\infty)$ may not be finite, so this fails.

The fix is to split at $t=0$ and use a *different* representation of ϕ on each piece, chosen so that after Fubini the FTC only runs over a finite interval. For $t<0$, anchor at $-\infty$: $\phi(t)=\int_{-\infty}^{t}\phi'(x)\,dx$. For $t>0$, anchor at $+\infty$: since $\phi(+\infty)=0$, the FTC gives $\phi(t)=-\int_t^\infty \phi'(x)\,dx$. In each case, after Fubini the inner integral only reaches 0.

$$\mathbb{E}\left[f'(Z)\right] = \int_{-\infty}^{0} f'(t)\phi(t)\,dt + \int_{0}^{\infty} f'(t)\phi(t)\,dt$$

Substitute the appropriate representation of $\phi(t)$ on each piece:

$$= \int_{-\infty}^{0} f'(t) \int_{-\infty}^{t} \phi'(x)\,dx\,dt + \int_{0}^{\infty} f'(t) \int_{t}^{\infty} -\phi'(x)\,dx\,dt$$

Apply Fubini to each piece. For the left piece the region $\{-\infty < x < t < 0\}$ becomes $\{-\infty < x < 0,\ x < t < 0\}$; for the right piece the region $\{0 < t < x < \infty\}$ becomes $\{0 < x < \infty,\ 0 < t < x\}$:

$$= \int_{-\infty}^{0} \phi'(x) \int_{x}^{0} f'(t)\,dt\,dx + \int_{0}^{\infty} (-\phi'(x)) \int_{0}^{x} f'(t)\,dt\,dx$$

Apply the FTC (valid since f is absolutely continuous) to the inner integrals:

$$\begin{aligned}
&= \int_{-\infty}^{0} \phi'(x)\left(f(0) - f(x)\right)dx - \int_{0}^{\infty} \phi'(x)\left(f(x) - f(0)\right)dx \\
&= f(0)\left(\int_{-\infty}^{0} \phi'(x)dx + \int_{0}^{\infty} \phi'(x)dx\right) \\
&\quad - \int_{-\infty}^{0} \phi'(x)f(x)dx - \int_{0}^{\infty} \phi'(x)f(x)dx
\end{aligned}$$

Since $\phi'(x) = -x\phi(x)$ and $\mathbb{E}[Z] = 0$, the first term vanishes:

$$\begin{aligned}
&= f(0)\int_{-\infty}^{\infty} -x\phi(x)\,dx + \int_{-\infty}^{\infty} x\phi(x)f(x)\,dx \\
&= \int_{-\infty}^{\infty} x\phi(x)f(x)\,dx \\
&= \mathbb{E}\left[Zf(Z)\right]. \qquad \square
\end{aligned}$$

Aside 5.3.2. Alternatively, we can derive Stein's identity via integration by parts if we don't see the clever representation trick. Recall $\phi'(t) = -t\phi(t)$. Then

$$\begin{aligned}
\mathbb{E}\left[Zf(Z)\right] &= \int tf(t)\phi(t)\,dt \\
&= -\int f(t)\phi'(t)\,dt \quad \text{since } t\phi(t) = -\phi'(t) \\
&= -\left(f(t)\phi(t)\big|_{-\infty}^{\infty} - \int f'(t)\phi(t)\,dt\right) \quad \text{(integration by parts)} \\
&= \mathbb{E}\left[f'(Z)\right],
\end{aligned}$$

where the boundary term vanishes because $\phi(t) \to 0$ exponentially fast and $\mathbb{E}\left[f(Z)\right] < \infty$ ensures $f(t)\phi(t) \to 0$ as $t \to \pm\infty$.

This proof technique generalizes integration by parts: the key step is multiplying by the integrating factor $e^{-w^2/2}$ to convert Stein's ODE into an exact derivative.

Proposition 5.3.3 (Converse of Stein's identity).

Suppose W is a random variable such that for every absolutely continuous f with

$$\mathbb{E}\left[f(W)\right] < \infty, \quad \mathbb{E}\left[|Wf(W)|\right] < \infty$$

and

$$\mathbb{E}\left[f'(W)\right] = \mathbb{E}\left[Wf(W)\right],$$

we have $W \sim N\left(0, 1\right)$.

Proof of Proposition 5.3.3. Consider *Stein's equation*:

$$f'(w) - wf(w) = \mathbb{1}_{w \le t} - \Phi(t).$$

If this differential equation has a solution for each t, then $W \sim N(0,1)$. To see why, note that for $W \sim P$, we can take an expectation of Stein's equation and get

$$\mathbb{E}\left[f'(W) - Wf(W)\right] = P\left[W \le t\right] - \Phi(t),$$

so if $\mathbb{E}\left[f'(W)\right] = \mathbb{E}\left[Wf(W)\right]$ (our hypotheses), we have

$$P\left[W \le t\right] = \Phi(t),$$

so $W \sim N(0,1)$. (Since agreement on the half-intervals $(-\infty, t]$, which generate $\mathscr{B}(\mathbb{R})$, implies agreement on all of $\mathscr{B}(\mathbb{R})$.)

Using the integrating factor $e^{-\frac{w^2}{2}}$, we can indeed solve the differential equation:

$$\begin{aligned} e^{-\frac{w^2}{2}} f'(w) - we^{-\frac{w^2}{2}} f(w) &= e^{-\frac{w^2}{2}} \left(\mathbb{1}_{w \le t} - \Phi(t)\right) \\ \frac{d}{dw}\left(e^{-\frac{w^2}{2}} f(w)\right) &= e^{-\frac{w^2}{2}} \left(\mathbb{1}_{w \le t} - \Phi(t)\right) \\ \implies e^{-\frac{w^2}{2}} f(w) &= \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(\mathbb{1}_{x \le t} - \Phi(t)\right) dx + C \\ f(w) &= e^{\frac{w^2}{2}} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(\mathbb{1}_{x \le t} - \Phi(t)\right) dx + Ce^{\frac{w^2}{2}}. \end{aligned}$$

Taking $C = 0$ gives the standard admissible solution used in Stein's method. $\square$

Remark 5.3.4. In our proof, where did we use that Φ was the Gaussian CDF? Couldn't we have replaced Φ by a different CDF G, then gotten that $W \sim G$?

No. A Gaussian RV is the only one that will satisfy $\mathbb{E}\left[f'(Z)\right] = \mathbb{E}\left[Zf(Z)\right]$

for all admissible f. After setting $C = 0$, we require

$$\begin{aligned}\infty &> \mathbb{E}\left[Zf(Z)\right] \\ &= \int_{-\infty}^{\infty} w \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}} f(w) dw \\ &= \int_{-\infty}^{\infty} w \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(\mathbb{1}_{x \le t} - G(t)\right) dx\, dw.\end{aligned}$$

This necessitates that $\lim\limits_{w \to \infty} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(\mathbb{1}_{x \le t} - G(t)\right) dx = 0$. (Otherwise, we would have linear growth coming from the w term in the outer integral.)

Therefore,

$$G(t) = \frac{\int_{-\infty}^{\infty} \mathbb{1}_{x \le t} e^{-\frac{x^2}{2}} dx}{\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx} = \Phi(t).$$

Aside 5.3.5. The converse implies that we can use discrepancy in Stein's equality as a criterion for convergence to $N(0,1)$: a random variable W is standard normal if and only if $\mathbb{E}\left[f'(W)\right] - \mathbb{E}\left[Wf(W)\right] = 0$ for all admissible f. So to show $W_n = \frac{1}{\sqrt{n}} \sum X_i \rightsquigarrow N(0,1)$, it suffices to show this discrepancy vanishes as $n \to \infty$.

This gives a different strategy from Lindeberg swapping. In Lindeberg swapping, we bounded the direct distance between distributions:

$$\left| \mathbb{E}\left[f\left(\sum_i \frac{X_i}{\sqrt{n}} \right) \right] - \mathbb{E}\left[f\left(N(0,1) \right) \right] \right|.$$

With Stein's method, we instead show W_n satisfies the Gaussian characterization by bounding

$$\left| \mathbb{E}\left[f'(W_n)\right] - \mathbb{E}\left[W_n f(W_n)\right] \right|.$$

For Theorem (5.3.8), we require the following lemma and definition.

Lemma 5.3.6.

We have that $\|f_h\|_\infty$, $\|f_h'\|_\infty$, $\|f_h''\|_\infty \le 2\|h'\|_\infty$.

Proof of Lemma 5.3.6. We just prove that $\|f_h\|_\infty \le 2\,\mathrm{Lip}(h)$; the derivative bounds are standard and follow by differentiating Stein's equation and using the same Gaussian-tail estimates. Recall

$$f_h(w) = e^{\frac{w^2}{2}} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(h(x) - Nh\right) dx.$$

When $w > 0$, the identity

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \left(h(x) - Nh\right) dx = 0$$

lets us rewrite
$$f_h(w) = -e^{\frac{w^2}{2}} \int_w^\infty e^{-\frac{x^2}{2}} (h(x) - Nh)\, dx.$$
Hence
$$\begin{aligned} |f_h(w)| &\le e^{\frac{w^2}{2}} \int_w^\infty e^{-\frac{x^2}{2}} |h(x) - h(0)|\, dx + e^{\frac{w^2}{2}} |h(0) - Nh| \int_w^\infty e^{-\frac{x^2}{2}} dx \\ &\le \mathrm{Lip}(h) e^{\frac{w^2}{2}} \int_w^\infty x e^{-\frac{x^2}{2}} dx + \mathrm{Lip}(h) \mathbb{E}\left[|Z|\right] e^{\frac{w^2}{2}} \int_w^\infty e^{-\frac{x^2}{2}} dx. \end{aligned}$$
Using
$$\int_w^\infty x e^{-\frac{x^2}{2}} dx = e^{-\frac{w^2}{2}}$$
and the standard Gaussian-tail bound
$$e^{\frac{w^2}{2}} \int_w^\infty e^{-\frac{x^2}{2}} dx \le \sqrt{\frac{\pi}{2}},$$
we obtain
$$|f_h(w)| \le \mathrm{Lip}(h) \left(1 + \sqrt{\frac{\pi}{2}} \mathbb{E}\left[|Z|\right]\right) = 2\,\mathrm{Lip}(h).$$
By symmetry, the same bound holds for $w < 0$, hence $\|f_h\|_\infty \le 2\,\mathrm{Lip}(h)$. ☐

Definition 5.3.7 (Wasserstein distance).

Suppose we have two distributions $W \sim P$ and $Z \sim Q$. The Wasserstein distance is defined between distributions (but often notated as the distance between random variables) and is
$$(\mathscr{W}(W, Z) =)\, \mathscr{W}(P, Q) := \sup_{\mathrm{Lip}(h) \le 1} \left|\mathbb{E}\left[h(W)\right] - \mathbb{E}\left[h(Z)\right]\right|.$$
We see that convergence in Wasserstein distance implies weak convergence.

We can prove the CLT with Stein's method instead of Lindeberg swapping. The key step is obtaining an explicit bound on the Wasserstein distance $\mathscr{W}(W_n, N(0,1))$ where $W_n = (X_1 + \cdots + X_n)/\sqrt{n}$; the CLT then follows by a triangular array argument.

Theorem 5.3.8 (Wasserstein distance bound; Wasserstein CLT).

Suppose $X_1, \ldots, X_n$ are independent with $\mathbb{E}\left[X_i\right] = 0$ and $\mathbb{E}\left[X_i^2\right] = 1$. Then
$$\mathscr{W}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, N(0,1)\right) \le \frac{3}{\sqrt{n}} \frac{1}{n} \sum_{i=1}^n \mathbb{E}\left[|X_i|^3\right].$$

In particular, if $\frac{1}{n^{\frac{3}{2}}} \sum_{i=1}^n \mathbb{E}\left[|X_i|^3\right] \to 0$, then

$$\frac{\sum_{i=1}^n X_i}{\sqrt{n}} \rightsquigarrow N(0,1).$$

Proof of Theorem 5.3.8. Consider the more general form of Stein's equation, where we replace the indicator with some function $h(\cdot)$ of w:

$$f'(w) - wf(w) = h(w) - Nh,$$

where $Nh := \mathbb{E}[h(Z)]$ for $Z \sim N(0,1)$.

We can follow the previous derivation to get a solution (dependent on h) as

$$f_h(w) = e^{\frac{w^2}{2}} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} (h(x) - Nh)\, dx.$$

The definition of Wasserstein distance between W and Z is

$$\mathscr{W}(W,Z) = \sup_{\mathrm{Lip}(h) \le 1} \left|\mathbb{E}[h(W)] - \mathbb{E}[h(Z)]\right|.$$

Since Stein's equation $f_h'(w) - wf_h(w) = h(w) - Nh$ holds pointwise for every w, we may take expectations to get

$$\mathbb{E}\left[f_h'(W) - Wf_h(W)\right] = \mathbb{E}[h(W) - Nh] = \mathbb{E}[h(W)] - \mathbb{E}[h(Z)].$$

Therefore

$$\begin{aligned} \mathscr{W}(W,Z) &= \sup_{\mathrm{Lip}(h) \le 1} \left|\mathbb{E}\left[f_h'(W) - Wf_h(W)\right]\right| \\ &\le \sup_{\substack{\|f\|_\infty \le 2 \\ \|f'\|_\infty \le 2 \\ \|f''\|_\infty \le 2}} \left|\mathbb{E}\left[f'(W)\right] - \mathbb{E}[Wf(W)]\right|. \end{aligned}$$

Suppose $X_1, \ldots, X_n$ are independent[2] with $\mathbb{E}[X_i] = 0$, $\mathrm{Var}(X_i) = 1$, and $\mathbb{E}\left[|X_i|^3\right] < \infty$. Define

$$W = \frac{X_1 + \cdots + X_n}{\sqrt{n}}, \quad W_i = W - \frac{X_i}{\sqrt{n}},$$

which gives $X_i \perp\!\!\!\perp W_i$.

[2] Notably, not necessarily identically distributed.

Then

$$\begin{aligned}
\mathbb{E}\left[Wf(W)\right] &= \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\mathbb{E}\left[X_i f(W)\right]\\
\mathbb{E}\left[X_i f(W)\right] &= \mathbb{E}\left[X_i\left(f(W)-f(W_i)\right)\right] + \underbrace{\mathbb{E}\left[X_i f(W_i)\right]}_{=\mathbb{E}[X_i]f(W_i)=0}\\
&= \mathbb{E}\left[X_i\left(f(W)-f(W_i)\right)\right]\\
&= \mathbb{E}\left[X_i\left(f(W)-f(W_i)-f'(W_i)(W-W_i)\right)\right] + \mathbb{E}\left[X_i f'(W_i)(W-W_i)\right]\\
&= \frac{1}{2}\mathbb{E}\left[X_i f''(\xi_i)\left(W-W_i\right)^2\right] + \frac{1}{\sqrt{n}}\mathbb{E}\left[X_i^2 f'(W_i)\right]\\
&= \frac{1}{2n}\mathbb{E}\left[X_i^3 f''(\xi_i)\right] + \frac{1}{\sqrt{n}}\mathbb{E}\left[f'(W_i)\right]
\end{aligned}$$

by independence of X_i and W_i and $(W-W_i)=X_i$.

Put

$$A_n = \frac{1}{2n\sqrt{n}}\sum_{i=1}^{n}\mathbb{E}\left[X_i^3 f''(\xi_i)\right], \qquad B_n = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[f'(W_i)\right].$$

Then

$$\mathbb{E}\left[Wf(W)\right] = A_n + B_n.$$

Putting

$$m_1 = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|\right], \qquad m_3 = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right],$$

we bound A_n and $\mathbb{E}[f'(W)] - B_n$ separately. For A_n, using $\|f''\|_\infty \le 2$:

$$|A_n| \le \frac{\|f''\|_\infty}{2n\sqrt{n}}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right] \le \frac{m_3}{\sqrt{n}}.$$

For $\mathbb{E}[f'(W)] - B_n$, the mean value theorem and $W - W_i = X_i/\sqrt{n}$ give $|f'(W) - f'(W_i)| \le \|f''\|_\infty |W - W_i| = 2\,|X_i|/\sqrt{n}$, so

$$\left|\mathbb{E}\left[f'(W)\right] - B_n\right| \le \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left|f'(W)-f'(W_i)\right| \le \frac{2}{n\sqrt{n}}\sum_{i=1}^{n}\mathbb{E}\left|X_i\right| = \frac{2m_1}{\sqrt{n}}.$$

Combining via the triangle inequality:

$$\left|\mathbb{E}\left[f'(W)\right] - \mathbb{E}\left[Wf(W)\right]\right| \le \frac{1}{\sqrt{n}}\left(2m_1+m_3\right) \le \frac{3m_3}{\sqrt{n}}.$$

Note that in bounding the first moment by the third moment of X_i, we use

$$\mathbb{E}\left[|X_i|\right] \le \left(\mathbb{E}\left[|X_i|^3\right]\right)^{\frac{1}{3}},$$

which follows by Jensen's inequality, since $t \mapsto t^3$ is convex on $[0, \infty)$. By assumption,

$$\mathbb{E}[X_i^2] = 1 \leq \left(\mathbb{E}\left[|X_i|^3\right]\right)^{2/3},$$

so $\mathbb{E}\left[|X_i|^3\right] \geq 1$; plugging this back into the previous equation shows $\mathbb{E}\left[|X_i|\right] \leq \mathbb{E}\left[|X_i|^3\right]$.

Substituting m_3 into the bound on the distance,

$$\mathscr{W}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, N(0,1)\right) \leq \frac{3}{\sqrt{n}} \cdot \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[|X_i|^3\right].$$

This bound holds for every fixed n and is non-asymptotic.

To deduce the CLT, put the problem into triangular-array notation: for each n, let $X_{n,1}, \ldots, X_{n,n}$ be independent with $\mathbb{E}[X_{n,i}] = 0$, $\mathbb{E}[X_{n,i}^2] = 1$, and set $W_n = n^{-1/2} \sum_{i=1}^{n} X_{n,i}$. Note that

$$S_n = \sqrt{\sum_{i=1}^{n} \mathbb{E}\left[X_{n,i}^2\right]} = \sqrt{\sum_{i=1}^{n} 1} = \sqrt{n},$$

so for $\delta = 1$,

$$\frac{1}{S_n^{2+\delta}} \sum_{i=1}^{n} \mathbb{E}\left[|X_{n,i}|^{2+\delta}\right] = \frac{1}{n^{\frac{3}{2}}} \sum_{i=1}^{k} \mathbb{E}\left[|X_{n,i}|^3\right].$$

If this expression goes to 0 as $n \to \infty$, then the Wassterstein bound gives

$$\mathscr{W}\left(W_n, N(0,1)\right) \to 0,$$

[3]and since Wasserstein convergence implies weak convergence, $W_n \xrightarrow{d} N(0,1)$.

In the classical i.i.d. case with $X_1, X_2, \ldots$ i.i.d., $\mathbb{E}[X_i] = 0$, $\mathbb{E}[X_i^2] = 1$, $\mathbb{E}[|X_i|^3] < \infty$: set $X_{n,i} = X_i$ for all n, so $L_n = \mathbb{E}[|X_1|^3]/\sqrt{n} \to 0$, giving $\mathscr{W}(W_n, N(0,1)) \leq 3\mathbb{E}[|X_1|^3]/\sqrt{n} \to 0$. $\square$

Definition 5.3.9 (Kolmogorov-Smirnov distance).

Suppose $W \sim P$ and $Z \sim Q$. Then we define the KS distance as

$$\begin{aligned} \mathrm{KS}(P,Q) &:= \sup_t |P(-\infty, t] - Q(-\infty, t]| \\ &= \sup_t |\mathbb{P}[W \leq t] - \mathbb{P}[Z \leq t]| \end{aligned}.$$

[3]The Lyapunov condition with $\delta = 1$ is satisfied. We only presented this as a Lyapunov condition to relate this way of proving the CLT to a way that we already know. The convergence condition required for the Wasserstein bound to go to 0 is the same as a Lyapunov convergence condition.

Proposition 5.3.10.

Suppose $Z \sim N(0,1)$. Then for every $\alpha > 0$,

$$\mathrm{KS}\,(W,Z) = \sup_t |\mathbb{P}\,[W \le t] - \mathbb{P}\,[Z \le t]| \le \frac{1}{\alpha}\mathscr{W}\,(W,Z) + \alpha.$$

Hence,

$$\mathrm{KS}\,(W,Z) \le 2\sqrt{\mathscr{W}\,(W,Z)}.$$

The KS distance can be bounded by twice the square root of the Wasserstein distance. (Details to follow in next lecture.)

Proof of Proposition 5.3.10. Fix $t \in \mathbb{R}$, and define $h_{t,\alpha}$ by

$$h_{t,\alpha}(w) = \begin{cases} 1, & w \le t \\ 1 - \frac{w-t}{\alpha}, & t < w < t+\alpha \\ 0, & w \ge t+\alpha \end{cases}.$$

Then $\mathrm{Lip}(h_{t,\alpha}) = \frac{1}{\alpha}$ and

$$\mathbb{1}_{w \le t} \le h_{t,\alpha}(w) \le \mathbb{1}_{w \le t+\alpha}.$$

Hence

$$\begin{aligned} \mathbb{P}\,[W \le t] - \mathbb{P}\,[Z \le t] &\le \mathbb{E}\,[h_{t,\alpha}(W)] - \mathbb{E}\,[h_{t,\alpha}(Z)] + \mathbb{P}\,[t < Z \le t+\alpha] \\ &\le \frac{1}{\alpha}\mathscr{W}\,(W,Z) + \alpha. \end{aligned}$$

Since the same bound holds with W and Z interchanged, taking absolute values and then the supremum over t yields the first claim. The second follows by choosing $\alpha = \sqrt{\mathscr{W}\,(W,Z)}$. □

Corollary 5.3.11.

Under the same setting,

$$\mathrm{KS}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, N(0,1)\right) \le 2\sqrt{\frac{3}{\sqrt{n}}\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right]}.$$

The Berry–Esseen bound is uniform across all t. Furthermore, the existence of third moment is necessary for this uniform convergence instead of just weak convergence, which does not require existence of the third moment.

We can get a sharper bound on the KS distance instead of the previous bound, which used Stein's method.

Proposition 5.3.12 (Berry–Esseen).

We have

$$\mathrm{KS}\left(\frac{X_1+\cdots+X_n}{\sqrt{n}}, N(0,1)\right) \leq \frac{C}{\sqrt{n}}\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right].$$

Poisson Limit Theorem 6

We eventually want to show that

$$\mathrm{Bin}\,(n,p) \rightsquigarrow \mathrm{Poi}(\lambda)$$

when $np \to \lambda$.

6.1 Coupling

We introduce the notion of *coupling*. Suppose

$$\begin{aligned} X &: (\Omega_1, \mathscr{F}_1, \mathbb{P}_1) \to (\mathbb{R}, \mathscr{B}(\mathbb{R})) \\ Y &: (\Omega_2, \mathscr{F}_2, \mathbb{P}_2) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}}). \end{aligned}$$

It would be nice if X and Y lived on the same space.

Definition 6.1.1 (Coupling).

The pair $\left(\hat{X}, \hat{Y}\right)$ is a coupling of (X, Y) if $\hat{X}$ and $\hat{Y}$ are on the same space and

$$\hat{X} \overset{\mathrm{d}}{=} X, \quad \hat{Y} \overset{\mathrm{d}}{=} Y.$$

Example 6.1.2.

Suppose $X \sim N(0,1)$ and $Y \sim N(1,1)$. One coupling is

$$\hat{X} = X, \quad \hat{Y} = X + 1.$$

There are many possible couplings for the above example. Are any couplings more notable than others?

Definition 6.1.3 (Maximal coupling).

We say $(\hat{X}, \hat{Y})$ is a maximal coupling of (X, Y) if

$$\mathbb{P}\left[\hat{X} = \hat{Y}\right]$$

is maximized.

Clearly, the previous example has $\mathbb{P}\left[\hat{X} = \hat{Y}\right] = 0$. How do we construct the maximal coupling?

Theorem 6.1.4 (Construction of maximal coupling, discrete RVs).

Suppose X, Y are both discrete and notate

$$\mathbb{P}\left[X = i\right] = p_i, \quad \mathbb{P}\left[Y = i\right] = q_k.$$

The maxmimal coupling satisfies

$$\mathbb{P}\left[\hat{X} \neq \hat{Y}\right] = \sum_i p_i \wedge q_i.$$

Proof of Theorem 6.1.4. Step I (Showing an upper bound). Define $\alpha = \sum_i p_i \wedge q_i$. For any coupling $(\hat{X}, \hat{Y})$ of (X, Y), define

$$A = \left\{i : p_i < q_i\right\}.$$

Then we can write

$$\begin{aligned}
\mathbb{P}\left[\hat{X} = \hat{Y}\right] &= \mathbb{P}\left[\hat{X} = \hat{Y} \in A\right] + \mathbb{P}\left[\hat{X} = \hat{Y} \in A^c\right] \\
&\leq \mathbb{P}\left[\hat{X} \in A\right] + \mathbb{P}\left[\hat{Y} \in A^c\right] \\
&= \mathbb{P}\left[X \in A\right] + \mathbb{P}\left[Y \in A^c\right] \\
&= \sum_{i \in A} p_i + \sum_{i \in A^c} q_i \\
&= \sum_i p_i \wedge q_i = \alpha.
\end{aligned}$$

The same argument applied to continuous RVs if we work with densities; take 383.

The LHS is called total variation affinity/distance.

Step II (Showing the upper bound is achieved). Define

$$\begin{cases} b_i &= \frac{p_i \wedge q_i}{\alpha} \\ c_i &= \frac{p_i - p_i \wedge q_i}{1-\alpha} \\ d_i &= \frac{q_i - p_i \wedge q_i}{1-\alpha} \end{cases}.$$

We see that all three of b_i, c_i, d_i are probabilities because they are nonnegative and individually sum to 1.

The denominators are scaling factors.

Generate independent random variables B, C, D such that

$$\mathbb{P}[B = i] = b_i, \quad \mathbb{P}[C = i] = c_i, \quad \mathbb{P}[D = i] = d_i.$$

Then generate a binary indicator $I \sim \text{Bernoulli}(\alpha)$.

If $I = 1$, then construct $\hat{X} = \hat{Y} = B$. If $I = 0$, then put $\hat{X} = C, \hat{Y} = D$. We have constructed $\hat{X}, \hat{Y}$ on the same space. We need to check that it is a coupling of (X, Y) and that it is indeed maximal.

If $I = 1$, which happens with probability α, $\hat{X}$ and $\hat{Y}$ agree. Otherwise, with probability $1 - \alpha$, they disagree, as if the probability of $C = i$, c_i is nonzero, the probability od $D = i$, d_i is zero.

We can write

$$\begin{aligned} \mathbb{P}\left[\hat{X} = i\right] &= \mathbb{P}\left[\hat{X} = i \mid I = 1\right] \mathbb{P}[I = 1] + \mathbb{P}\left[\hat{X} = i \mid I = 0\right] \mathbb{P}[I = 0] \\ &= \mathbb{P}[B = i] \mathbb{P}[I = 1] + \mathbb{P}[C = i] \mathbb{P}[I = 0] \\ &= \frac{p_i \wedge q_i}{\alpha} \alpha + \frac{p_i - p_i \wedge q_i}{1 - \alpha}(1 - \alpha) \\ &= p_i = \mathbb{P}[X = i]. \end{aligned}$$

When making a maximal coupling, we begin with marginal distributions of X and Y and want to find the joint distribution such that they are as dependent as possible.

Similarly,

$$\mathbb{P}\left[\hat{Y} = i\right] = q_i = \mathbb{P}[Y = i].$$

Therefore,

$$\mathbb{P}\left[\hat{X} = \hat{Y}\right] \geq \mathbb{P}[I = 1] = \alpha,$$

so $(\hat{X}, \hat{Y})$ is maximal. $\square$

6.2 LeCam and Stein–Chen Poisson Limit Theorems

Definition 6.2.1 (Total variation).

For random variables P, Q on the same measure space $(\Omega, \mathscr{F})$, define

$$\text{TV}(P, Q) = \sup_{B \in \mathscr{F}} |P(B) - Q(B)|.$$

Theorem 6.2.2.

Suppose P, Q are discrete random variables on $(\Omega, \mathscr{F})$. The maximial coupling $(\hat{X}, \hat{Y})$ of (P, Q) satisfies

$$\mathbb{P}\left[\hat{X} \neq \hat{Y}\right] = \mathrm{TV}(P, Q).$$

Proof of Theorem 6.2.2. It suffices to show $\mathrm{TV}(P, Q) = 1 - \underbrace{\sum_i p_i \wedge q_i}_{\text{total variation affinity}}$.

Again, let $A = \{i : p_i < q_i\}$. Since $A \in \mathscr{F}$, the definition of total variation gives

$$\mathrm{TV}(P, Q) \geq |P(A) - Q(A)| = Q(A) - P(A).$$

For every B,

$$\begin{aligned}
|P(B) - Q(B)| &= \left| \sum_{i \in B} (p_i - q_i) \right| \\
&= \left| \sum_{i \in B \cap A} \underbrace{(p_i - q_i)}_{\leq 0} + \sum_{i \in B \cap A^c} \underbrace{(p_i - q_i)}_{\geq 0} \right| \\
&= \left| \sum_{i \in B \cap A^c} (p_i - q_i) - \sum_{i \in B \cap A} (q_i - p_i) \right| \\
&\leq \max \left\{ \sum_{i \in B \cap A^c} (p_i - q_i), \sum_{i \in B \cap A} (q_i - p_i) \right\} \\
&\leq \max \left\{ \sum_{i \in A} (p_i - q_i), \sum_{i \in A} q_i - p_i \right\} \\
&= \max \left\{ 1 - P(A) - (1 - Q(A)), Q(A) - P(A) \right\} \\
&= Q(A) - P(A).
\end{aligned}$$

Taking a sup over all $B \in \mathscr{F}$, we have

$$\mathrm{TV}(P, Q) \leq Q(A) - P(A).$$

Combining with the other direction we already showed, we have equality.

Since

$$\begin{aligned}
\mathrm{TV}(P, Q) &= Q(A) - P(A) \\
\sum_i p_i \wedge q_i &= \sum_{i \in A} p_i + \sum_{i \in A^c} q_i \\
&= P(A) + Q(A^c)
\end{aligned}$$

sum to 1, we see that

$$\mathrm{TV}(P,Q)+\sum_i p_i \wedge q_i = 1,$$

and subtract $\sum_i p_i \wedge q_i$ to conclude. □

Theorem 6.2.3 (LeCam Poisson Limit Theorem).

Suppose $X_i \sim \mathrm{Bernoulli}(p_i)$ independently for all $i = 1, \ldots, n$ and define $\lambda = \sum_{i=1}^n p_i$. Then

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poi}(\lambda)\right) \leq \sum_{i=1}^n p_i^2.$$

Like other notions of distance, the total variation between two random variables is really the total variation between their distributions.

Example 6.2.4.

Suppose $p_i = \frac{\lambda}{n}$. Then we have

$$\mathrm{TV}\left(\mathrm{Bin}(n, \frac{\lambda}{n}), \mathrm{Poi}(\lambda)\right) \leq \frac{\lambda^2}{n} \to 0.$$

The strength of this result is that it only relies on independence. We allow p_i to vary, have an explicit bound on the total variation, and have that total variation goes to zero, which implies weak convergence.[a]

[a] Total variation is the distance in measures across all sets, including half intervals. So we have convergence in CDF and thus weak convergence.

Proof of Theorem 6.2.3. Define $Z_i \sim \mathrm{Poi}(p_i)$ independently for all i.

Then

$$\sum_{i=1}^n Z_i \sim \mathrm{Poi}(\sum_{i=1}^n p_i) = \mathrm{Poi}(\lambda).$$

So it suffices to bound the total variation between $\sum_i X_i$ and $\sum_i Z_i$. By Theorem (6.2.2), this is equal to the probability that the maximal coupling variables are not equal.

Let $\left(\hat{X}_i, \hat{Z}_i\right)$ be the maximal coupling of (X_i, Z_i). Then by our theorem,

$$\begin{aligned}
\mathbb{P}\left[\hat{X}_i = \hat{Z}_i\right] &= 1 - TV\left(X_i, Z_i\right) \\
&= \sum_j \mathbb{P}\left[X_i = k\right] \wedge \mathbb{P}\left[Z_i = k\right] \\
\mathbb{P}\left[Z_i = k\right] &= \frac{p_i^k}{k!} e^{-p_i}, \quad \mathbb{P}\left[X_i = k\right] = \begin{cases} p_i, & k = 1 \\ 1 - p_i, & k = 0 \end{cases} \\
\implies \mathbb{P}\left[\hat{X}_i = \hat{Z}_i\right] &= \min\left\{1 - p_i, e^{-p_i}\right\} + \min\left\{p_i, p_i e^{-p_i}\right\} \\
&= 1 - p_i + p_i e^{-p_i} \\
&\geq 1 - p_i + p_i(1 - p_i) \\
&= 1 - p_i^2.
\end{aligned}$$

Note that

$$\left(\sum_{i=1}^n \hat{X}_i, \sum_{i=1}^n \hat{Z}_i\right)$$

is a coupling of

$$\left(\sum_{i=1}^n X_i, \sum_{i=1}^n Z_i\right).$$

It need not be maximal, but it suffices for the first inequality in

$$\begin{aligned}
\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poi}(\lambda)\right) &= \mathrm{TV}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n Z_i\right) \\
&= \mathbb{P}\left[\widehat{\sum_{i=1}^n X_i} \neq \widehat{\sum_{i=1}^n Z_i}\right] \\
&\leq \mathbb{P}\left[\sum_{i=1}^n \hat{X}_i \neq \sum_{i=1}^n \hat{Z}_i\right] \\
&\leq \sum_{i=1}^n \mathbb{P}\left[\hat{X}_i \neq \hat{Z}_i\right] \\
&\leq \sum_{i=1}^n p_i^2.
\end{aligned}$$

$\square$

Lemma 6.2.5 (Stein-Chen).

We have that

$$Z \sim \mathrm{Poi}(\lambda) \iff \forall\ f, \mathbb{E}\left[Z f(Z)\right] = \lambda \mathbb{E}\left[f(Z+1)\right].$$

Proof of Lemma 6.2.5. ($\Longrightarrow$) Since we have for every $k = 0, 1, 2, \ldots$

$$\mathbb{P}[Z = k] = \frac{\lambda^k}{k!} e^{-\lambda},$$

we know

$$\begin{aligned}
\mathbb{E}[Zf(Z)] &= \sum_{k=0}^{\infty} kf(k) \frac{\lambda^k}{k!} e^{-\lambda} \\
&= \sum_{k=1}^{\infty} kf(k) \frac{\lambda^k}{k!} e^{-\lambda} \\
&= \lambda \sum_{k=1}^{\infty} f(k) \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} \\
&= \lambda \sum_{k=0}^{\infty} f(k+1) \frac{\lambda^k}{k!} e^{-\lambda} = \lambda \mathbb{E}[f(Z+1)].
\end{aligned}$$

The forward direction lets f be any function such that the expectations exist. A minimal assumption $|Zf(Z)| < \infty$. The backwards direction requires the identity hold for all bounded $f : \mathbb{N}_0 \to \mathbb{R}$.

($\Longleftarrow$) We use a difference equation instead of the differential equation in Stein's method. Consider for each $k = 0, 1, 2, \ldots$ and the equation

$$\lambda f(k+1) - kf(k) = \mathbb{1}_{k \in A} - \mathbb{P}[Z \in A] \tag{6.1}$$

for some set A. If we can show that this equation has a solution f_A, then we know for a random variable W with distribution satisfying the Stein–Chen identity for all f, we can plug in $f = f_A$ to get

$$\mathbb{P}[W \in A] = \mathbb{P}[Z \in A],$$

so $W \sim \mathrm{Poi}(\lambda)$.

If W satisfies the Stein–Chen identity for every f, then plugging in the solution f_A forces $\mathbb{P}(W \in A) = \mathbb{P}(Z \in A)$ for every set A, hence $W \sim \mathrm{Poi}(\lambda)$.

We would like to get equation (6.1) into a form of $g(k+1) - g(k) = [\cdots]$, since then we can sum the right hand side to get g. Multiplying each side of (6.1) by $\frac{\lambda^k}{k!}$, we have that for $k = 1, 2, \ldots$,

$$\frac{\lambda^{k+1}}{k!} f(k+1) - \frac{\lambda^k}{(k-1)!} f(k) = \frac{\lambda^k}{k!} \left(\mathbb{1}_{k \in A} - \mathbb{P}[Z \in A]\right), \tag{6.2}$$

and for $k = 0$,

$$\lambda f(1) = \mathbb{1}_{0 \in A} - \mathbb{P}[Z \in A].$$

For $g(k) = \frac{\lambda^k}{(k-1)!} f(k)$, we have that

$$\begin{cases} g(k+1) - g(k) = \frac{\lambda^k}{k!} \left(\mathbb{1}_{k \in A} - \mathbb{P}[Z \in A]\right), & k = 1, 2, \ldots \\ g(1) = \mathbb{1}_{0 \in A} - \mathbb{P}[Z \in A] \end{cases},$$

so we sum to get

$$g(k+1) = \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}_{j\in A} - \mathbb{P}\left[Z \in A\right]\right), \quad k = 0, 1, 2, \ldots.$$

Therefore, our solution to the original difference equation is

$$f_A(k+1) = \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}_{j\in A} - \mathbb{P}\left[Z \in A\right]\right), \quad k = 0, 1, 2, \ldots.$$

Note that we don't actually need the solution $f(0)$, since the right hand side of the goal equation does not involve $f(0)$ and the left hand side multiplies it by 0.

Review why the term $\mathbb{1}_{j\in A} - \mathbb{P}\left[Z \in A\right]$ is necessary (the sum must be 0).

Let $Z \sim \mathrm{Poi}(\lambda)$. Suppose that for some random variable W, for any f,

$$\mathbb{E}\left[W f(W)\right] = \lambda \mathbb{E}\left[f(W+1)\right].$$

Plugging in our f_A, we know

$$\begin{aligned} 0 &= \mathbb{E}\left[\lambda f_A(W+1) - W f_A(W)\right] \\ &= \mathbb{E}\left[\mathbb{1}_{W\in A} - \mathbb{P}\left[Z \in A\right]\right] \\ &= \mathbb{P}\left[W \in A\right] - \mathbb{P}\left[Z \in A\right]. \end{aligned}$$

Since A is arbitrary, we know $W \sim \mathrm{Poi}(\lambda)$. □

Lemma 6.2.6.

The solution f_A to the Stein-Chen equation is

$$f_A(k+1) = \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}_{j\in A} - \mathbb{P}\left[Z \in A\right]\right),$$

and it satisfies, for integers k, l,

$$\left|f_A(k) - f_A(l)\right| \le \min\left\{1, \frac{1}{\lambda}\right\} |k - l|.$$

Proof of Lemma 6.2.6. The formula for f_A was established in the proof of the ($\Longleftarrow$) direction of the Stein–Chen lemma above. It remains to prove the Lipschitz bound.

Write $f_A(k+1) = \sum_{i\in A} f_i(k+1)$, where $f_i := f_{\{i\}}$. Substituting $A = \{i\}$ into the formula and using the Poisson probability $\mathbb{P}[Z = j] = \frac{\lambda^j}{j!}e^{-\lambda}$,

$$\begin{aligned} f_i(k+1) &= \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}_{j=i} - \mathbb{P}[Z = i]\right) \\ &= \frac{k!}{\lambda^{k+1}} \cdot \frac{\lambda^i}{i!} \left(\mathbb{1}_{i \le k} - \mathbb{P}[Z \le k]\right) \\ &= \frac{\mathbb{P}[Z = i]}{\lambda \mathbb{P}[Z = k]} \left(\mathbb{1}_{i \le k} - \mathbb{P}[Z \le k]\right), \end{aligned}$$

where the second line uses $\sum_{j=0}^{k} \frac{\lambda^j}{j!} \mathbb{1}_{j=i} = \frac{\lambda^i}{i!} \mathbb{1}_{i \le k}$ and

$$\mathbb{P}[Z = i] \sum_{j=0}^{k} \frac{\lambda^j}{j!} = \frac{\lambda^i}{i!} e^{-\lambda} \cdot e^{\lambda} \mathbb{P}[Z \le k] = \frac{\lambda^i}{i!} \mathbb{P}[Z \le k],$$

and the third uses

$$\frac{k!}{\lambda^{k+1}} \cdot \frac{\lambda^i}{i!} = \frac{e^{-\lambda}}{\lambda \mathbb{P}[Z = k]} \cdot \frac{\mathbb{P}[Z = i]}{e^{-\lambda}} = \frac{\mathbb{P}[Z = i]}{\lambda \mathbb{P}[Z = k]}.$$

We claim:

(i) $f_k(k+1) \ge f_k(k)$,

(ii) For $i \ne k$, $f_i(k+1) \le f_i(k)$.

With these, we could bound $f_A(k+1) - f_A(k)$ from above: adding k to A increases the sum by (i), and each remaining $i \ne k$ term is non-positive by (ii), so

$$\begin{aligned} f_A(k+1) - f_A(k) &\le f_{A \cup \{k\}}(k+1) - f_{A \cup \{k\}}(k) \\ &\le f_k(k+1) - f_k(k). \end{aligned}$$

We evaluate using $\frac{\mathbb{P}[Z=k]}{\mathbb{P}[Z=k-1]} = \frac{\lambda}{k}$:

$$\begin{aligned} f_k(k+1) &= \frac{\mathbb{P}[Z > k]}{\lambda}, \\ f_k(k) &= -\frac{\mathbb{P}[Z = k]\mathbb{P}[Z \le k-1]}{\lambda \mathbb{P}[Z = k-1]} = -\frac{\mathbb{P}[Z \le k-1]}{k}, \end{aligned}$$

so

$$f_k(k+1) - f_k(k) = \frac{\mathbb{P}[Z > k]}{\lambda} + \frac{\mathbb{P}[Z \le k-1]}{k}.$$

For each $0 \le j \le k-1$, the Poisson recurrence gives $\mathbb{P}[Z = j] = \frac{j+1}{\lambda}\mathbb{P}[Z = j+1]$, so $\frac{\mathbb{P}[Z=j]}{k} = \frac{j+1}{k} \cdot \frac{\mathbb{P}[Z=j+1]}{\lambda} \le \frac{\mathbb{P}[Z=j+1]}{\lambda}$ since $j+1 \le k$. Summing,

$$\frac{\mathbb{P}[Z \le k-1]}{k} \le \frac{1}{\lambda} \sum_{j=0}^{k-1} \mathbb{P}[Z = j+1] = \frac{\mathbb{P}[1 \le Z \le k]}{\lambda}.$$

Therefore,

$$f_k(k+1) - f_k(k) \le \frac{\mathbb{P}[Z > k] + \mathbb{P}[1 \le Z \le k]}{\lambda} = \frac{\mathbb{P}[Z \ge 1]}{\lambda} = \frac{1 - e^{-\lambda}}{\lambda}.$$

Since $1 - e^{-\lambda} \le 1$ and $e^{-\lambda} \ge 1 - \lambda$ gives $1 - e^{-\lambda} \le \lambda$,

$$f_k(k+1) - f_k(k) \le \frac{1 - e^{-\lambda}}{\lambda} \le \min\left\{1, \frac{1}{\lambda}\right\}.$$

For the other direction, note that $\mathbb{P}[Z \in \mathbb{N}_0] = 1$, so applying the formula gives $f_{\mathbb{N}_0} \equiv 0$ and thus $f_{A^c} = -f_A$. Thus

$$f_A(k) - f_A(k+1) = f_{A^c}(k+1) - f_{A^c}(k) \le \min\left\{1, \frac{1}{\lambda}\right\},$$

by the same argument applied to A^c. Combining both directions,

$$\left|f_A(k+1) - f_A(k)\right| \le \min\left\{1, \frac{1}{\lambda}\right\},$$

and the triangle inequality extends this to $|f_A(k) - f_A(l)| \le \min\{1, \frac{1}{\lambda}\}|k - l|$.

It remains to prove (i) and (ii). From the formulas above, $f_k(k+1) = \frac{\mathbb{P}[Z>k]}{\lambda} \ge 0$ and $f_k(k) = -\frac{\mathbb{P}[Z\le k-1]}{k} \le 0$, so $f_k(k+1) \ge 0 \ge f_k(k)$, proving (i).

For (ii), when $i < k$ both numerators involve the tail:

$$f_i(k+1) = \frac{\mathbb{P}[Z = i]\mathbb{P}[Z > k]}{\lambda\mathbb{P}[Z = k]}, \qquad f_i(k) = \frac{\mathbb{P}[Z = i]\mathbb{P}[Z > k-1]}{\lambda\mathbb{P}[Z = k-1]},$$

so $f_i(k+1) \le f_i(k)$ iff $\frac{\mathbb{P}[Z>k]}{\mathbb{P}[Z=k]} \le \frac{\mathbb{P}[Z>k-1]}{\mathbb{P}[Z=k-1]}$. Writing

$$\frac{\mathbb{P}[Z > k]}{\mathbb{P}[Z = k]} = \sum_{l>0} \lambda^l \frac{k!}{(l+k)!},$$

each term $\frac{k!}{(l+k)!} = \frac{1}{(k+1)\cdots(k+l)}$ is decreasing in k, so the sum is decreasing in k, which gives $f_i(k+1) \le f_i(k)$.

When $i > k$ both numerators involve the CDF:

$$f_i(k+1) = -\frac{\mathbb{P}[Z = i]\mathbb{P}[Z \le k]}{\lambda\mathbb{P}[Z = k]}, \qquad f_i(k) = -\frac{\mathbb{P}[Z = i]\mathbb{P}[Z \le k-1]}{\lambda\mathbb{P}[Z = k-1]},$$

so $f_i(k+1) \le f_i(k)$ iff $\frac{\mathbb{P}[Z\le k]}{\mathbb{P}[Z=k]} \ge \frac{\mathbb{P}[Z\le k-1]}{\mathbb{P}[Z=k-1]}$. Writing

$$\frac{\mathbb{P}[Z \le k]}{\mathbb{P}[Z = k]} = \sum_{l=0}^{k} \lambda^{-l} \frac{k!}{(k-l)!},$$

both the number of terms and each factor $\frac{k!}{(k-l)!} = k(k-1)\cdots(k-l+1)$ increase in k, so the ratio is increasing in k, giving $f_i(k+1) \leq f_i(k)$. □

Proposition 6.2.7.

For $Z \sim \mathrm{Poi}(\lambda)$ and a random variable W, the total variation is

$$\begin{aligned}\mathrm{TV}\left(W, Z\right) &= \sup_A \left|\mathbb{P}\left[W \in A\right] - \mathbb{P}\left[Z \in A\right]\right| \\ &= \sup_A \left|\mathbb{E}\left[W f_A(W)\right] - \lambda \mathbb{E}\left[f_A(W+1)\right]\right| \\ &\leq \sup_{\mathrm{Lip}(f) \leq \min\left\{1, \frac{1}{\lambda}\right\}} \left|\mathbb{E}\left[W f(W) - \lambda \mathbb{E}\left[f(W+1)\right]\right]\right|.\end{aligned}$$

Proof of Proposition 6.2.7. The first equality is the definition of total variation, the second comes from the Stein–Chen identity, and the third from the Lipschitz bound we just derived. □

Theorem 6.2.8 (Stein-Chen).

Suppose we have independent $X_i \sim \mathrm{Bernoulli}(p_i)$ for $i = 1, \ldots, n$. Put $\lambda = \sum_{i=1}^n p_i$. Then we have

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poi}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^n p_i^2.$$

We have a stronger result than LeCam's result because of the min term. When $\lambda \to \infty$ and $\sum_{i=1}^n p_i^2$ is controlled, we get that $\sum_{i=1}^n X_i \rightsquigarrow \mathrm{Poi}(\lambda)$.

Proof of Theorem 6.2.8. Let $W = X_1 + \cdots + X_n$, with $X_i \sim \mathrm{Bernoulli}(p_i)$ independent for $i = 1, \ldots, n$. Write $W_i = W - X_i$ and put $\lambda = \sum_{i=1}^n p_i$.

Step I (Decomposing the discrepancy). We decompose the Stein–Chen discrepancy as

$$\lambda \mathbb{E}\left[f(W+1)\right] - \mathbb{E}\left[W f(W)\right] = \sum_{i=1}^n p_i \mathbb{E}\left[f(W+1)\right] - \sum_{i=1}^n \mathbb{E}\left[X_i f(W_i + X_i)\right].$$

Since $X_i \in \{0, 1\}$ and $X_i \perp\!\!\!\perp W_i$, conditioning on W_i yields

$$\begin{aligned}\mathbb{E}\left[X_i f(W_i + X_i)\right] &= \mathbb{E}\left[\mathbb{E}\left[X_i f(W_i + X_i) \mid W_i\right]\right] \\ &= \mathbb{E}\left[p_i \cdot f(W_i + 1) + (1 - p_i) \cdot 0\right] \\ &= p_i \mathbb{E}\left[f(W_i + 1)\right],\end{aligned}$$

where the $X_i = 0$ term vanishes because $X_i \cdot f(W_i + X_i) = 0 \cdot f(W_i) = 0$ when

$X_i = 0$. Substituting back,

$$\lambda\mathbb{E}\left[f(W+1)\right] - \mathbb{E}\left[Wf(W)\right] = \sum_{i=1}^{n} p_i \left(\mathbb{E}\left[f(W+1)\right] - \mathbb{E}\left[f(W_i+1)\right]\right).$$

Step II (Applying the Lipschitz bound). Since $W = W_i + X_i$ and $X_i \in \{0,1\}$, we have $|W - W_i| = X_i$. The Lipschitz bound on f_A from the lemma above then gives $|f(W+1) - f(W_i+1)| \le \min\left\{1, \frac{1}{\lambda}\right\} X_i$, so

$$\begin{aligned}
|\lambda\mathbb{E}\left[f(W+1)\right] - \mathbb{E}\left[Wf(W)\right]| &\le \sum_{i=1}^{n} p_i \mathbb{E}\left[|f(W+1) - f(W_i+1)|\right] \\
&\le \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i \mathbb{E}\left[X_i\right] \\
&= \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i^2.
\end{aligned}$$

The total variation bound then follows from the proposition above. $\square$

Example 6.2.9.

Consider the typical example $p_i = \frac{\lambda}{n}$. Then we get the Law of Small Numbers.

Proposition 6.2.10 (TV convergence implies weak convergence).

If $\mu_n \to \mu$ in total variation, i.e. $\mathrm{TV}(\mu_n, \mu) \to 0$, then for every bounded measurable function f,

$$\int f\, d\mu_n \to \int f\, d\mu.$$

In particular, $\mu_n \rightsquigarrow \mu$.

Proof of Proposition 6.2.10. For any set $A \in \mathscr{F}$,

$$\left|\int \mathbf{1}_A\, d\mu_n - \int \mathbf{1}_A\, d\mu\right| = |\mu_n(A) - \mu(A)| \le \mathrm{TV}(\mu_n, \mu) \to 0,$$

so the result holds for indicators. By linearity it extends to simple functions. For general bounded measurable f, given $\varepsilon > 0$ choose a simple function s with $\|f - s\|_\infty < \varepsilon$. Then

$$\begin{aligned}
\left|\int f\, d\mu_n - \int f\, d\mu\right| &\le \left|\int (f-s)\, d\mu_n\right| + \left|\int s\, d\mu_n - \int s\, d\mu\right| + \left|\int (s-f)\, d\mu\right| \\
&\le 2\varepsilon + \left|\int s\, d\mu_n - \int s\, d\mu\right|.
\end{aligned}$$

The second term tends to 0 as $n \to \infty$. Since ε was arbitrary, $\int f\, d\mu_n \to \int f\, d\mu$. Weak convergence follows because bounded continuous functions are bounded and measurable. $\square$

Example 6.2.11.

The boundedness assumption cannot be dropped. Let μ_n assign mass $\frac{1}{n}$ to $\{n\}$ and $1 - \frac{1}{n}$ to $\{0\}$, and let $\mu(x) = \mathbb{1}_0$. The only points with positive probability are 0 and n, so $\mathrm{TV}(\mu_n, \mu) = \frac{1}{n} \to 0$. However, taking $f(x) = x$ shows

$$\int f\, d\mu_n = n \cdot \frac{1}{n} + 0 \cdot \left(1 - \frac{1}{n}\right) = 1 \neq 0 = \int f\, d\mu$$

for every n, so $\int f\, d\mu_n \not\to \int f\, d\mu$.

6.3 Dependence

What if $X_1, \ldots, X_n$ are dependent?

Definition 6.3.1 (Dependency neighborhood).

For each $i \in \mathbb{N}$, consider $\{N_i\}$ where

$$i \in N_i \subseteq [n] := \{1, 2, \ldots, n\}$$

such that

$$X_i \perp\!\!\!\perp \{X_j\}_{j \notin N_i}.$$

We call N_i the dependency neighborhood of X_i.

Theorem 6.3.2 (Stein-Chen, dependent variables).

Suppose $X_i \sim \mathrm{Bernoulli}(p)$ for $i = 1, \ldots, n$, where the X_is are not necessarily independent. Put $\lambda = \sum_{i=1}^n p_i$. Then for dependency neighborhoods N_i, we have the total variation bound

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poi}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\} \left(\sum_{i=1}^n \sum_{j \in N_i \setminus \{i\}} \mathbb{E}\left[X_i X_j\right] + \sum_{i=1}^n \sum_{j \in N_i} p_i p_j\right).$$

Proof of Theorem 6.3.2. Fix a Stein solution f with

$$|f(k+1) - f(k)| \leq \min\left\{1, \frac{1}{\lambda}\right\}.$$

For each i, write

$$\begin{aligned}W &= X_i + \sum_{j \in N_i \setminus \{i\}} X_j + \sum_{j \notin N_i} X_j \\ &=: X_i + \sum_{j \in N_i \setminus \{i\}} X_j + V_i.\end{aligned}$$

By assumption, $X_i \perp\!\!\!\perp V_i$.

Now decompose

$$\begin{aligned}\mathbb{E}\left[W f(W)\right] - \lambda \mathbb{E}\left[f(W+1)\right] &= \sum_{i=1}^{n} \left(\mathbb{E}\left[X_i f(W)\right] - p_i \mathbb{E}\left[f(W+1)\right]\right) \\ &= \text{(I)} + \text{(II)},\end{aligned}$$

where

$$\begin{aligned}\text{(I)} &= \sum_{i=1}^{n} \left(\mathbb{E}\left[X_i f(W)\right] - \mathbb{E}\left[X_i f(V_i+1)\right]\right), \\ \text{(II)} &= \sum_{i=1}^{n} \left(\mathbb{E}\left[X_i f(V_i+1)\right] - p_i \mathbb{E}\left[f(W+1)\right]\right).\end{aligned}$$

Since $X_i \in \{0,1\}$,

$$\mathbb{E}\left[X_i f(V_i + X_i)\right] = \mathbb{E}\left[X_i f(V_i+1)\right] = p_i \mathbb{E}\left[f(V_i+1)\right],$$

simplifying (II).

Writing $f(W) - f(V_i+1) \le |f(W) - f(V_i+1)|$,

$$\begin{aligned}|\text{(I)}| &\le \sum_{i=1}^{n} \mathbb{E}\left[X_i \left|f(W) - f(V_i+1)\right|\right] \\ &\le \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \mathbb{E}\left[X_i \sum_{j \in N_i \setminus \{i\}} X_j\right] \\ &= \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \sum_{j \in N_i \setminus \{i\}} \mathbb{E}\left[X_i X_j\right],\end{aligned}$$

where the second inequality follows from the Lipschitz bound on f. Also,

$$\begin{aligned}
|(\mathrm{II})| &\leq \sum_{i=1}^{n} p_i \left|\mathbb{E}\left[f(V_i+1)\right] - \mathbb{E}\left[f(W+1)\right]\right| \\
&\leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i \mathbb{E}\left[|W - V_i|\right] \\
&= \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i \mathbb{E}\left[\sum_{j \in N_i} X_j\right] \\
&= \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \sum_{j \in N_i} p_i p_j.
\end{aligned}$$

This gives the stated bound. $\square$

Example 6.3.3.

Consider the undirected Erdös–Rényi graph with n nodes and edges $A_{ij} \in \{0, 1\}$ with $A_{ij} = A_{ji} \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$.
We can count the number of edges with the CLT (for large n) or the independent version of the PLT (small n).
More interestingly, we can count the number of triangles:

$$T := \sum_{1 \leq i < j < k \leq n} A_{ij} A_{jk} A_{ki}.$$

We can use the dependency neighborhood condition. To make indexing easier, consider the dependency neighborhoods N_α, so $\alpha \in N_\alpha$ and $X_\alpha \perp\!\!\!\perp \{X_\beta\}_{\beta \notin N_\alpha}$.
The dependency version of Stein-Chen PLT then says for $X_\alpha \sim \text{Bernoulli}(p_\alpha)$ and $\lambda = \sum_\alpha p_\alpha$, we have

$$\begin{aligned}
\mathrm{TV}\left(\sum_\alpha X_\alpha, \mathrm{Poi}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\} &\sum_\alpha \sum_{\beta \in N_\alpha \setminus \{\alpha\}} \mathbb{E}\left[X_\alpha X_\beta\right] \\
+ \min\left\{1, \frac{1}{\lambda}\right\} &\sum_\alpha \sum_{\beta \in N_\alpha} p_\alpha p_\beta
\end{aligned}.$$

Write $\alpha = (i, j, k)$, so $X_\alpha = A_{ij} A_{jk} A_{ki}$. Since a triangle with a shared node shares exactly one edge, and each of the three edges of a triangle can be shared with triangles constructed by connecting the two nodes of the edge with $n - 3$ different nodes, we see that when N_α are as small as possible, $|N_\alpha| = 3(n-3) + 1$. (The dependency neighborhood of a triangle includes itself.)

We see

$$\begin{aligned}
\text{(II)} &= \sum_{\alpha} \sum_{\beta \in N_\alpha} p_\alpha p_\beta \\
&= O\left(n^3 \cdot n \cdot p^6\right) \\
&= O\left(n^4 p^6\right) \\
\text{(I)} &= \sum_{\alpha} \sum_{\beta \in N_\alpha \setminus \{\alpha\}} \mathbb{E}\left[X_\alpha X_\beta\right] \\
&= \sum_{\alpha} \sum_{\beta \in N_\alpha \setminus \{\alpha\}} \mathbb{E}\left[A_{ab} A_{bc} A_{cd} A_{de} A_{ea}\right] \\
&= O\left(n^4 p^5\right) \\
\lambda &= \binom{n}{3} p^3 = O\left(n^4 p^5, np^2\right).
\end{aligned}$$

Therefore,

$$\mathrm{TV}\left(T, \mathrm{Poi}\left(\binom{n}{3} p^3\right)\right) = O\left(\min\left\{n \cdot n^3 p^3, np^2\right\}\right),$$

which tends to 0 when $p \ll \frac{1}{\sqrt{n}}$.

Remark 6.3.4. We can prove this dependent PLT using the martingale CLT instead of Stein's method.

Conditional Expectation 7

We will talk about conditional expectation from the perspective of the Radom–Nikodym derivative.

7.1 Signed Measure and the Radom–Nikodym Derivative

Definition 7.1.1 (Signed measure).

Suppose we have a measurable space $(\Omega, \mathscr{F})$. ϕ is a signed measure on $(\Omega, \mathscr{F})$ if:

(i) $\phi(\varnothing) = 0$

(ii) For disjoint countable $\{A_n\} \subseteq \mathscr{F}$,

$$\phi\left(\bigcup_n A_n\right) = \sum_n \phi(A_n).$$

(Implicitly, the sum is absolutely convergent.)

The only difference between a measure and signed measure is that a signed measure can take any value, not just nonnegative ones.

With nonnegative measures, there was no trouble with sets having measure infinity. But there is a potential collision with signed measure if two sets have measures infinity and negative infinity.

However, the following proposition rules this out.

Proposition 7.1.2.

Either

$$-\infty < \phi(A) \leq \infty, \quad \forall\, A \in \mathscr{F}$$

or

$$-\infty \le \phi(A) < \infty, \quad \forall\, A \in \mathscr{F}.$$

Proof of Proposition 7.1.2. Suppose not, so there are $A, B \in \mathscr{F}$ such that

$$\phi(A) = \infty, \quad \phi(B) = -\infty.$$

By countable additivity, we can write

$$\begin{aligned} \phi(A \cup B) &= \phi(A) + \phi(B \setminus A) = \infty \\ \phi(A \cup B) &= \phi(B) + \phi(A \setminus B) = -\infty. \end{aligned}$$

The definition of a signed measure imposes that the measure of any set in $\mathscr{F}$ is well-defined.

□

Theorem 7.1.3 (Hahn decomposition).

Given a signed measure ϕ, there exist $\Omega^+, \Omega^- \in \mathscr{F}$ such that they partition Ω:

$$\Omega^+ \cup \Omega^- = \Omega, \quad \Omega^+ \cap \Omega^- = \varnothing$$

and for any $A \in \mathscr{F}$,

$$\phi(A \cap \Omega^-) \le 0 \le \phi(A \cap \Omega^+).$$

Furthermore, this partition is unique in the sense that if we have Hahn decompositions $\left(\Omega_1^+, \Omega_1^-\right)$ and $\left(\Omega_2^+, \Omega_1^-\right)$,

$$\begin{aligned} A \subseteq \Omega_1^+ \Delta \Omega_2^+ &\implies \phi(A) = 0, \quad \forall\, A \in \mathscr{F} \\ A \subseteq \Omega_1^- \Delta \Omega_2^- &\implies \phi(A) = 0. \end{aligned}$$

Proposition 7.1.4 (Jordan decomposition).

From a signed measure ϕ, we can recover measures ϕ^+, ϕ^- by putting

$$\begin{aligned} \phi^+(A) &= \phi(A \cap \Omega^+) \\ \phi^-(A) &= \phi(A \cap \Omega^-). \end{aligned}$$

Note that Jordan decomposition is unique; starting with different Hahn decompositions yields the same Jordan decomposition. There is a remark if the Jordan decomposition $\phi = \phi^+ - \phi^-$ imposes mutual singularity.

Proof of Proposition 7.1.4. Immediate (or maybe not). □

Example 7.1.5.

Given a measure space $(\Omega, \mathscr{F}, \mu)$, where $f : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, we have that

$$\phi(A) := \int_A f d\mu$$

is a signed measure with the Hahn decomposition

$$\Omega^+ = \{f \geq 0\}, \quad \Omega^- = \{f < 0\}.$$

Note if f is nonnegative, ϕ is a measure.
The Jordan decomposition is

$$\phi^{\pm}(A) = \int_A f^{\pm} d\mu.$$

Definition 7.1.6 (Absolute continuity).

Given a signed measure ϕ and measure μ on $(\Omega, \mathscr{F})$, $\phi \ll \mu$[a] if for any $A \in \mathscr{F}$ such that $\mu(A) = 0$, $\phi(A) = 0$.

[a]Read as ϕ is absolutely continuous with respect to μ.

Remark 7.1.7. The signed measure ϕ as an integral of f over μ is absolutely continuous with respect to μ.

Theorem 7.1.8 (Radom–Nikodym).

Suppose ϕ is a signed measure and μ is a σ-finite measure on $(\Omega, \mathscr{F})$. If $\phi \ll \mu$, then there exists a function f (unique up to a measure zero set on $(\Omega, \mathscr{F}, \mu)$) such that

$$\phi(A) = \int_A f d\mu$$

for every $A \in \mathscr{F}$.
f is called the Random–Nikodym derivative (or the derivative between the two measures) and is also notated

$$f = \frac{d\phi}{d\mu}.$$

Example 7.1.9.

Consider $(\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ and $\mathbb{P} = N(0,1)$, with λ as Lebesgue measure. Then $\frac{d\mathbb{P}}{d\lambda}$ is the density of a standard Gaussian distribution.
For $\left(\mathbb{N}, 2^{\mathbb{N}}\right)$ with $\mathbb{P}$ as the Poisson distribution and λ the counting measure, $\frac{d\mathbb{P}}{d\lambda}$ is the probability mass function of the Poisson distribution.

7.2 Motivation of Conditional Expectation

For random variables X, Y, what should $\mathbb{E}[X \mid Y]$ be? In the simplest case, where $X = \mathbb{1}_A$ and $Y = \mathbb{1}_B$, we want

$$\mathbb{E}[X \mid Y] = \mathbb{P}[A \mid \mathbb{1}_B].$$

If $\mathbb{1}_B = 1$, then $\mathbb{P}[A \mid \mathbb{1}_B] = \mathbb{P}[A \mid B]$ and if $\mathbb{1}_B = 0$, then $\mathbb{P}[A \mid \mathbb{1}_B] = \mathbb{P}[A \mid B^c]$. So,

$$\mathbb{E}[X \mid Y] = \mathbb{P}[A \mid B]\,\mathbb{1}_B + \mathbb{P}[A \mid B^c]\,\mathbb{1}_{B^c}, \tag{7.1}$$

and the conditional expectation is measurable with respect to $\mathscr{G} = \{\varnothing, \Omega, B, B^c\}$.

What if $Z = a\mathbb{1}_B + b\mathbb{1}_{B^c}$ with $a \neq b$? Then

$$\mathbb{E}[X \mid Z] = \mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \mathscr{G}],$$

as $\sigma(Z) = \{\varnothing, \Omega, B, B^c\} = \sigma(Y)$.

For any $D \in \mathscr{G} = \{\varnothing, \Omega, B, B^c\}$, we have that by equation (7.1),

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[X \mid Y]\,\mathbb{1}_D] &= \mathbb{P}[A \mid B]\,\mathbb{P}[B \cap D] + \mathbb{P}[A \mid B^c]\,\mathbb{P}[B^c \cap D] \\
&= \begin{cases} 0, \quad D & = \varnothing \\ \mathbb{P}[A], \quad D & = \Omega \\ \mathbb{P}[A \cap B], \quad D & = B \\ \mathbb{P}[A \cap B^c], \quad D & = B^c \end{cases} \\
&= \mathbb{E}[X\mathbb{1}_D].
\end{aligned}$$

We have shown that the tower property holds *in this example.*

Proposition 7.2.1 (Tower Property).

For $D \in \mathscr{G}$ (D is measurable with respect to $\mathscr{G}$),

$$\mathbb{E}[\mathbb{E}[X \mid \mathscr{G}]\,\mathbb{1}_D] = \mathbb{E}[X\mathbb{1}_D].$$

Summarizing our three motivating examples, we see that $\mathbb{E}[X \mid Y]$ should:

(i) Equal $\mathbb{E}[X \mid \mathscr{G}]$ with $\mathscr{G} = \sigma(Y)$,

(ii) Be measurable with respect to $\mathscr{G}$, and

(iii) Satisfy the tower property.

It can be shown that there is a unique function satisfying all of these requirements. We call this conditional expectation.

Definition 7.2.2 (Conditional expectation with respect to a σ-field).

Suppose $(\Omega, \mathscr{F}, \mathbb{P})$ is a probability space, $\mathscr{G} \subseteq \mathscr{F}$ is a σ-field, and $X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ is a random variable such that $\mathbb{E}[|X|] < \infty$.
$\mathbb{E}[X \mid \mathscr{G}]$ is defined as the unique[a] $\mathscr{G}$-measurable function such that for every $A \in \mathscr{G}$,

$$\int_A \mathbb{E}[X \mid \mathscr{G}]\, d\mathbb{P} = \int_A X d\mathbb{P}.$$

[a]Up to null sets in $\mathscr{G}$, *not* in $\mathscr{F}$.

7.3 Existence and Uniqueness

Why does $\mathbb{E}[X \mid \mathscr{G}]$ exist and why is it unique? Define

$$\phi(A) = \int_A X d\mathbb{P}$$

for every $A \in \mathscr{G}$. ϕ is a signed measure and $\mathbb{P}$ is a measure on the smaller measurable space $(\Omega, \mathscr{G})$ and $\phi \ll \mathbb{P}$. By the Radon–Nikodym theorem applied to $(\Omega, \mathscr{G})$, there exists a unique function $\frac{d\phi}{d\mathbb{P}}$ such that

$$\int_A X d\mathbb{P} = \phi(A) = \int_A \frac{d\phi}{d\mathbb{P}} d\mathbb{P},$$

which we define to be $\mathbb{E}[X \mid \mathscr{G}]$.

Example 7.3.1.

If $\mathscr{G} = \{\varnothing, \Omega\}$, then

$$\mathbb{E}[X \mid \mathscr{G}] = \mathbb{E}[X].$$

Returning to one of our motivating examples, if $X = \mathbb{1}_A$ and $\mathscr{G} = \{\varnothing, \Omega, B, B^c\}$, then

$$\begin{aligned}\mathbb{E}[X \mid G] &= a\mathbb{1}_B + b\mathbb{1}_{B^c} \\ &= \begin{cases} a\mathbb{P}[B] = \mathbb{P}[A \cap B], & D = B \\ b\mathbb{P}[B^c] = \mathbb{P}[A \cap B^c], & D = B^c \\ a\mathbb{P}[B] + b\mathbb{P}[B^c] = \mathbb{P}[A], & D = \Omega \\ a \cdot 0 + b \cdot 0 = 0, & D = \varnothing \end{cases}.\end{aligned}$$

The third and fourth cases are redundant. If $\mathbb{P}[B] \neq 0$, then we have the solution

$$a = \begin{cases} \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]} = \mathbb{P}[A \mid B], & \mathbb{P}[B] \neq 0, \\ \text{arbitrary}, & \mathbb{P}[B] = 0, \end{cases}$$

$$b = \begin{cases} \mathbb{P}[A \mid B^c], & \mathbb{P}[B^c] \neq 0 \\ \text{arbitrary}, & \mathbb{P}[B^c] = 0 \end{cases}.$$

If $\mathscr{G} = \mathscr{F}$, then

$$\mathbb{E}[X \mid \mathscr{G}] = X.$$

Lemma 7.3.2 (Doob–Dynkin).

Suppose $Y : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ and $Z : (\Omega, \sigma(Y)) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$. Then there exists $f : (\mathbb{R}, \mathscr{B}_{\mathbb{R}}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ such that $Z = f(Y)$.

Proof of Lemma 7.3.2. The proof is standard: first check the property for Z simple, then for nonnegative Z, and finally for general Z. We only check the simple-function case.

The conditions of this lemma imply $\sigma(Z) \subseteq \sigma(Y)$.

We can write $Z = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i}$. Since $\sigma(Z) \subseteq \sigma(Y)$, we have

$$A_i \in \sigma(Y) = \left\{Y^{-1}B : B \in \mathscr{B}_{\mathbb{R}}\right\}.$$

Thus there exists $B_i \in \mathscr{B}_{\mathbb{R}}$ such that $A_i = Y^{-1}B_i$, which lets us write

$$\begin{aligned} Z(\omega) &= \sum_{i=1}^{n} a_i \mathbb{1}_{\omega \in A_i} \\ &= \sum_{i=1}^{n} a_i \mathbb{1}_{\omega \in Y^{-1}(B_i)} \\ &= \sum_{i=1}^{n} a_i \mathbb{1}_{Y(\omega) \in B_i} \\ &= f(Y), \end{aligned}$$

with $f(t) = \sum_{i=1}^{n} a_i \mathbb{1}_{t \in B_i}$. $\square$

Definition 7.3.3 (Conditional expectation with respect to a random variable).

For random variables X, Y define

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)].$$

By Doob–Dynkin, this is a function of Y.

Note that the statement

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathscr{G}]) \mathbb{1}_A] = 0, \quad \forall\, A \in \mathscr{G}$$

is equivalent to

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathscr{G}]) Y] = 0, \quad \forall\, Y \text{ s.t. } \sigma(Y) \subseteq \mathscr{G}.$$

This motivates some sort of orthogonality property — the conditional expectation $\mathbb{E}[X \mid \mathscr{G}]$ is a projection of X onto $\{Y : \sigma(Y) \subseteq \mathscr{G}\}$.

Similarly, since $\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)]$, we have

$$\mathbb{E}[(X - \mathbb{E}[X \mid Y]) g(Y)] = 0, \forall \text{ functions } g \text{ of } Y.$$

We have a condition (from where?) that $\mathbb{E}[(X - \mathbb{E}[X \mid Z]) Z] = 0$ for all Z such that $\sigma(Z) \subseteq \sigma(Y)$. By Doob–Dynkin, this is equivalent to the proposition.

We state these formally:

Proposition 7.3.4 (Conditional expectation as projection).

$$\mathbb{E}[X \mid \mathscr{G}] = \min_{Y : \sigma(Y) \subseteq \mathscr{G}} \mathbb{E}\left[(Y - X)^2\right],$$

while

$$\mathbb{E}[X \mid Y] = \min_{g} \mathbb{E}\left[(g(Y) - X)^2\right].$$

Remark 7.3.5. We have shown that if the second moment of X exists, we can define conditional expectation from the projection perspective. But the orthogonality condition is more general, as it does not require existence of the second moment.

7.4 Properties of Conditional Expectation

Proposition 7.4.1 (Properties of conditional expectation).

(i) If $\sigma(X) \subseteq \mathscr{G}$, then $\mathbb{E}[X \mid \mathscr{G}] = X$

(ii) If $\sigma(X) \perp\!\!\!\perp \mathscr{G}$, then $\mathbb{E}[X \mid \mathscr{G}] = \mathbb{E}[X]$

(iii) If $\mathscr{G}_0 \subseteq \mathscr{G}$, then

$$\mathbb{E}[\mathbb{E}[X \mid \mathscr{G}_0] \mid \mathscr{G}] = \mathbb{E}[\mathbb{E}[X \mid \mathscr{G}] \mid \mathscr{G}_0] = \mathbb{E}[X \mid \mathscr{G}_0].$$

(iv) If $X \leq Y$, then

$$\mathbb{E}[X \mid \mathscr{G}] \leq \mathbb{E}[Y \mid \mathscr{G}].$$

(v) $\mathbb{E}[aX + bY \mid \mathscr{G}] = a\mathbb{E}[X \mid \mathscr{G}] + b\mathbb{E}[Y \mid \mathscr{G}]$

(vi) Suppose $\{X_n\}$ is a increasing sequence of nonnegative functions converging to X almost everywhere. Then

$$\mathbb{E}[X_n \mid \mathscr{G}] \uparrow \mathbb{E}[X \mid \mathscr{G}].$$

(vii) For $X_n \geq 0$,

$$\mathbb{E}\left[\liminf_{n\to\infty} X_n \mid \mathscr{G}\right] \leq \liminf_{n\to\infty} \mathbb{E}[X_n \mid \mathscr{G}].$$

(viii) Suppose $X_n \to X$ almost everywhere, $|X_n| \leq Y$ almost everywhere, and $\mathbb{E}[Y] < \infty$. Then

$$\mathbb{E}[X_n \mid \mathscr{G}] \to \mathbb{E}[X \mid \mathscr{G}].$$

(ix) If $\sigma(Y) \subseteq \mathscr{G}$, then

$$\mathbb{E}[XY \mid \mathscr{G}] = Y\mathbb{E}[X \mid \mathscr{G}].$$

Proof of Proposition 7.4.1. Left as exercises; these are all routine. $\square$